AC – 15.06.2026
Item No. – 2.11(N)

As Per NEP 2020

TERNA ENGINEERING COLLEGE

Bachelor of Engineering of

Computer Engineering

First Year Engineering (F.E.) Syllabus

Semester I to II
for
Academic Year 2026-27

Prepared by: Board of Studies: Date: 14.06.2026
for
Science and Humanities

Approved By: Academic Council: Date:15.06.2026
of
Terna Engineering College

Rev-2026

Ref: As per GR dated 4th July 2023 for Credit Structure of UG
(As per AICTE & NEP 2020 Guidelines)

PREAMBLE

"Think. Design. Innovate. – Learning that Leads the Future."

Terna Engineering College is proud to be among the oldest institutions to attain academic autonomy. This achievement strengthens our commitment to becoming a world-class centre for socioeconomic development and quality education. It is with great pride and purpose that we introduce the restructured curriculum under academic autonomy at **Terna Engineering College**. This initiative marks a significant step toward transforming engineering education by equipping our graduates with strong technical foundations, research acumen, interdisciplinary perspective, and an innovation-driven mindset is essential for success in today's evolving professional landscape.

Curriculum Design: Theme-Based, Experiential, and Outcome-Oriented. The autonomous curriculum adopts a bottom-up, learner-centric approach aligned with all 11 Program Outcomes. Each semester is driven by a central societal or technological theme, with problem statements rooted in local needs, socio-economic priorities, and the **UN Sustainable Development Goals (UNSDGs)**. Students engage with these challenges through individual projects in the **NIRMAN LAB,** integrating social science methodologies with engineering solutions to foster innovation, contextual learning, and socially responsible problem-solving.

Purpose of Autonomy: Empowering Innovation and Real-World Learning. Our pursuit of autonomy is rooted in bridging the gap between academic learning and industry relevance. By integrating undergraduate research through the **NIRMAN LAB**—our dedicated innovation and problem-solving hub—students engage with real-world challenges, apply interdisciplinary knowledge, and develop socially impactful solutions. This approach nurtures engineers who are not only technically sound but also responsible and creative thinkers.

Alignment with National Education Policy-2020: Aligned with NEP 2020, the curriculum emphasizes multidisciplinary, hands-on learning. Through theme-based projects, electives, value-added courses, and flexible assessments, **Terna Engineering College** ensures holistic student development driven by real-world application and academic rigor. **Terna Engineering College**, as a part of **TPCTs**, benefits from being in a network of HEIs. All Higher Education Institutions under **TPCTs** can collectively contribute to curriculum development, promoting multidisciplinary studies.

Innovative Pedagogy Initiative by IQAC: As part of its transformation strategy, IQAC has introduced an innovative pedagogy model centered on interdisciplinary research and contextual problem-solving. The **NIRMAN LAB** enables students to apply subject knowledge through socially relevant projects, supported by design thinking and faculty mentorship. A two-tier assessment framework—covering both project execution and subject-wise application—ensures academic depth, continuous mentoring, and quality assurance.

Empowering Faculty as Mentors and Innovators: Terna Engineering College believes that the success of academic autonomy lies in the hands of its faculty. By empowering educators to think beyond conventional boundaries, we enable them to instill Knowledge, Skills, and Attitudes (KSA) in students while enriching their own professional growth. Faculty serve as research mentors and project guides, supported through structured training in innovation pedagogy and assessment.

Sincerely

Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-Academic Council

Dr. Deven Shah
Principal
Chairman-Academic Council

PREFACE

It gives me immense pleasure to present the Curriculum Structure and Examination Scheme for the First year Engineering Programme of Terna Engineering College (Autonomous), effective from the Academic Year 2026–27.

First year Engineering plays a crucial role in shaping the academic foundation, professional attitude, and problem-solving abilities of students. Keeping this in view, the curriculum has been carefully designed in alignment with the National Education Policy (NEP) 2020, Outcome-Based Education (OBE) principles, and the evolving requirements of industry, society, and emerging technologies.

This curriculum provides a strong grounding in Mathematics, Basic Sciences, Computing, Electrical Engineering, Engineering Graphics, and Communication Skills while simultaneously fostering innovation, ethical values, and interdisciplinary thinking. Special emphasis has been placed on experiential learning through laboratory work, project-based activities, workshops, and the Nirman Lab, enabling students to transform theoretical concepts into practical solutions.

Recognizing the transformative impact of Artificial Intelligence and digital technologies across engineering disciplines, the curriculum incorporates courses such as Professional Communication Skills for Engineers in the Age of AI, Programming for Problem Solving, Data Structures, Network Science, and Web Technology. These courses are designed to equip students with computational thinking, analytical skills, and technological competence necessary for the future workforce.

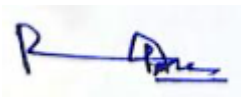
The curriculum also promotes self-learning, critical thinking, creativity, social responsibility, and lifelong learning habits. Courses such as Social Innovation and Engineering Ethics, Indian Knowledge Systems, NSS & Civil Defense, and value-added skill enhancement modules aim to nurture responsible engineers who contribute meaningfully to society and sustainable development.

The examination scheme has been structured to ensure continuous assessment of student learning through Internal Assessment Tests, Continuous Evaluation components, practical examinations, oral examinations, and end-semester examinations. This balanced assessment framework encourages consistent engagement and holistic development of learners.

I sincerely appreciate the efforts of the Board of Studies members, faculty experts, industry professionals, alumni, and academic stakeholders whose valuable contributions have helped in designing this contemporary and learner-centric curriculum.

I am confident that this curriculum will provide students with a strong academic foundation, professional competence, ethical values, and innovative thinking required to excel in higher education, research, entrepreneurship, and industry.

I extend my best wishes to all students and faculty members for successful implementation of this curriculum and for achieving excellence in engineering education.



Prof.R.D.Patane
BoS-Chairman
Science and Humanities



Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-Academic Council



Dr. Deven Shah
Principal
Chairman-Academic Council

Contents

Sr. No.	Item
1	Abbreviations
2	Curriculum Scheme for First Year Engineering (F.E.)
3	F.E. Syllabus

Abbreviations

AEC	Ability Enhancement Course
AECL	Ability Enhancement Course Laboratory
AU	Audit Course
BSC	Basic Science Course
BSCL	Basic Science Course Laboratory
CC	Co-Curricular Courses
CEP	Community Engagement Project
ELC	Experiential Learning Course
ESC	Engineering Science Course
ESCL	Engineering Science Course Laboratory
FP	Field Project
BESC	Basic Engineering Science Course
BESL	Basic Engineering Science Laboratory
HSSM	Humanities Social Sciences and Management
IKS	Indian Knowledge System
INTR	Internship
L	Lecture
LC	Laboratory Course
LLC	Liberal Learning Course
MDMC	Multidisciplinary Minor Course
MDML	Multidisciplinary Minor Laboratory
MJP	Major project
MP	Mini Project
NL	Notional Learning
OE	Open Elective
OJT	On Job Training
P	Practical
PCC	Program Core Course
PCL	Program Core Laboratory
PEC	Program Elective Course
PEL	Program Elective Laboratory
RP	Research Project
SBL	Skill-Based Laboratory
SEC	Skill Enhancement Course
SL	Self-Learning
T	Tutorial
VEC	Value Education Course
VSEC	Vocational and Skill Enhancement Course
VSC	Vocational Skill Course

Curriculum Structure and Examination Scheme
for
First Year of Computer Engineering

Terna Engineering College(With Effect from 2026-2027)

Semester-I

Course Code	Course Type	Course Description	Teaching & Learning Scheme								
			Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per Semester)					Total Credit (C) Total Hours (Notional Learning Hours)/30
			L	T	P	L	T	P	SL	Notional Learning Hour	
1111101	BESC101	Computational Engineering Mathematics - I	2	--	--	30	--	--	30	60	2
1111102	BESC102	Basic Electrical & Electronics Engineering	3	--	–	45	--	--	45	90	3
1111103	BESC103	Applied Physics for Computer Technology	2	--	–	30	--	--	30	60	2
1111104	BESC104	Programming for Problem Solving Using C	2	--	--	30	--	--	30	60	2
2991105	BESC105	Social innovation and Engineering Ethics	2	--	--	30	--	--	30	60	2
1111106	BEST101	Computational Engineering Mathematics - I Tutorial	--	2	--	--	30	--	–	30	1
1111107	BESL101	Applied Physics Lab	--	--	2	--	--	30	--	30	1
1111108	BESL102	Basic Electrical & Electronics Engineering Lab	--	--	2	--	--	30	--	30	1
1111109	BESL103	Programming for Problem Solving Using C	--	--	2	--	--	30	--	30	1
1111110	BESCL104	Nirman Lab-I	--	---	2	--	--	30	--	30	1
1111401	VSEC101	Workshop I (InnoTech Workshop)	--	--	2*+2	--	--	60	--	60	2
2991501	AEL101	Professional Communication Skills for Engineers in the Age of AI	--	--	2*+2	--	--	60	--	60	2
	Total		**11**	**02**	**16**	**165**	**30**	**240**	**165**	**600**	**20**

* Two hours of practical class to be conducted for the full class as a demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Note:

1) **AEC101, AEL101 and BESC105** will be common to all branches.

2) **BSC, ESC** and the corresponding lab are Group Specific.

3) **Questions based on self-learning** must not be asked in the internal assessment and the end-of-semester exam.

4) * **Nirman Lab**: The Nirman Lab fosters individual interdisciplinary projects through theme-based, subject-integrated, experiential learning, assessed via standardised rubrics for innovation, application, and documentation."

Notional Learning Hours: In higher education, Notional Learning Hours (NLH) indicates the estimated time that a typical learner, at a specific academic level, is expected to invest in order to achieve the set learning outcomes of a course or module. This time includes a range of structured and independent learning activities, each contributing to the development of knowledge and competencies within a module.

Self-Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

Course Code	Course Type	Course Description	Examination scheme							
			Continuous Assessment (CA)			End Sem. Exam Marks (ESE)	End Sem. Exam Duration of Theory (Hrs)	Term Work (Tw)	Pract & Oral	Total
			IAT-I	IAT-II	Total =IAT-I+IAT-II					
1111101	BESC101	Computational Engineering Mathematics - I	20	20	40	60	02	--	--	100
1111102	BESC102	Basic Electrical & Electronics Engineering	20	20	40	60	02	--	--	100
1111103	BESC103	Applied Physics for Computer Technology	20	20	40	60	02	--	--	100
1111104	BESC104	Programming for Problem Solving Using C	20	20	40	60	02	--	--	100
2991105	BESC105	Social innovation and Engineering Ethics	20	20	40	60	02	--	--	100
1111106	BEST101	Computational Engineering Mathematics - I Tutorial	--	--	--	--	--	25	--	25
1111107	BESL101	Applied Physics Lab	--	--	--	--	--	25	–	25
1111108	BESL102	Basic Electrical & Electronics Engineering Lab	--	--	--	--	--	25	–	25
1111109	BESL103	Problem Solving Using C Lab	--	--	--	--	--	25	25	50
1111110	BESCL104	Nirman Lab-I	--	--	--	--	--	25	25	50
1111401	VSEC101	Workshop I (InnoTech Workshop)	--	--	--	--	--	25	25	50
2991501	AEL101	Professional Communication Skills for Engineers in the Age of AI	--	--	--	--	--	25	25	50
	Total		100	**100**	**200**	**300**	**10**	**175**	**100**	**775**

Curriculum Structure and Examination Scheme
for
First Year of Computer Engineering

Terna Engineering College(With Effect from 2026-2027)

Semester-II

Course Code	Course Type	Course Description	Teaching & Learning Scheme								
			Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per Semester)					Total Credit (C) Total Hours (Notional Learning Hours)/30
			L	T	P	L	T	P	SL	Notional Learning Hour	
1112101	BESC201	Computational Engineering Mathematics - II	2	--	--	30	--	--	30	60	2
1112102	BESC202	Data Structure for Engineering Science	3	--	–	45	--	--	45	90	3
1112103	BESC203	Engineering Mechanics & Graphics	2	--	--	30	--	--	30	60	2
1112104	BESC204	Foundation of Network Science	3	--	--	45	--	--	45	90	3
1112105	PCC201	Web Technology	2	--	--	30	--	--	30	60	2
1112106	BEST201	Computational Engineering Mathematics - II Tutorial	--	2	--	--	30	--	–	30	1
1112107	BESL202	Data Structure Lab	--	--	2	--	--	30	--	30	1
1112108	BESL203	Engineering Design Graphics Lab	--	--	2	--	--	30	--	30	1
1112109	BESL204	Network Science Lab	--	--	2	--	--	30	--	30	1
1112110	BESL205	Nirman Lab-II	--	--	2	--	--	30	--	30	1
1112111	PCL201	Web Technology Lab	--	--	2	--	--	30	--	30	1
1112401	VSEC201	Workshop -II(FAB Lab)	--	--	2*+2	--	--	60	--	60	2
2992601	CC201	NSS & Civil Defense/Yoga for Adolescents	--	--	2*+2	--	--	60	--	60	2
2992501	IKS201	Indian Wisdom for Modern Challenges	--	--	2*+2	--	--	60	--	60	2
	Total		**12**	**02**	**22**	**180**	**30**	**330**	**180**	**720**	**24**

* Two hours of practical class to be conducted for the full class as a demo/discussion.

Theory / Tutorial 1 credit for 1 hour and Practical 1 credit for 2 hours

Note:

1) **IKS201** will be common to all branches.

2) **BSC, ESC** and the corresponding lab are Group Specific.

3) **Questions based on self-learning** must not be asked in the internal assessment and the end-of-semester exam.

4) *** Nirman Lab**: The Nirman Lab fosters individual interdisciplinary projects through theme-based, subject-integrated,

experiential learning, assessed via standardised rubrics for innovation, application, and documentation."

Notional Learning Hours: In higher education, Notional Learning Hours (NLH) indicates the estimated time that a typical learner, at a specific academic level, is expected to invest in order to achieve the set learning outcomes of a course or module. This time includes a range of structured and independent learning activities, each contributing to the development of knowledge and competencies within a module.

Self-Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

Course Code	Course Type	Course Description	Examination scheme							
			Continuous Assessment (CA)			End Sem. Exam Marks (ESE)	End Sem. Exam Duration of Theory (Hrs)	Term Work (Tw)	Pract & Oral	Total
			IAT-I	IAT-II	Total =IAT-I+IAT-II					
1112101	BESC201	Computational Engineering Mathematics - II	20	20	40	60	02	--	--	100
1112102	BESC202	Data Structure for Engineering Science	20	20	40	60	02	--	--	100
1112103	BESC203	Engineering Mechanics & Graphics	20	20	40	60	02	--	--	100
1112104	BESC204	Foundation of Network Science	20	20	40	60	02	--	--	100
1112105	PCC201	Web Technology	20	20	40	60	02	--	--	100
1112106	BEST201	Computational Engineering Mathematics - II Tutorial	--	--	--	--	--	25	--	25
1112107	BESL202	Data Structure Lab	--	--	--	--	--	25	25	50
1112108	BESL203	Engineering Design Graphics Lab	--	--	--	--	--	25	–	25
1112109	BESL204	Network Science Lab	--	--	--	--	--	25	25	50
1112110	BESL205	Nirman Lab-II	--	--	--	--	--	25	25	50
1112111	PCL201	Web Technology Lab	--	--	--	--	--	25	25	50
1112401	VSEC201	Workshop -II(FAB Lab)	--	--	--	--	--	50	--	50
2992601	CC201	NSS & Civil Defense/Yoga for Adolescents	--	--	--	--	--	25	--	25
2992501	IKS201	Indian Wisdom for Modern Challenges	--	--	--	--	--	25	--	25
	Total		100	**100**	**200**	**300**	**10**	**250**	**100**	**850**

F.E. Semester – I

Bachelor of
Computer Engineering

Syllabus

Semester I

Theme:Digital Bharat for Inclusive Learning

Aligned with UNSDG

- SDG 4: Quality Education
- SDG 9: Industry, Innovation and Infrastructure
- SDG 10: Reduced Inequalities
- SDG 11: Sustainable Cities and Communities
- SDG 17: Partnerships for the Goals

Keywords

Digital Inclusion, Smart Education, Accessibility, Digital Literacy, Community Learning, Technology for Society, Inclusive Development, Citizen Empowerment, Lifelong Learning, Educational Equity

Theme Description

India's digital transformation has created unprecedented opportunities for education, skill development, and knowledge dissemination. However, ensuring that these opportunities reach every section of society remains a significant challenge. In rapidly growing urban regions like Navi Mumbai, diverse populations including students, working professionals, senior citizens, migrant communities, and economically disadvantaged groups require accessible, affordable, and inclusive learning opportunities.

The theme **"Digital Bharat for Inclusive Learning"** focuses on leveraging digital technologies to create equitable access to education, information, skills, and opportunities. It encourages students to explore innovative approaches that bridge digital divides, promote lifelong learning, and support community development through technology-enabled solutions.

Generalized Problem Statements

1. Digital Accessibility and Inclusive Education

Many learners face barriers in accessing educational resources due to differences in language, economic background, physical abilities, and digital literacy levels. There is a need to improve accessibility and inclusiveness in learning opportunities across Navi Mumbai.

2. Community-Based Digital Literacy and Skill Development

Despite increasing digital adoption, several sections of society lack the necessary digital skills required for education, employment, and participation in modern society. Strengthening digital literacy remains a critical challenge.

3. Equitable Access to Learning Resources

Learners from different socio-economic backgrounds often experience unequal access to educational content, learning tools, and technology-enabled learning environments. Mechanisms are required to ensure equitable learning opportunities for all.

4. Technology-Enabled Citizen Awareness and Empowerment

Citizens require access to reliable information regarding education, public services, health, governance, and skill development. Improving awareness and engagement through digital means remains an important societal need.

5. Lifelong Learning for Sustainable Community Development

Rapid technological and societal changes require continuous learning and skill enhancement. Communities need effective approaches to support lifelong learning and knowledge sharing across different age groups.

6. Digital Inclusion for Smart and Sustainable Navi Mumbai

As Navi Mumbai continues to grow as a modern urban center, there is a need to ensure that digital transformation benefits all sections of society, promotes social inclusion, and contributes to sustainable community development.

These six problem statements are intentionally broad so they can be mapped to:

- Engineering Graphics
- Basic Electrical Engineering
- Engineering Physics/Chemistry
- Programming Fundamentals
- Communication Skills
- Mathematics
- Environmental Studies
- Community Engagement Activities
- Innovation and Project-Based Learning

while maintaining a common first-year theme of **"Digital Bharat for Inclusive Learning."**

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1111101	Computational Engineering Mathematics - I	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111101	Computational Engineering Mathematics I	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The course provides fundamental knowledge of engineering mathematics essential for analytical modeling, computational techniques, and problem-solving in engineering applications. It develops mathematical proficiency through the study of matrices, complex numbers, partial differentiation, and numerical methods, which form the basis for data analysis, optimization, intelligent systems, and computational algorithms. The course further enhances logical reasoning, quantitative analysis, and systematic thinking required in the fields of Artificial Intelligence, Computer Engineering, and Information Technology.

Course Objectives:

Sr. No.	Course Objectives
1	To apply matrix algebra and linear systems in solving computational and engineering problems.
2	To analyze complex numbers, De Moivre's theorem, and related expansions for engineering applications.
3	To evaluate circular, hyperbolic, and logarithmic functions of complex variables in mathematical modeling.
4	To differentiate multivariable functions using partial differentiation techniques for engineering analysis.
5	To formulate and optimize mathematical models using maxima-minima and successive differentiation methods.
6	To implement numerical techniques for solving algebraic, transcendental, and system-based engineering problems.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Apply matrix operations and rank methods to obtain solutions of systems of linear equations arising in engineering applications. *(Apply)*	L3
2	Analyze and compute powers, roots, and expansions of complex numbers using De Moivre's theorem. *(Analyze)*	L4
3	Evaluate circular, hyperbolic, and logarithmic functions of complex variables for mathematical and engineering analysis. *(Evaluate)*	L5
4	Differentiate functions of two and three variables using partial differentiation techniques. *(Apply)*	L3
5	Optimize multivariable functions using maxima-minima and successive differentiation methods. *(Evaluate)*	L5
6	Implement numerical methods to solve algebraic, transcendental, and linear system equations in engineering computations. *(Implement)*	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Algebraic operations and factorization, Trigonometric identities and equations, Coordinate geometry and graphs, Elementary differential calculus and functions ,Complex numbers and basic matrix operations.	02	-
1	**Matrices**	1.1 Properties of Matrices: Hermitian, Skew Hermitian, Unitary, Orthogonal Matrices. 1.2 The rank of a Matrix using Echelon form, reduction to normal form, and PAQ form, The rank of a Matrix reduction to PAQ form. 1.3 System of homogeneous and non – non-homogeneous equations, their consistency, and solutions **Self- Learning topic**- Review of Inverse of a matrix, addition, multiplication, and transpose of a matrix, symmetric, skew-symmetric Matrix. **Online Reference:** https://nptel.ac.in/courses/111106051?utm_source https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/?utm_source	**05**	CO1

2	**Complex Numbers**	2.1 D Moivre's Theorem, 2.2 Expansion of functions, Expansion of sin^n θ, cos^n θ in terms of sines and cosines of multiples of θ; Expansion of sinnθ, cosnθ in powers of sinθ, cosθ. 2.3 Powers and Roots of a complex number **Self- Learning topic**- Review of Complex Numbers - Algebra of Complex Numbers, Cartesian, polar and exponential form of complex number. **Online Reference:** https://nptel.ac.in/courses/111103070?utm_source	**04**	CO2
3	**Hyperbolic Functions & Logarithms of Complex Numbers**	3.1 Definition of Circular functions of complex numbers and Hyperbolic functions. 3.2 Inverse Circular and Inverse Hyperbolic Functions, Separation of real and imaginary parts of all types of Functions. 3.3 Logarithm of Complex Number (Simple examples). **Self- Learning topic**- Application of complex numbers, Rotation and transformation using complex numbers, Logarithm of complex numbers in cyber security. **Online Reference:** https://onlinecourses.nptel.ac.in/noc24_ma20/preview?utm_source	**04**	CO3
4	**Partial Differentiation**	4.1 Basic of derivatives , Partial Differentiation: Function of two and three variables, Partial derivatives of first and higher order. 4.2 Differentiation of composite function. 4.3 Euler's Theorem on Homogeneous functions with two independent variables (with proof), Deductions from Euler's Theorem (without proof). **Self- Learning topic**- Total Differentiation, implicit functions, Euler's theorem for three independent variables. **Online Reference:** https://nptel.ac.in/courses/111104229?utm_soure https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/?utm_source	**05**	CO4
5	**Applications of Partial Differentiation and Successive Differentiation**	5.1 Maxima and Minima of a function of two independent variables. 5.2 Successive differentiation: nth derivative of standard functions, Leibnitz's Theorem (without proof) and simple examples	**04**	CO5

		Self- Learning topic- Jacobian's of two and three independent variables (simple problems) Lagrange's Multiplier method. **Online Reference:** https://nptel.ac.in/courses/111107122?utm_source		
6	**Numerical methods**	6.1 Solution of Transcendental Equations: Solution by Newton Raphson method, Solution by Regula –Falsi method 6.2 Solution of a system of linear algebraic equations, by (1) Gauss Jacobi Iteration Method, by (2) Gauss Seidel Iteration Method. **Self- Learning topic**- Indeterminate forms , L-Hospital Rule, Gauss Elimination Method, Gauss Jordan Method. **Online Reference:** https://nptel.ac.in/courses/111105090?utm_soure https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-fall-2010/?utm_source	**05**	CO6
			Total Hours	30

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References :

1. Bali N.P and Goyal M. "A textbook of Engineering Mathematics" Laxmi Publication, 10th Ed. 2022.
2. Gupta C.B. Singh S.R and Mukesh Kumar : "Engineering Mathematics for semester I and II", Mc- Graw Hill education ,2015
3. Pal S. & Bhunia S.C: "Engineering Mathematics " Oxford University Press, 3rd Ed, 2016.
4. Higher Engineering Mathematics, Dr.B.S.Grewal, Khanna Publication
5. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Narosa publication
6. Matrices, Shanti Narayan, S. Chand publication.
7. Elementary Linear algebra with application by Howard Anton and Christ Rorres. 6th edition.
8. Numerical Methods for Scientific and Engineering Computation ,M. K. Jain, S. R. K. Iyengar and R. K. Jain , New Age International Publishers.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://nptel.ac.in/
2	MIT OpenCorseWare	https://ocw.mit.edu/courses
3	Wolfram Mathworld	http://mathworld.wolfram.com

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1111102	Basic Electrical &Electronics Engineering	03	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111102	Basic Electrical &Electronics Engineering	20	20	40	60	2	—	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: The theme "Digital Bharat for Inclusive Learning" aims to promote the use of technology for accessible, affordable, and sustainable education and community development. Basic Electrical Engineering provides the foundation of electrical systems, electronic devices, digital technologies, and energy-efficient solutions used in modern digital infrastructure.

Course Objectives:

Sr. No.	Course Objectives
1	To provide knowledge on fundamentals of DC circuits
2	To provide knowledge of single phase and three phase AC circuits.
3	To inculcate fundamental knowledge of the 1-Φ transformer.
4	To provide basic knowledge on fundamentals of DC and AC machines.
5	To provide knowledge of fundamentals of Digital Systems.
6	To provide knowledge of fundamentals of batteries and energy Storage systems.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Apply various network theorems and circuit analysis techniques to determine the behavior of DC circuits.	L3
2	Analyze the performance of single-phase and three-phase AC circuits using electrical circuit concepts and phasor relationships.	L4
3	Explain the construction, operation, and applications of single-phase transformers.	L2
4	Illustrate the working principles and applications of DC motors, single-phase induction motors, and three-phase induction motors.	L2
5	Apply the fundamentals of digital systems and logic gates in basic digital circuit analysis.	L3

6	Explain the fundamentals of batteries and energy storage systems used in modern electrical and electronic applications.	L2

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Series and parallel connections of resistance, concepts of voltage, current, power and energy and its units.	01	-
1	DC Circuits	1.1 Kirchhoff's Laws, Ideal and Practical Voltage and Current Sources, Source Transformation, Mesh analysis, Star-Delta / Delta-Star Transformations, 1.2 Superposition Theorem, Thevenin's Theorem, Norton's Theorem and Maximum Power Transfer Theorem. **Self-learning topics**:Application of Kirchhoff's Laws in household and automotive electrical circuits, Simulation-based verification of network theorems using open-source tools https://onlinecourses-archive.nptel.ac.in/noc18_ee18/preview?utm_source	8	CO1
2	AC Circuits	2.1 Generation of alternating voltage, basic definitions, average and RMS values for sinewave only, 2.2 phasor and phase difference, Single-phase ac series and parallel circuits consisting of R, L, C, RL, RC, RLC combinations, definitions - real, reactive and apparent power, 2.3 Generation of Three-Phase Voltages, voltage & current relationships in Star and Delta Connections. **Self-learning topics:** Function of phase, neutral, and earth wire, Power rating of appliances,Monthly electricity consumption **Online Reference:** https://onlinecourses.nptel.ac.in/noc23_ee81/preview	8	**CO2**
3	Single Phase Transformer	3.1 Working principle of single-phase transformer, types of single-phase transformer, transformation ratio, actual (practical) and ideal transformer, Transformer losses, efficiency, applications of transformer. **Self-learning topics**: Build foundation for power systems, electric drives, and renewable energy applications. **Online Reference:** https://nptel.ac.in/courses/108105155	04	**CO3**
4	Electrical Machines	4.1 principle of operation, constructional details, classification and applications of DC Motor, three-phase induction motor and Single-Phase induction Motor, Introduction to BLDC Motor. **Self-learning topics**: Develop skills in circuit connections, testing, and troubleshooting. **Online Reference:** https://nptel.ac.in/courses/108105155	04	**CO4**
5	Fundamentals of Digital Systems	5.1 Number System: Binary, octal, decimal,Hexadecimal, their conversion. Combinational circuit design using 3 and 4 variable K-Map.	04	**C05**

		Self-learning topics: Logic Gates and truth table, universal gate and truth table. **Online Reference:** https://archive.nptel.ac.in/content/syllabus_pdf/117106086.pdf?utm_source		
6	Electrical Energy Storage	6.1 Introduction to types of batteries, generalized battery parameters such as SoC, DoD, energy and power densities, battery C rating, comparison of Charging and discharging characteristics. **Self-learning topics**: Lithium-Ion Battery Chemistries, Sustainable Energy Storage Technologies, **Online Reference:** https://onlinecourses-archive.nptel.ac.in/noc19_ee18/preview?utm_source	02	**CO6**
		Total Hours	30	

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References :

1. B.L. Theraja “Electrical Engineering “ Vol-I and II
2. D C Kulshreshtha,” Basic Electrical Engineering”
3. Patrick T. Moseley, Jürgen Garche,”Electrochemical Energy Storage for Renewable Sources and Grid Balancing”
4. Hughes ,”Electrical and Electronics Technology”

Online References:

Sr. No.	Website Name	URL
1	All About Circuits	https://www.allaboutcircuits.com
2	Circuit Lab	https://www.circuitlab.com
3	Tinker cad	https://www.tinkercad.com

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage

of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1111103	Applied Physics for Computer Technology	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR/ OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111103	Applied Physics for Computer Technology	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: The course *Applied Physics for Computer Technology* is designed as per the principles of NEP 2020 with emphasis on multidisciplinary, skill-based, experiential, and industry-oriented learning aligned to the semester theme *"Digital Bharat for Inclusive Learning."* The course provides a scientific foundation for understanding modern computing technologies and digital systems essential in the era of Artificial Intelligence, Data Science, intelligent computing, IoT, and smart technologies. It develops analytical thinking, problem-solving ability, and scientific aptitude required for emerging technological domains. The curriculum includes concepts such as semiconductor physics, optical fibre communication, lasers, sensors, nanomaterials, quantum physics, smart materials, and electronic devices that form the basis for advanced subjects in computer technology and intelligent systems. Through laboratory experiments, simulations, and activity-based learning using modern and open-source tools, students gain practical exposure to scientific measurement, data analysis, and electronic instrumentation. The course supports experiential learning and enhances technical, computational, and decision-making skills relevant to industry applications.

The knowledge gained through this course is applicable in healthcare diagnostics, smart agriculture, digital banking, e-governance, intelligent transportation, smart cities, and sustainable digital infrastructure. The course contributes to India's digital transformation initiatives and aligns with UNSDG 4: Quality Education and UNSDG 9: Industry, Innovation and Infrastructure by promoting technology-enabled learning, innovation, and sustainable development.

Course Objectives:

Sr. No.	Course Objectives
1	To introduce the fundamental concepts of quantum mechanics and their role in the development of modern technological systems.
2	To develop understanding of modern physics concepts and semiconductor devices used in electronic and engineering applications.
3	To explain the principles and applications of lasers and optical fibers in communication systems, computers, and modern technologies.
4	To provide knowledge of acoustics and ultrasonic wave generation and their applications in medical, industrial, and electronic equipment..

5	To develop understanding of electromagnetism and its applications in sensors and modern engineering devices.
6	To introduce the concepts of nanotechnology and smart materials and explore their applications in advanced engineering systems.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Learners will be able to relate the foundation of Quantum mechanics with the development of modern technology.	L2
2	Understand fundamental concepts of modern physics and semiconductor devices.	L2
3	Apply the principles of lasers and Fiber optics in computers and modern communication technology.	L3
4	Understand the principles of acoustics and ultrasonic generation and apply them in computer and medical equipment	L3
5	Understand the principles of electromagnetism and apply them in various sensors.	L3
6	Explore the applications of nanotechnology and smart materials in modern engineering systems	L4

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Basic knowledge of optics, atomic structure Wavefront and Huygens principle, reflection and refraction, Refractive index of a material, Snell's law. Basics of vector algebra, partial differentiation concepts, Dual nature of radiation, Photoelectric effect, matter waves. Intrinsic and extrinsic semiconductors, electrical resistivity and conductivity concepts	02	-
1	**Quantum Physics**	1.1 Wave-particle duality: Wave-particle duality, de Broglie's hypothesis of matter waves, Properties of matter waves. de Broglie wavelength for electron 1.2 Heisenberg's uncertainty principle: Statement of Heisenberg's uncertainty principle, nonexistence of electron in nucleus. 1.3 Photoelectric effect: Statement of Photoelectric effect, Numerical Problems 1.4 Quantum computing: Classical versus quantum computing, limitations of classical computers, qubits, superposition, and measurement principles. Qubits and quantum gates: Mathematical representation of qubits, probability amplitudes, normalization and basic quantum gates (X,Z, and Hadamard). 1.5 Quantum Circuits and entanglement: Multi Qubit systems, entanglement, Bell states, CNOT gate, and construction of simple Quantum circuits.	05 05	CO1

		Self-Learning Topics: Davisson-Germer experiment. Equation of Wave function, Operators in Quantum Mechanics (basic idea). Applications and future pathways: Introduction to Quantum algorithms, applications in cryptography, AI, materials science, and roadmap to advanced quantum computing concepts. Online reference: **https://nptel.ac.in/courses/122106034** **https://nptel.ac.in/courses/106106232**		
2	**Semiconductor Devices**	2.1 Energy band structure in solids: Energy bands in solids, Conductors, semiconductors, insulators, Intrinsic and 2.2 Concept of Femi level: Fermi level in Intrinsic and extrinsic semiconductors. 2.3 PN junction diode: Formation of PN junction diode, Forward and reverse biased diode with IV characteristics. 2.4 Important diodes: LED, solar cell. Zener diode as voltage regulator. 2.5 Light sensors: LDR, Photo diode and photo transistor-characteristics, Analogue to digital converter (only basic concepts) **Self-learning Topics:** Direct and indirect band gap semiconductors. Applications of Semiconductor devices in computers and mobile systems Sensors and digital electronics. Micro controller connection basics. https://nptel.ac.in/courses/108106181	05 05	CO2
3	**Laser and Fiber Optics**	3.1 Quantum processes in Laser Technology: Spontaneous and stimulated emission, population inversion, pumping, active medium & active centre, resonance cavity. 3.2 Characteristics and Applications of lasers: Characteristics of lasers. Application in Holography, medical and industrial Applications. 3.3 Basic Principles of Optical Fiber: Total internal reflection, Critical angle. 3.4 Optical Fiber Parameters: acceptance angle, acceptance cone, numerical aperture, propagation of light. 3.5 Applications of Optical Fiber: Applications of Optical fibres in structural and health Monitoring. Fiber optic communication, Numerical problems. **Self-learning Topics:** Applications of optical fibres in medicine and industry. Optical networking, Laser links connecting servers, switches, and storage systems https://archive.nptel.ac.in/courses/115/102/115102124/\	05 05	CO3
4	**Acoustics and Ultrasonics**	4.1 Fundamentals of Acoustics: Basics of acoustics, Sound wave generation and propagation	05	CO4

		4.2 Sound Phenomena: Sound interference, Doppler effect, Numerical problems on sound waves and Doppler effect 4.3 Architectural Acoustics and Noise Control: Principles of sound absorbers, Noise generation and its impact Reverberation and Sabine's formula, Significance of reverberation, Numerical problems on reverberation 4.4 Generation of Ultrasonic Waves: Magnetostriction method, Piezoelectric method 4.5 Applications of Ultrasonics: Medical imaging, Smart devices, Industrial and engineering applications of ultrasonics **Self-learning Topics:** Applications of ultrasonics in non-destructive testing , Ultrasonic flaw detector , Applications in SONAR https://nptel.ac.in/courses/121108458	05	
5	**Electromagnetism and Sensors**	5.1 Fundamentals of Electromagnetism: Gauss's law in electricity and magnetism, Faraday's law of electromagnetic induction, Ampere's circuital law, Statement of Maxwell's equations 5.2 Hall Effect: Hall effect statement, Derivation of Hall voltage, Numerical problems on Hall effect 5.3 Magnetic Materials: Diamagnetic materials, Paramagnetic materials, Ferromagnetic materials, Medical and industrial applications of magnetic materials 5.4 Physics of Sensors and Transducers: Introduction to sensors and transducers, Resistive sensors, Temperature measurement using PT100, PT100 construction and calibration 5.5 Piezoelectric Sensors and Applications: Concept of piezoelectric effect, Piezoelectric transducer for distance measurement, Applications of piezoelectric sensors in engineering systems **Self-learning Topics:** Applications of various sensors; Motion sensors, Smart devices, Industrial automation, Healthcare monitoring systems. Basics of IoT sensors, Optical sensor, pyroelectric sensor and their applications. Analogue pressure sensor, construction, working and calibration and applications. Piezoelectric transducers for liquid and air velocity measurement https://nptel.ac.in/courses/117103065 https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_ee13	05 05	CO5
6	**Advanced Materials and Nanotechnology**	6.1 Introduction to Nanomaterials: Types of nanomaterials, Properties of nanomaterials, Optical properties, Electrical properties, Magnetic properties, Structural properties, Mechanical properties	05	CO6

		6.2 Carbon Nanotubes: Structure of carbon nanotubes, Properties of carbon nanotubes, Applications of carbon nanotubes 6.3 Smart Materials: Introduction to smart materials, Shape memory alloys, Piezoelectric nanomaterials, Quartz crystal and its applications 6.4 Superconductivity and Superconductors: Introduction to superconductivity, Superconductors, Critical temperature, Critical magnetic field, Meissner effect 6.5 Supercapacitors: Principle of supercapacitors, Construction of supercapacitors, Materials used in supercapacitors, Applications of supercapacitors **Self-learning Topics:** Magnetostrictive Materials. Electrochromic materials and their applications of piezoelectric nanomaterials. Self-healing materials. Nano sensors. Conducting polymers. Applications in nanoelectronics, smart system and modern computing hardware. Batteries as energy storage systems; Li battery, Zn carbon batteries Comparison of batteries with super capacitors. https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_mm38	05	
			Total Hours	30

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Dr. M. N. Avadhanulu, Dr. P. G. Kshirsagar, “A Textbook of Engineering Physics”S. Chand ,2018 edition
2. A. S. Vasudeva, S. Chand, “Modern Engineering Physics” Revised Edition Year varies with edition (commonly available revised editions around 2005–2012).
3. D. K. Bhattacharya, Poonam Tandon, “Engineering Physics”Oxford Higher Education 2011
4. R. K. Gaur, S. L. Gupta, “Engineering Physics”, Dhanpat Rai Publications 2012.
5. V. Rajendran, “Engineering Physics”, McGraw Hill Education 2013
6. T. Pradeep, “A Textbook of Nanoscience and Nanotechnology”, Tata McGraw Hill Education Pvt. Ltd.2007.

Online References:

Sr. No.	Website Name	URL
1	NPTEL – Optical Engineering	https://archive.nptel.ac.in/courses/115/102/115102124/
2	NPTEL – Engineering Electromagnetics	https://archive.nptel.ac.in/courses/115/102/115102025/
3	NPTEL – Introduction to Nanoscience and Nanotechnology	https://archive.nptel.ac.in/courses/115/105/115105132/

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	NL	
1111104	Problem Solving Using C	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111104	Problem Solving Using C	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The course is designed in alignment with the principles of NEP 2020 with emphasis on skill-based, experiential, multidisciplinary, and industry-oriented learning. Aligned with the Computer Engineering theme “Digital India”, the course introduces students to the fundamentals of C programming with focus on logical reasoning, algorithm development, and problem-solving skills. Students will learn structured and modular programming concepts using control structures, arrays, functions, pointers, structures, and file handling.

The curriculum integrates real-world applications such as banking systems, healthcare utilities, inventory management, smart billing systems, and digital record management to promote practical and application-based learning. Emphasis is also given to coding standards, debugging techniques, and secure programming practices used in modern software industries. The course supports the vision of Digital India by developing computational and technology-driven problem-solving abilities for future engineering applications. It also contributes toward UNSDG 4 (Quality Education), UNSDG 8 (Decent Work and Economic Growth), UNSDG 9 (Industry, Innovation and Infrastructure), UNSDG 11 (Sustainable Cities and Communities), and UNSDG 16 (Peace, Justice and Strong Institutions).

Course Objectives:

Sr. No.	Course Objectives
1	To introduce students to the fundamentals of programming using C language.
2	To develop logical thinking and problem-solving skills.
3	To enable students to write simple and structured C programs.
4	To familiarize students with decision making, loops, arrays, functions, and pointers.
5	To develop confidence in beginners for coding and debugging
6	To prepare students for advanced programming and software development courses.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Understand the basic structure and syntax of the C programming language.	L2
2	Develop C programs using operators, conditional statements, and loops.	L3
3	Implement programs using arrays and strings	L3
4	Design modular programs using functions for code reusability, parameter passing, and problem solving.	L6
5	Apply pointers and structures in problem solving.	L3
6	Perform file handling operations in C.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Introduction to computers, software and hardware, programming languages, algorithm and flowchart, basics of logic building, introduction to compiler and execution of program.		-
1	Computational Thinking and C Fundamentals	**1.1 Introduction to Problem Solving and Computational Thinking:** Problem Solving Concepts, Computational Thinking Basics, Algorithms, Flowcharts, Pseudocode **1.2 Fundamentals of C Programming:** Overview and Features of C Language, Structure of a C Program, Compilation and Execution Process, Tokens, Keywords, Identifiers, Variables, Constants, Data Types, Operators, Input-Output Functions **Self-Learning Topics:** Write-up on history and evolution of programming languages , Practicing Non-Formatted Inp& Output functions – getchar( ), putchar( ), getch( ), getche( ), putch( ), gets( ), puts( ) , Practicing Formatted Input & Output functions – printf( ) **Online References:** https://www.programiz.com/c-programming , https://www.geeksforgeeks.org/c-programming-language/	04	CO_01
2	Decision Making and Iterative Problem Solving	**2.1 Conditional statements** :if, if-else, nested if, switch-case, **2.2 Iterative statements** :for, while, do-while, nested loops, jump statements, menu-driven programming and problem solving using control structures. **Self-Learning:** Pattern programs (Pyramids , Star , Square , Upper triangular matrix , Lower triangular matrix) **Online References:**	06	CO_02

		https://www.tutorialspoint.com/cprogramming/c_decision_making.htm , https://www.programiz.com/c-programming/c-for-loop		
3	Arrays and Strings	**3.1 Introduction to Arrays** : Declaration ,definition ,accessing array elements,One-dimensional and two-dimensional arrays, matrix operations,array counting. **3.2 Strings:** Declaration, Initialization, String Operations, Array of Strings, String manipulation Functions in string.h **Self-Learning:** Sorting an array, Binary searching , Declaration and Initialization of a Multidimensional Array , Arrays of strings: Two-dimensional character array , Passing Arrays to Functions **Online References:** https://www.geeksforgeeks.org/arrays-in-c-cpp/ , https://www.programiz.com/c-programming/c-arrays	06	CO_03
4	Modular Programming with Functions	**4.1 Function:** Defining a Function, accessing a Function, Passing Arguments to a Function, call by value,call by Reference ,Recursion **4.2 Storage classes** : auto,static,register,extern. **Self-Learning:** Implement recursive solutions (e.g., Fibonacci, Tower of Hanoi) , Comparison of recursion and iteration (mini presentation) , Industry application: Use of arrays/strings in data processing. **Online References:** https://www.geeksforgeeks.org/functions-in-c/ , s	06	CO_04
5	Pointers and Structure	**5.1 Pointers:** Basics, address operator (&), Declaring & initializing Pointer Variables, Indirection Operator and Dereferencing, Pointer Expressions and Pointer Arithmetic, Types of Pointers, Pointer to Pointers, Pointers and Arrays, Call. by value vs Call by address,malloc, calloc, realloc, free. **5.2 Structures and Unions** :Declaring Structures and Structure Variables, Accessing the members of a structure, Initialization of Structures, typedef, Nested Structures, Arrays of Structures **Self-Learning:** Call by address using pointers, returning more than one value from a function, returning pointer from a function, structures and pointers, passing structure to function ,Structure versus Union, Enumeration types , Processing binary files	05	CO_05

		Online References: https://www.programiz.com/c-programming/c-pointer **nptel.ac.in/courses/106104128**		
6	File Handling	**6.1 Basic File Operations**: Opening, reading, writing, and closing a .txt file using fopen(), fscanf(), fprintf(), and fclose(). **Self-Learning:** Understand the purpose of file handling in C programming,Learn the difference between text files and binary files,File modes, **Online References:** https://www.programiz.com/c-programming/c-file-input-output https://www.geeksforgeeks.org/basics-file-handling-c/	03	CO_06
			Total Hours	**30**

Note:

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2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 8th Edition / 2019.
2. Brian W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Pearson Education, 2nd Edition / 2015.
3. Yashavant Kanetkar, "Let Us C", BPB Publications, 18th Edition / 2021.
4. Reema Thareja, "Programming in C", Oxford University Press, 2nd Edition / 2016.

Online References:

Sr. No.	Website Name	URL
1	Coursera – C Programming Specialization	https://www.coursera.org/specializations/c programming
2	NPTEL – Programming in C	https://onlinecourses.nptel.ac.in/e learning/preview/noc19_cs42
3	Udemy – C Programming Courses	https://www.udemy.com/topic/c-programming

Assessment:

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Question paper format:

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- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2991105	Social Innovation &Engineering Ethics	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2991105	Social Innovation &Engineering Ethics	20	20	40	60	2	—	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

_Engineering education today must prepare students not only for technical excellence but also for **social responsibility, ethical decision-making, sustainability, digital citizenship, and responsible use of technology and AI**. Computer Engineering students need early exposure to real-life social problems, community needs, environmental concerns, waste management, NGO/community linkage, and ethical challenges in engineering practice.

The course **Social Innovation and Engineering Ethics** introduces students to the role of engineers as socially responsible problem-solvers. It encourages students to identify real community problems, understand human needs, apply empathy and design thinking, explore waste-to-wealth practices, understand engineering ethics, and use technology responsibly for public good.

The course is aligned with contemporary educational priorities such as **sustainable development, experiential learning, social responsibility, AI ethics, digital safety, community engagement, and innovation for society**. The United Nations describes the 17 Sustainable Development Goals as an urgent call for action by all countries in a global partnership, making SDG awareness highly relevant for engineering education.

Course Objectives:

Sr. No.	Course Objectives
1	Understand the concept, need, and importance of social innovation in engineering education.
2	Recognize the role of engineers in addressing social, environmental, technological, and community challenges.
3	Understand basic principles of engineering ethics, professional responsibility, integrity, safety, and public welfare.

4	Conduct basic community study and identify real-time social problems through observation, survey, and NGO/community interaction.
5	Apply sustainable thinking and waste-to-wealth practices for solving local and campus-based problems.
6	Develop a simple community-based social innovation project, prototype, campaign, survey report, or awareness model.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the concept, need, and importance of social innovation and engineering ethics in contemporary society.	L2
2	Identify social, environmental, and community-based problems relevant to engineering solutions.	L3
3	Analyze ethical dilemmas related to technology, AI, sustainability, safety, privacy, and professional responsibility.	L4
4	Apply design thinking, empathy mapping, and problem-solving tools to propose socially useful engineering solutions.	L3
5	Develop a simple prototype, model, campaign, or project idea addressing a real social challenge.	L6
6	Demonstrate teamwork, communication, ethical reasoning, social responsibility, and reflective learning through project-based activities.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Basic awareness of society, community issues, environmental sustainability, communication skills, ethical values, digital literacy, and problem-solving ability. Familiarity with responsible use of technology and willingness to engage in community-based learning activities.	0	-

1	**Module 01:** **Social Innovation, Society and Engineering Responsibility**	1.1 Meaning and importance of social innovation 1.2.Role of engineers in social development 1.3.Relationship between society, technology and innovation 1.4.Human needs and community behaviour 1.5.Technology as a tool for social change 1.6.Inclusive and people-centred innovation 1.7.Local, rural and urban community issues 1.8.Engineering solutions for everyday problems 1.9.Social responsibility of engineers **Real-Time Problems** Lack of civic awareness >.< Poor access to basic digital services >.< Difficulty faced by elderly people in using technology >.< Communication gap between citizens and local authorities >.< Lack of awareness about government schemes >.< Low participation of youth in community improvement **Suggested Examples:** Low-cost water filter >.< Solar street light >.< Smart dustbin >.< Assistive device for differently abled people >.< Low-cost health monitoring device >.< E-waste collection system >.< Campus safety solution **Self learning Topics** **A. Understanding & Reflection** Write a short note on how engineering can solve one local community problem. >.< Compare invention, innovation, entrepreneurship, and social innovation with examples. >.< Identify 3 social problems in your campus or locality and suggest possible causes. **B. Problem-Solving Activities** Design a simple concept for a *low-cost solution* to any one real-time problem listed in the module. Propose an engineering solution for improving digital access for elderly people. >.< Suggest improvements for communication between citizens and local authorities using technology. **C. Creativity & Design Thinking** Design a concept for a *smart dustbin / solar street light / e-waste system*. >.< Draw a basic model	10	CO1

		(block diagram) of a community-based solution system. >.< Develop an idea for a campus safety or accessibility improvement system. **D. Research-Based Activities** Study one real-life social innovation project and prepare a short report. >.< Explore how engineers contribute to achieving UNSDG goals (choose any 2 goals). >.< Analyze a successful Indian social innovation case study. **E. AI-Based Learning Activities** Use AI tools to brainstorm solutions for a social problem and refine the best idea. >.< Ask an AI agent to simulate community feedback for your proposed solution. >.< Use AI to generate design ideas and then critically evaluate their feasibility. **Online References** **Social Innovation & Ethics** ● Stanford Social Innovation Review ● MIT Open Learning – Ethics & Society ● Coursera Social Innovation Courses **Engineering & Sustainability** ● UN Sustainable Development Goals (UNSDGs) ● NASA Earth & Sustainability Resources ● World Bank Development Projects **Engineering Learning Platforms** ● NPTEL Engineering Courses ● edX Engineering & Innovation Courses ● MIT OpenCourseWare Engineering **Design Thinking & Innovation** ● IDEO Design Thinking Guide ● Google Design Sprint Guide		
2	**Module 2:** **Fundamentals of Engineering Ethics**	2.1 Meaning of ethics, morals, values, and integrity 2.2.Personal ethics vs professional ethics 2.3.Engineering ethics 2.4.Duties of engineers towards society 2.5 Safety, welfare, and public interest 2.6 Honesty and accountability	10	CO2

		2.7 Conflict of interest 2.8 Ethical decision-making process 2.9 Case studies of engineering failures 2.10 Whistleblowing and professional responsibility **Suggested Case Studies:** Bridge collapse due to poor quality >.< Data privacy breach >.< Unsafe construction practices Fake certificates / plagiarism >.< Poor waste disposal by industry >.< Misuse of technology **Self learning Topics** **A. Understanding & Reflection** Write a short note on the importance of ethics in engineering practice. >.< Differentiate between personal ethics and professional ethics with suitable examples. >.< Explain how honesty and accountability influence public trust in engineers. **B. Problem-Solving Activities** Analyze a case where safety standards were ignored and suggest preventive measures. >.< Suggest ethical solutions to avoid plagiarism and fake certification in academics. >.< Propose methods to ensure data privacy and responsible digital behaviour in engineering projects. **C. Creativity & Critical Thinking** Prepare an ethical decision-making flowchart for engineers. >.< Design a poster or infographic on "Engineering Ethics and Public Safety." >.< Create a code of conduct for engineering students working on community projects. **D. Research-Based Activities** Study one famous engineering failure and identify ethical mistakes involved. >.< Explore the role of professional engineering bodies in maintaining ethics. >.< Analyze a real-life whistleblowing case related to technology or engineering. **E. AI-Based Learning Activities** Use AI tools to generate ethical solutions for a technology misuse problem. >.< Ask an AI chatbot to simulate an ethical dilemma and analyze your response. >.< Use AI to compare ethical and unethical engineering practices in real-world scenarios.		

		Online References **Ethics & Professional Responsibility** **1.** NSPE Code of Ethics for Engineers **2.** IEEE Ethics Resources **3.** MIT Ethics Case Studies **Engineering Safety & Society** ● ASCE Ethics Resources ● Coursera Engineering Ethics Courses		
3	**Module 03:** **Community Study, NGO Linkage and Real-Time Problem Identification**	3.1.Meaning of community and stakeholders. 3.2.Importance of NGO/community linkage in engineering education. 3.3.Types of NGOs and social organizations. 3.4.Field observation and community interaction. 3.5.Interview and basic data collection. 3.6.Stakeholder mapping 3.7.Root cause analysis 3.8.Problem prioritization 3.9.Documentation of field findings 3.10.Ethical behaviour during community interaction **Real-Time Problems** Waste segregation not followed in residential areas >.< Lack of awareness about emergency helpline numbers >.< Poor cleanliness in public spaces >.< E-waste disposal issues >.< Water wastage in homes, hostels and college campus >.<Lack of first-aid awareness >.< Traffic congestion near college gate >.< Plastic use in canteen and local shops >.< Lack of digital literacy among elderly people **Self learning Topics** **A. Understanding & Reflection** Explain the importance of community interaction in engineering education. >.< Write a short note on the role of NGOs in solving social problems. >.< Identify key stakeholders involved in one community issue near your locality. **B. Problem-Solving Activities**	10	CO3

		Prepare a simple survey questionnaire for waste management awareness. >.< Suggest solutions for reducing traffic congestion near a college campus. >.< Develop a basic awareness campaign idea for digital literacy among elderly people. **C. Creativity & Design Thinking** Design a stakeholder map for a community cleanliness project. >.< Draw a block diagram for a smart complaint management system for citizens. >.< Develop an idea for improving emergency helpline awareness in local communities. **D. Research-Based Activities** Study one successful NGO-led community project in India. >.< Analyze how surveys and interviews help in identifying real community problems. >.< Prepare a short report on a local environmental or social issue. **E. AI-Based Learning Activities** Use AI tools to create survey questions for a community study. >.< Ask an AI chatbot to simulate responses from different stakeholders. >.< Use AI to analyze possible root causes of a selected community problem. **Online References** **Community & NGO Learning** ● UNICEF Community Engagement Resources ● NGO Darpan India ● PRIA Participatory Research in Asia **Survey & Research Skills** ● Google Forms Learning Center ● Questionnaire Design Guide by Pew Research		
4	**Module 04.** **Sustainable Social Innovation and**	4.1.Concept of sustainable development 4.2.Sustainable Development Goals: basic introduction 4.3.Environment, health, safety and society 4.4Cleanliness, hygiene and public health	10	CO4

	Waste-to-Wealth Practices	4.5.Waste management and waste segregation 4.6.E-waste management 4.7.Water conservation 4.8.Energy conservation 4.9.Green campus and eco-friendly practices 4.10.Community safety and disaster awareness 4.11.Waste to wealth concept 4.12.Best out of waste as a social innovation model **Real-Time Problems** Improper waste disposal on campus >.< Lack of cleanliness in public areas >.< Excessive use of plastic bottles and disposable items >.< Poor awareness about e-waste collection >.< Water leakage and wastage >.< Lack of tree plantation and plant adoption >.< Low awareness about disaster preparedness >.< Lack of basic first-aid knowledge >.< Food waste from canteen >.< Paper waste from classrooms and offices **Suggested Practical Areas:** Campus waste audit >.< Water usage survey >.< E-waste collection drive >.< Energy-saving awareness >.< Accessibility audit of college campus >.< Plastic-free campus campaign >.<Green festival model **Engineering Relevance:** This module connects directly with Civil, Mechanical, Computer, IT, AI, Electronics, EXTC, and Electrical branches. **Self Learning Topics** **A. Understanding & Reflection** Explain the concept of sustainable development with examples. >.< Write a short note on the importance of waste segregation. >.< Identify environmental problems present in your campus or locality. **B. Problem-Solving Activities** Suggest methods to reduce paper waste in educational institutions. >.< Propose an engineering solution for water conservation in hostels or homes. >.< Develop ideas for reducing plastic usage in canteens and public places.		

		C. Creativity & Design Thinking Design a “Best Out of Waste” product using reusable materials. >.< Draw a basic model of an e-waste collection and recycling system. >.< Create a concept for a green campus initiative. **D. Research-Based Activities** Study one successful waste-to-wealth project in India or abroad. >.< Explore how engineering contributes to environmental sustainability. >.< Analyze the role of renewable energy in sustainable development. **E. AI-Based Learning Activities** Use AI tools to generate eco-friendly innovation ideas for campuses. >.< Ask an AI chatbot to prepare a waste audit analysis for a college campus. >.< Use AI to compare different waste management techniques and their effectiveness. **Online References** **Sustainability & Environment** ● UN Sustainable Development Goals ● UNEP Environmental Resources ● Swachh Bharat Mission **Waste Management & Green Innovation** ● Central Pollution Control Board India ● National Geographic Sustainability Resources ● edX Sustainability Courses		
5	**Engineering Ethics, Responsible Technology and AI Ethics**	6.1.Selection of real-time community problems. 6.2.NGO/community partner identification 6.3.Field survey and stakeholder interaction 6.4.Problem statement writing 6.5.Root cause analysis 6.6.Solution design 6.7.Prototype / product / model / awareness tool 6.8.Start-up or social enterprise possibility 6.9.Cost and resource analysis 6.10.Social and environmental impact analysis 6.11.Ethical considerations 6.12.Reflection and future scope 6.13.Project documentation **Real-Time Project Areas** Students may select any one problem from:	10	CO5

		Waste management >.< E-waste collection and disposal >.< Cleanliness and hygiene >.< First-aid awareness >.< Emergency helpline awareness >.< Water conservation >.< Energy conservation>.< Road safety >.< Digital literacy >.< Cyber safety awareness >.< Tree plantation and plant adoption >.< Campus safety >.< Health and hygiene awareness >.< Plastic reduction Accessibility for elderly or differently abled people >.< Best out of waste / waste to wealth products **Self Learning Topics** A. Understanding & Reflection Write a short note on the importance of community-based projects in engineering education. >.< Explain the role of stakeholder interaction in identifying real problems. >.< Identify one local issue that can be solved through engineering innovation. B. Problem-Solving Activities Prepare a problem statement for a selected community issue. >.< Suggest a low-cost prototype idea for solving a social problem. >.< Develop a simple cost and resource analysis for your proposed solution. C. Creativity & Design Thinking Design a prototype concept for waste management / water conservation / campus safety. >.< Draw a block diagram of a community-based innovation system. >.< Create an awareness tool (poster/app idea/video concept) for a selected social issue. D. Research-Based Activities Study one successful student-led social innovation project. >.< Analyze the social and environmental impact of a community engineering solution. >.< Explore startup or social enterprise opportunities based on community needs. E. AI-Based Learning Activities Use AI tools to brainstorm innovative project ideas for social problems. >.< Ask an AI chatbot to simulate feedback from community members about your project. >.< Use AI to improve your		

		project design and evaluate feasibility, cost, and impact. **Online References** Social Innovation & Community Projects ● Ashoka Social Innovation Network ● Atal Innovation Mission India ● UNDP India Innovation Projects Project Development & Entrepreneurship ● Startup India ● MIT D-Lab Community Innovation ● IDEO Design Thinking Resources		
6	**Community-Based Social Innovation Project**	**Content** 6.1. Selection of real-time community problem. 6.2. NGO/community partner identification 6.3. Field survey and stakeholder interaction 6.4.Problem statement writing 6.5.Root cause analysis 6.6.Solution design 6.7.Prototype / product / model / awareness tool 6.8.Start-up or social enterprise possibility 6.9.Cost and resource analysis 6.10.Social and environmental impact analysis 6.11.Ethical considerations 6.12.Reflection and future scope 6.13.Project documentation **Real-Time Project Areas** **Students may select any one problem from:** Waste management >.< E-waste collection and disposal >.< Cleanliness and hygiene >.< First-aid awareness >.< Emergency helpline awareness >.< Water conservation >.< Energy conservation >.< Road safety >.< Digital literacy >.< Cyber safety awareness >.< Tree plantation and plant adoption >.< Campus safety >.< Health and hygiene awareness >.< Plastic reduction >.< Accessibility for elderly or differently abled people >.< Best out of waste / waste to wealth products **Self Learning Topics** A. Understanding & Reflection Write a short note on the importance of community-based projects in engineering education. >.< Explain the role of stakeholder interaction in identifying real problems. >.<	10	CO6

		Identify one local issue that can be solved through engineering innovation. B. Problem-Solving Activities Prepare a problem statement for a selected community issue. >.< Suggest a low-cost prototype idea for solving a social problem. >.< Develop a simple cost and resource analysis for your proposed solution. C. Creativity & Design Thinking Design a prototype concept for waste management / water conservation / campus safety. >.< Draw a block diagram of a community-based innovation system. >.< Create an awareness tool (poster/app idea/video concept) for a selected social issue. D. Research-Based Activities Study one successful student-led social innovation project. >.< Analyze the social and environmental impact of a community engineering solution. >.< Explore startup or social enterprise opportunities based on community needs. E. AI-Based Learning Activities Use AI tools to brainstorm innovative project ideas for social problems. >.< Ask an AI chatbot to simulate feedback from community members about your project. >.< Use AI to improve your project design and evaluate feasibility, cost, and impact. **Online References** Social Innovation & Community Projects ● Ashoka Social Innovation Network ● Atal Innovation Mission India ● UNDP India Innovation Projects Project Development & Entrepreneurship ● Startup India ● MIT D-Lab Community Innovation ● IDEO Design Thinking Resources		
			Total Hours	60

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short

reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References Books:

1. Virginia Dignum, "Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way", Springer, 1st Edition, 2019.
2. Mark Coeckelbergh, "AI Ethics", MIT Press, 1st Edition, 2020.
3. Peter F. Drucker, "Innovation and Entrepreneurship", Harper Business, Revised Edition, 2006.
4. Eric Ries, "The Lean Startup", Crown Business, 1st Edition, 2011.
5. Nigel Cross, "Design Thinking: Understanding How Designers Think and Work", Bloomsbury Publishing, 2nd Edition, 2023.
6. Ian G. Barbour, "Ethics in an Age of Technology", HarperCollins Publishers, 1st Edition, 1993.
7. Charles E. Harris Jr., Michael S. Pritchard and Michael J. Rabins, "Engineering Ethics: Concepts and Cases", Cengage Learning, 5th Edition, 2019.
8. R. R. Gaur, R. Sangal and G. P. Bagaria, "Human Values and Professional Ethics", Excel Books, 2nd Edition, 2010.
9. M. Govindarajan, S. Natarajan and V. S. Senthil Kumar, "Engineering Ethics", PHI Learning Pvt. Ltd., 2nd Edition, 2017.
10. Geoff Mulgan, "Social Innovation: How Societies Find the Power to Change", Policy Press, 1st Edition, 2019.
11. Don Norman, "The Design of Everyday Things", Basic Books, Revised and Expanded Edition, 2013.
12. Jennifer A. Elliott, "An Introduction to Sustainable Development", Routledge, 5th Edition, 2023.
13. Erach Bharucha, "Textbook of Environmental Studies for Undergraduate Courses", Universities Press, 3rd Edition, 2020.
14. Idris Mootee, "Design Thinking for Strategic Innovation", Wiley, 1st Edition, 2013.

Online References:

Sr. No.	Website Name	URL
1	Stanford Social Innovation Review	https://ssir.org
2	United Nations Sustainable Development Goals	https://sdgs.un.org/goals
3	Atal Innovation Mission, NITI Aayog	https://aim.gov.in
4	National Society of Professional Engineers – Code of Ethics	https://www.nspe.org/resources/ethics/code-ethics
5	IEEE Ethics and Member Conduct Resources	https://www.ieee.org/about/corporate/governance/p7-8.html
6	MIT Ethics Case Studies	https://ethics.mit.edu
7	NGO Darpan (Government of India)	https://ngodarpan.gov.in
8	UNICEF Community Engagement Resources	https://www.unicef.org/community-engagement

9	United Nations Environment Programme (UNEP)	https://www.unep.org
10	Swachh Bharat Mission	https://swachhbharatmission.ddws.gov.in
11	Central Pollution Control Board (CPCB)	https://cpcb.nic.in
12	UNESCO Recommendation on the Ethics of Artificial Intelligence	https://www.unesco.org/en/artificial-intelligence/recommendation-ethics
13	OECD AI Principles	https://oecd.ai/en/ai-principles
14	IDEO Design Thinking Resources	https://designthinking.ideo.com
15	Stanford d.school Resources	https://dschool.stanford.edu/resources
16	NPTEL	https://nptel.ac.in
17	SWAYAM	https://swayam.gov.in
18	Coursera	https://www.coursera.org
19	edX	https://www.edx.org
20	MIT OpenCourseWare	https://ocw.mit.edu

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	S L	Notional Learning Hour	
1111106	Computational Engineering Mathematics - I Tutorial	--	02	–	--	30	–	–	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111106	Computational Engineering Mathematics- I Tutorial	---	---	---	---	---	25	-	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Engineering Mathematics forms the foundation for problem-solving, modeling, computation, and analytical thinking required in Computer Engineering, Information Technology, and Artificial Intelligence & Data Science. The tutorial component enables students to apply mathematical concepts through systematic problem-solving techniques, matrix computations, numerical methods, complex variable analysis, and multivariable calculus. These skills are essential for algorithm design, machine learning, data analysis, scientific computing, optimization, and engineering applications.

Tutorial Objectives:

Sr. No.	Tutorial Objectives
1	Apply elementary matrix transformations to determine the rank of matrices and solve systems of linear equations.
2	Solve problems involving trigonometric expansions, complex numbers, hyperbolic functions, and inverse functions.
3	Analyze functions of two and three variables using partial differentiation and composite function differentiation.

4	Apply Euler's theorem and optimization techniques to determine maxima and minima of multivariable functions.
5	Compute higher-order derivatives using successive differentiation and Leibnitz's theorem.
6	Implement numerical methods for solving transcendental equations and systems of linear algebraic equations.

Tutorial Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Tutorial Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Determine the rank of a matrix using echelon form, normal form, and PAQ form, and solve homogeneous and non-homogeneous systems of equations.	L3
2	Solve engineering problems involving trigonometric expansions, roots of complex numbers, hyperbolic functions, and inverse functions.	L3
3	Evaluate real and imaginary parts of complex functions and compute logarithms of complex numbers.	L3
4	Apply partial derivatives, higher-order derivatives, and Euler's theorem to analyze multivariable functions.	L3
5	Solve optimization problems involving maxima and minima of functions of two independent variables.	L4
6	Apply Newton-Raphson, Regula-Falsi, Gauss-Jacobi, and Gauss-Seidel methods to obtain numerical solutions of engineering problems.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Algebraic operations and factorization, Trigonometric identities and equations, Coordinate geometry and graphs, Elementary differential calculus and functions ,Complex numbers and basic matrix operations.	02	-

1	**Matrices**	1.1 Properties of Matrices: Hermitian, Skew Hermitian, Unitary, Orthogonal Matrices. 1.2 The rank of a Matrix using Echelon form, reduction to normal form, and PAQ form, The rank of a Matrix reduction to PAQ form. 1.3 System of homogeneous and non – non-homogeneous equations, their consistency, and solutions **Self- Learning topic**- Review of Inverse of a matrix, addition, multiplication, and transpose of a matrix, symmetric, skew-symmetric Matrix. Link : https://nptel.ac.in/courses/111106051?utm_source https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/?utm_source	05	LO1
2	**Complex Numbers**	2.1 D Moivre's Theorem, 2.2 Expansion of functions, Expansion of sin^ n θ, cos^n θ in terms of sines and cosines of multiples of θ; Expansion of sinnθ, cosnθ in powers of sinθ, cosθ, 2.3 Powers and Roots of a complex number **Self- Learning topic**- Review of Complex Numbers - Algebra of Complex Numbers, Cartesian, polar and exponential form of complex number. Link: https://nptel.ac.in/courses/111103070?utm_source	04	LO2
3	**Hyperbolic Functions & Logarithms of Complex Numbers**	3.1 Definition of Circular functions of complex numbers and Hyperbolic functions. 3.2 Inverse Circular and Inverse Hyperbolic Functions, Separation of real and imaginary parts of all types of Functions. 3.3 Logarithm of Complex Number (Simple examples). **Self- Learning topic**- Application of complex numbers, Rotation and transformation using complex numbers,	04	LO3

		Logarithm of complex numbers in cyber security. **Link :** https://onlinecourses.nptel.ac.in/noc24_ma20/preview?utm_source		
4	**Partial Differentiation**	4.1 Basic of derivatives , Partial Differentiation: Function of two and three variables, Partial derivatives of first and higher order. 4.2 Differentiation of composite function. 4.3 Euler's Theorem on Homogeneous functions with two independent variables (with proof), Deductions from Euler's Theorem (without proof). **Self- Learning topic**- Total Differentiation, implicit functions, Euler's theorem for three independent variables. **Link :** https://nptel.ac.in/courses/111104229?utm_source https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/?utm_source	05	LO4
5	**Applications of Partial Differentiation and Successive Differentiation**	5.1 Maxima and Minima of a function of two independent variables. 5.2 Successive differentiation: nth derivative of standard functions, Leibnitz's Theorem (without proof) and simple examples **Self- Learning topic**- Jacobian's of two and three independent variables (simple problems) Lagrange's Multiplier method. Link: https://nptel.ac.in/courses/111107122?utm_source	05	LO5
6	**Numerical methods**	6.1 Solution of Transcendental Equations: Solution by Newton Raphson method, Solution by Regula –Falsi method 6.2 Solution of a system of linear algebraic equations, by (1) Gauss Jacobi Iteration Method, by (2) Gauss Seidel Iteration Method.	05	LO6

		Self- Learning topic- Indeterminate forms , L-Hospital Rule, Gauss Elimination Method, Gauss Jordan Method. Link : https://nptel.ac.in/courses/111105090?utm_source https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-fall-2010/?utm_source		
		Total	**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References Books:

1. Dr.B.S.Grewal, "Higher Engineering Mathematics" ,Khanna Publication
2. R. K. Jain and S. R. K. Iyengar, "Advanced Engineering Mathematics", Narosa publication
3. Shanti Narayan, "Matrices", S. Chand publication.
4. Howard Anton and Christ Rorres, "Elementary Linear algebra with application", 6th edition.
5. M. K. Jain, S. R. K. Iyengar and R. K. Jain , "Numerical Methods for Scientific and Engineering Computation" New Age International Publishers.
6. Bali N.P and Goyal M. "A textbook of Engineering Mathematics" Laxmi Publication, 10th Ed. 2022.
7. Gupta C.B. Singh S.R and Mukesh Kumar : "Engineering Mathematics for semester I and II", Mc- Graw Hill education ,2015
8. Pal S. & Bhunia S.C: "Engineering Mathematics " Oxford University Press, 3rd Ed, 2016.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://nptel.ac.in/
2	MIT OpenCorseWare	https://ocw.mit.edu/courses
3	Wolfram Mathworld	http://mathworld.wolfram.com

List of Tutorials: .

Sr No	List of Assignments / Tutorials	Hrs
01	Finding the rank of a Matrix using Echelon form, reduction to normal form, and PAQ form, The rank of a Matrix reduction to PAQ form	02
02	Problems on System of homogeneous and non –homogeneous equations, their consistency, and solutions.	01

03	Problems on expansion of sin^ n θ, cos^n θ, sinn θ, cosn θ in terms of sines and cosines of multiples of θ and roots of a complex number.	02
04	Problems on Hyperbolic functions, Inverse Circular and Inverse Hyperbolic Functions.	01
05	Problems on Separation of real and imaginary,Problems on Logarithm of Complex Number.	02
06	Problems on Function of two and three variables, Partial derivatives of first and higher order, Differentiation of composite function	01
07	Euler's Theorem on Homogeneous functions with two independent variables,Problems on Maxima and Minima of a function of two independent variables.	02
08	Successive differentiation: nth derivative of standard functions, Leibnitz's Theorem.	01
09	Problem based on Solution of Transcendental Equations: Newton Raphson method , Regula –Falsi method.	02
10	Problem based on Solution of a system of linear algebraic equations, by (1) Gauss Jacobi Iteration Method, by (2) Gauss Seidel Iteration Method.	01
	Total	**15**

Assessment:

- **Term Work:** Term Work shall consist of at least 10 Tutorials based on the above list.
- **Term Work Marks:** 25 Marks (Total) = 10 Marks (Tutorials) + 10 Marks (Mini Project or Quiz) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1111107	Applied Physics for Computer Technology Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR/OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111107	Applied Physics for Computer Technology Lab	---	---	---	---	---	25		25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: Applied Physics provides the scientific foundation required for understanding modern engineering systems, materials, structures, instrumentation, and emerging technologies. The subject helps Civil and Mechanical Engineering students develop analytical thinking, problem-solving ability, and practical knowledge of engineering applications. The topics included in the syllabus directly support industrial, constructional, manufacturing, and technological requirements of these branches.

Lab Objectives:

Sr. No.	Lab Objectives
1	To develop scientific understanding of the physics concepts.
2	To develop the ability to explain the processes and applications related to science subjects.
3	To apply skills and knowledge in real life situations.
4	To improve the knowledge about the theory concepts of Physics learned in the class.
5	To improve ability to analyse experimental results and write laboratory report.
6	To develop understanding about inferring and predicting.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	**Measure and analyse** the divergence of a He–Ne laser beam and relate the results to the properties of laser radiation.	L4
2	**Determine the numerical aperture of an optical fibre** and explain light propagation through total internal reflection.	L3
3	**Evaluate Planck's constant using a photocell** (semiconductor diode) and verify the principles of the photoelectric effect.	L5

4	**Determine the energy band gap of a semiconductor** by analysing the variation of electrical characteristics with temperature.	L4
5	**Study and interpret ultrasonic wave behaviour** using ultrasonic distance meter/interferometer techniques for measurement applications	L4
6	**Determine carrier concentration using the Hall effect** and relate Hall voltage measurements to semiconductor properties.	L3,L4

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Contents: **Basic concepts of electricity and electronics** Understanding of current, voltage, resistance, power supply, and simple electrical circuits. **Knowledge of wave motion and optics** Fundamentals of light waves, reflection, refraction, interference, diffraction, and sound waves. **Basic semiconductor physics** Understanding of conductors, semiconductors, energy bands, electrons, holes, and p-n junction concepts. **Familiarity with electromagnetic principles** Basic ideas of magnetic fields, electromagnetic force, and interaction of charges with magnetic fields. **Measurement and instrumentation skills** Use of laboratory instruments such as multimeter, voltmeter, ammeter, micrometer screw gauge, and graph plotting techniques. **Mathematical and analytical skills** Basic algebra, logarithms, graph interpretation, unit conversion, and error calculation methods.	04	-
1	Laser and Fiber Optics	1.1 Fundamentals of Lasers: Basic principles of laser. Monochromatic light, Coherent light 1.2 Optical Measurement Techniques: Measurement using scale, Measurement using vernier instruments, Plotting and interpretation of graphs 1.3 Wave Optics and Safety: Basics of wave optics, Laser safety precautions, Safe handling of optical instruments **Self-learning Topics:** Determination of wavelength using Diffraction Online reference: gratinghttps://www.youtube.com/watch?v=iCvDv3TQosw Practical List: **Study of divergence of laser beam using He–Ne Laser source**	03	LO1
2	Laser and Fiber Optics	2.1 Fundamentals of Optical Fibres: Total internal reflection, Acceptance angle, Propagation of light in optical fibres 2.2 Measurement and Calculations: Use of measuring instruments, Basic trigonometric calculations	02	LO2

		Self-Learning Topics: Study of input and output characteristics using optical fiber **https://www.focc-fiber.com/info/characteristics-of-optical-fibers-103226358.html** Practical List: **Determination of Numerical Aperture of an optical fibre**		
3	Diodes and Transistors	3.1 Photoelectric Effect Fundamentals: Photoelectric effect, Concept of work function, Stopping potential 3.2 Electrical Measurements and Semiconductor Devices: Electric circuits, Graph plotting and analysis, Basic semiconductor device operation Experiment: **Determination of 'h' using photo cell** **Self-learning Topics:** Determination of 'h' using LED https://www.youtube.com/watch?v=yIM0QwaQ06E	02	LO3
4	Semiconductor Devices	4.1 Semiconductor Fundamentals: Basic understanding of semiconductors, Energy band theory 4.2 Electrical Characteristics of Semiconductors: Temperature dependence of resistance, Semiconductor behaviour with temperature variation 4.3 Measurement Techniques and Calculations: Use of mustimeter, Use of power supply, Logarithmic calculations Experiment: **Determination of energy band gap of semiconductor** **Self-learning Topics:** I V Characteristics of p-n junction diode https://www.youtube.com/watch?v=Nds6Qrd6k40&t=237s	02	LO4
5	Acoustics and Ultrasonics	5.1 Fundamentals of Sound and Ultrasonic Waves: Understanding of sound waves, Ultrasonic wave generation, Propagation of ultrasonic waves 5.2 Wave Behaviour and Principles: Reflection principles, Interference principles, Applications of ultrasonic wave behaviour 5.3 Electronic Instruments and Measurements: Basic operation of electronic instruments, Use of ultrasonic measurement instruments, Experimental observation and interpretation Experiment: **Study of Ultrasonic distance meter / interferometer** **Self-learning Topics:** Determination of Reverberation time using Subine's formula. https://www.youtube.com/watch?v=bMVDPkwzL8I	02	LO5
6	Electromagnetism and Sensors	6.1 Fundamentals of Magnetism and Hall Effect: Magnetic fields, Lorentz force, Hall effect principle 6.2 Semiconductor Concepts: Basic semiconductor concepts, Charge carrier behaviour in semiconductors	03	LO3

		6.3 Electrical Measurements and Data Interpretation: Current-voltage measurements, Graph interpretation, analysis of experimental results Experiment: **Determination of Carrier concentration using Hall Effect** **Self-learning** Topics: Determination of type of semiconductor using Hall effect https://mpv-au.vlabs.ac.in/modern-physics/Hall_Effect_Experiment/		
7	Semiconductor Devices	7.1 Forward and reverse characteristics of semiconductor diode, 7.2 Breakdown mechanism: Avalanche and Zener breakdown. Experiment: **Forward characteristics of Si and Ge diodes** **Self-learning:** Characteristics of LED, difference between LED and Zener diode https://be-iitkgp.vlabs.ac.in/exp/voltage-regulator/simulation.html	03	LO6
8	Semiconductor Devices	8.1 Zener Diode Characteristics: Forward and reverse bias characteristics of Zener diode, Zener breakdown voltage and V–I characteristics 8.2 Applications and Measurements: Experimental measurements and graph plotting Experiment: **Characteristics of Zener diode** **Self-learning topics:** Study of Zener Diode as a Voltage Regulator https://ssp-amrt.vlabs.ac.in/exp/characteristics-zener-diode/	02	LO3
9	Semiconductor Devices	9.1 Fundamentals of RTD and PT100: • Principle of Resistance Temperature Detector (RTD), Construction and working of PT100, Temperature dependence of resistance, Positive temperature coefficient of resistance 9.2 Characteristics and Applications of PT100: Resistance–temperature relationship of PT100, Calibration and measurement methods, Advantages and limitations of RTD sensors, Industrial and engineering applications of PT100 sensors Experiment: **To understand the working principle of RTD for PT100** **Self-learning:** Calibration of PT100 Sensor Using Digital Temperature Indicator https://eil-iitg.vlabs.ac.in/Signal%20Conditioning%20Circuit%20for%20RTD%20%28Resistance%20Temperature%20Detector%29%28theory%29.html	03	LO3

10	Mini project	A group mini project is to be done related the course content	04	LO1, 2,3,4, 6
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References Books:

1. Dr. M. N. Avadhanulu, Dr. P. G. Kshirsagar, **"**A Textbook of Engineering Physics**",** S. Chand 2018
2. R. K. Gaur, S. L. Gupta, "Engineering Physics" Dhanpat Rai Publications,2014 edition
3. Francis A. Jenkins, Harvey E. White, "Fundamentals of Optics" McGraw Hill Publication 1957 (4th Edition)

Online References:

Sr. No.	Website Name	URL
1	Diffraction Grating Experiment / Interference of Light	https://www.youtube.com/watch?v=iCvDv3TQosw
2	NPTEL – Introduction to Optics	https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_ph51
3	NPTEL – Optics Course	https://nptel.ac.in/courses/115105120

List of Experiments.

Sr No	List of Experiments	Hrs
01	Study of divergence of laser beam using He–Ne Laser source	02
02	Determination of Numerical Aperture of an optical fibre	02
03	Determination of 'h' using photocell	02
04	Determination of energy band gap of semiconductor	02
05	To study the Characteristics of Zener diode	02
06	Determination of Carrier concentration using Hall Effect	02
07	Forward characteristics of Si and Ge diodes	02
08	To study the Characteristics of Zener diode	02
09	To understand the working principle of RTD for PT100	02
10	A group mini project is to be done related the course content	04

Assessment:

- Term Work: Term Work shall consist of at least 7-8 Practical based on the above list. In addition,
- Term Work Marks: 25 Marks (Total) = 10 Marks (Experiments) + 10 Marks (Mini project) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1111108	Basic Electrical Engineering Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111108	Basic Electrical Engineering Lab	---	---	---	---	---	25		25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: Digital technologies are transforming education, industry, and society. To participate effectively in this digital ecosystem, students require a strong foundation in electrical and electronic systems that support modern technologies. This course develops practical skills in electrical measurements, circuit analysis, energy systems, and emerging technologies, enabling students to contribute to a digitally empowered and inclusive society.

Lab Objectives:

Sr. No.	Lab Objectives
1	To develop practical understanding of electrical circuit laws, network theorems, and circuit analysis techniques through laboratory experiments.
2	To familiarize students with the operation and measurement techniques of DC and AC electrical circuits using standard electrical instruments.
3	To enable students to verify the characteristics and performance of single-phase transformers and electrical machines experimentally.
4	To provide hands-on experience in analyzing power, power factor, energy consumption, and efficiency in electrical systems.
5	To introduce students to basic battery technologies, charging-discharging characteristics, and important battery parameters.
6	To enhance problem-solving, teamwork, safety awareness, and technical report-writing skills through experimental investigations

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy

1	Apply electrical circuit laws, network theorems, and circuit analysis techniques to verify the behavior of DC electrical circuits experimentally.	Apply (L3)
2	Operate and use standard electrical instruments to measure voltage, current, power, and waveform parameters in DC and AC circuits.	Apply (L3)
3	Analyze the characteristics and performance of single-phase transformers and electrical machines through laboratory testing.	Analyze (L4)
4	Evaluate power, power factor, efficiency, and energy consumption of electrical systems using experimental observations and calculations.	Evaluate (L5)
5	Examine battery charging–discharging characteristics and determine important battery parameters through practical experimentation.	Analyze (L4)
6	Demonstrate problem-solving ability, teamwork, laboratory safety practices, and technical report-writing skills during experimental investigations.	Understand (L2)

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
1	Basic safety precautions	Basic safety precautions Introduction and use of measuring instruments - voltmeter, ammeter, multi-meter, oscilloscope. Real-life resistors, capacitors, and inductors **Self-learningTopics:** Online References: **https://onlinecourses.nptel.ac.in/noc21_ee107/preview?utm_source**	02	LO1
2	**DC Circuits**	Superposition Theorem. Thevenin's Theorem. **Self-learning Topics: KCL, KVL, Oham's Law** Online References: https://onlinecourses-archive.nptel.ac.in/noc18_ee18/preview?utm_source	04	LO2
3	AC Circuits	To find the resistance and inductance of a coil connected in series with a pure resistance using the voltmeter method. Voltage & current relationships in Star Connections. **Self-learning Topics: R, L, C** Online References: https://onlinecourses.nptel.ac.in/noc23_ee81/preview	04	LO3
4	Single phase Transformer& machine	Working of the single-phase transformer. Construction of a DC machine. **Self-learning Topics:** Online References: https://www.youtube.com/watch?v=oJtY6xn6dkQ&utm_source https://freevideolectures.com/course/5020/nptel-electrical-machines-i/22?utm_source	04	LO4

5	Fundamentals of Digital Systems	logic gates and universal NAND and NOR gates. **Self-learning Topics:** Online References: https://onlinecourses.nptel.ac.in/noc23_ee50/preview?utm_source	04	LO5
6	Electrical Energy Storage	Battery parameters, Soc, Dod **Self-learning Topics:** Online References: **https://www.nptel.ac.in/courses/113105102?utm_source=chatgpt.com**	04	LO6
		Mini Project Based on All Topics	08	ALL
Total			**30**	

List of Experiments:

Sr No	List of Experiments	LO Mapping	Hrs
01	Basic safety precautions. Introduction and use of measuring instruments - voltmeter, ammeter, multi-meter, oscilloscope. Real- life resistors, capacitors, and inductors.	LO1	02
02	Verification of Superposition Theorem.	LO2	02
03	Verification Thevenin's Theorem	LO2	02
04	To find the resistance and inductance of a coil connected in series with a pure resistance using the voltmeter method.	LO3	02
05	To measure the relationship between phase and line, currents and voltages in a three-phase system for star connection.	LO3	02
06	To demonstrate cut-out sections of the single-phase transformer	LO4	02
07	To demonstrate cut-out sections of the DC machine	LO4	02
08	To study and verify logic gates and to study universal NAND and NOR gates	LO5	04
09	Experimental study of charge and discharge characteristics of Lead -acid battery.	LO6	04
10	Mini project based on the theme **Digital Bharat for Inclusive Learning.**	All	08

Assessment:

- **Term Work:** Term Work shall consist of at least 8-10 Practicals based on the above list.
- **Term Work Marks:** 25 Marks (Total) = 20 Marks (Experiments) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	NL	
1111109	Problem Solving Using C Lab	-	-	2	-	-	30		30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111109	Problem Solving Using C Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: The Problem Solving Using C Laboratory is designed in alignment with the principles of NEP 2020, emphasizing experiential learning, skill development, critical thinking, and industry-oriented education. Complementing the theory course, the laboratory provides students with hands-on experience in implementing computational solutions using the C programming language. The course encourages learning by doing, enabling students to translate algorithms and logical designs into efficient and structured programs.

Aligned with the Computer Engineering theme "Digital India", the laboratory focuses on developing programming proficiency, logical reasoning, analytical thinking, and software development skills through real-world problem-solving activities. The laboratory adopts a problem-based learning approach through application-oriented experiments. The laboratory contributes to the realization of the Digital India vision by fostering technology-enabled problem-solving capabilities and digital competencies among future engineers. It also supports the achievement of UNSDG 4 (Quality Education), UNSDG 8 (Decent Work and Economic Growth), UNSDG 9 (Industry, Innovation and Infrastructure), UNSDG 11 (Sustainable Cities and Communities), and UNSDG 16 (Peace, Justice and Strong Institutions) through the development of practical, innovative, and socially relevant computing solutions.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand the fundamentals of computational thinking and develop problem-solving skills using the C programming language.
2	To implement decision-making, looping, and control structures for solving real-world applications.
3	To develop programs using arrays and strings for data analysis, record management, and validation systems.

4	To apply modular programming concepts using functions, pointers, and dynamic memory allocation for efficient and reusable code design.
5	To use pointers, dynamic memory allocation, and structures for efficient data storage and management in C applications.
6	To design and implement file handling applications for maintaining and processing records such as library systems and payroll processing.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Write, compile, and execute C programs to solve basic computational and mathematical problems.	L3
2	Apply conditional statements, loops, and iterative techniques to develop interactive and menu-driven applications.	L3
3	Use arrays and strings for processing, analyzing, and validating data efficiently.	L3
4	Design modular and reusable programs using functions in C.	L4
5	Implement applications using pointers, dynamic memory allocation, and structures for efficient data handling and management.	L4
6	Develop file handling applications for storing, retrieving, and processing records in systems.	L4

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:
Desktop Computer, RAM (Minimum 4 GB), Printer (Optional)	Operating System (Windows/Linux) ,GCC Compiler, Visual Studio Code.

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Introduction to compiler and execution of program.		-
1	**Computational Thinking and C Fundamentals**	1.1 Data types, 1.2 Operators 1.3 Input-output functions. **Self-Learning Topics:** Write-up on history and evolution of programming languages , Practicing Non-Formatted Inp& Output functions – getchar( ), putchar( ), getch( ), getche( ), putch( ), gets( ), puts( ) , Practicing Formatted Input & Output functions – printf( ) **Online Resources:** https://www.programiz.com/c-programming , https://www.geeksforgeeks.org/c-programming-language/ **Introduction to C**	04	LO_1
2	**Decision Making and Iterative Problem Solving**	2.1 Conditional statements (if, if-else, nested if, switch-case) 2.2 Iterative statements (for, while, do-while), 2.3 Nested loops 2.4 Menu-driven programming. **Self-learning Topics: Self-Learning:** Pattern programs (Pyramids , Star , Square , Upper triangular matrix , Lower triangular matrix) **Online Resources:** https://www.tutorialspoint.com/cprogramming/c_decision_making.htm , https://www.programiz.com/c-programming/c-for-loop, **If statements in C** **For, While and Do While Loops in C** **switch case in C**	6	LO_2
3	**Arrays and Strings**	3.1 Matrix operations 3.2 String handling functions **Self-Learning:**	6	LO_3

		Sorting an array, Binary searching , Declaration and Initialization of a Multidimensional Array **Online Resources:** https://www.geeksforgeeks.org/arrays-in-c-cpp/ , https://www.programiz.com/c-programming/c-arrays **Arrays in C**		
4	**Modular Programming with Functions**	4.1 Function declaration and definition 4.2 Parameter passing. **Self-Learning:** Passing Arrays to Functions, Implement recursive solutions (e.g., Fibonacci, Tower of Hanoi) , Industry application: Use of arrays/strings in data processing. **Online Resources:** https://www.geeksforgeeks.org/functions-in-c/ , s **Functions in C**	6	LO_4
5	**Structures and Pointers**	5.1 Structures and union 5.2 Arrays of structures 5.3 Pointers **Self-Learning:** Call by address using pointers, returning more than one value from a function, returning pointer from a function, structures and pointers, passing structure to function ,Structure versus Union, Enumeration types , Processing binary files **Online Resources:** https://www.programiz.com/c-programming/c-pointer **nptel.ac.in/courses/106104128** **Structures in C**	4	LO_5
6	File Handling	6.1 Basic File Operations **Self-Learning:** Understand the purpose of file handling in C programming,Learn the difference between text files and binary files,File modes,	4	LO_6

		Online Resources: https://www.programiz.com/c-programming/c-file-input-output https://www.geeksforgeeks.org/basics-file-handling-c/		
	Total		**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 8th Edition / 2019.
2. Brian W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Pearson Education, 2nd Edition / 2015.
3. Yashavant Kanetkar, "Let Us C", BPB Publications, 18th Edition / 2021.
4. Reema Thareja, "Programming in C", Oxford University Press, 2nd Edition / 2016.

Online References:

Sr. No.	Website Name	URL
1	Coursera – C Programming Specialization	https://www.coursera.org/specializations/c programming
2	NPTEL – Programming in C	https://onlinecourses.nptel.ac.in/e learning/preview/noc19_cs42
3	Udemy – C Programming Courses	https://www.udemy.com/topic/c-programming

List of Experiments:

Sr No	List of Experiments	Hrs
01	**Smart Electricity Bill Calculator:** Design a program that calculates monthly electricity bills for domestic consumers based on slab rates, fixed charges, and applicable discounts. Write a program to compute the future value of a specified principal amount, rate of interest, and a number of years.	2
02	Write a program to get the volume of a sphere.	2
03	**Student Performance Analyzer:** Design a program that accepts marks of students in multiple subjects and generates grade, percentage, pass/fail status, and class performance statistics.	2
04	**ATM Simulation System:** Create a menu-driven ATM application supporting balance inquiry, cash withdrawal and deposit.	2

05	**Sales Data Analytics Application:**Develop a system to analyze monthly sales data of products and generate highest sales, lowest sales, average sales, and ranking reports.	2
06	**Pattern Generator:** Develop a system that generates different patterns	2
07	**Smart Matrix Calculator:** Develop a menu-driven matrix utility that performs addition, subtraction, transpose, and multiplication of matrices based on user choice.	2
08	**Secure Password Validation System:** Create a password validation utility that checks password strength using conditions such as length, uppercase letters, digits, and special characters.	2
09	**Text Analytics Utility:** Develop a Text Analytics Utility that accepts a text string from the user and generates basic analytical information about the entered text.(number of words, characters, digits, and special symbols)	2
10	**Banking Interest and Loan Estimator:** Develop a modular application using functions to calculate EMI, loan eligibility, and interest for different banking schemes.	2
11	**Hospital Patient Monitoring Utility:**Design a modular healthcare application that stores patient vitals and identifies abnormal conditions using function-based processing.	2
12	**Loan Eligibility Assessment System :** Develop a modular banking application that accepts customer income, age, and credit score information and determines loan eligibility using separate functions for validation, calculation, and reporting.	2
13	**Employee Record Management:** Create a program to manage employee details such as ID, salary, department, and experience.	2
14	**Swap Two Numbers Using Pointers:** A user wants to interchange the values of two variables without directly modifying them in the main program. Develop a C program that uses pointers to swap two numbers and display the values before and after swapping.	2
15	**File Copy Utility:**Develop a C program that opens an existing file, reads its contents, and saves the contents into another file.	2

Sr No	List of Assignments
01	Design and implement a menu-driven Pattern Generator that allows users to generate Star Pyramid, Square Pattern, Upper Triangular Pattern, and Lower Triangular Pattern using nested loops. Analyze the logic used for each pattern.

02	Develop a Student Record Search System that stores student marks in an array, sorts the records in ascending order, and performs binary search to locate a specific student's marks. Compare linear search and binary search in terms of efficiency.
03	Implement the Fibonacci Series and Factorial computation using both recursion and iteration. Compare their execution steps, memory usage, advantages, and limitations through a short presentation or report.
04	Develop an Employee Management System using structures and pointers. Demonstrate how structures can be passed to functions and how pointers can be used to access structure members efficiently.
05	Prepare a comparative study of Structures and Unions in C. Illustrate memory allocation differences using suitable examples and identify scenarios where each is preferred.
06	Create a reference guide explaining different file modes in C (r, w, a, r+, w+, a+) and demonstrate their usage through suitable examples. Include a comparison between text files and binary files.

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	NL	
1111110	Nirman Lab - I	-	-	2	-	-	30		30	1

		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)	TW	PR/OR	Total
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111110	Nirman Lab - I	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: Aligned with the National Education Policy (NEP) 2020, the institution emphasizes experiential, interdisciplinary, and project-based learning through the Nirman Lab a central hub for hands-on innovation.To strengthen the undergraduate research ecosystem, the institution has adopted a theme-based academic model aligned with UNSGD. Each semester features six curated problem statements based on local need and aligned with core subjects in the same semester, enabling students to apply classroom knowledge to real-world challenges. Every student selects one problem and develops an individual, subject-integrated solution enhancing both academic understanding and research skills.The Nirman Lab supports this initiative with facilities for design thinking, prototyping, and product development. Students maintain a project logbook throughout the semester to track their progress and reflections

Course Objectives:

Sr. No.	Course Objectives
1	To promote experiential and project-based learning that bridges theoretical knowledge with real-world problem-solving.
2	To encourage interdisciplinary integration by enabling students to apply concepts from multiple subjects within a single cohesive project.
3	To develop innovation and design thinking skills through hands-on activities and iterative solution development.
4	To foster critical thinking and creativity by engaging students in open-ended problems with multiple solution pathways.
5	To enhance communication, collaboration, and documentation skills essential for professional engineering practice.
6	To build an entrepreneurial and research mindset by guiding students to develop scalable socially-relevant, and technically viable prototype.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Recall and articulate key concepts from core and allied subjects relevant to the assigned project	L1
2	Explain the interdisciplinary nature of the problem and the role of each subject in	L2

	addressing it	
3	Apply appropriate tools, techniques, and theoretical knowledge to develop project components	L3
	Analyze problem constraints and user requirements to structure a feasible and efficient solution	L4
5	Evaluate multiple design options and justify the chosen solution based on technical and practical considerations	L5
6	Create a functional prototype or solution that demonstrates innovation, utility, and integration of interdisciplinary knowledge	L6

Project Guidelines (Interdisciplinary Project Execution in Nirman Lab)

- Each student works on an individual interdisciplinary project aligned with the semester theme.
- Faculty in-charges mentor both the academic and technical aspects, and track weekly progress.
- Project assessment will be rubric-based, ensuring depth, innovation, documentation, and ownership.
- Students shall convert the best solution into a working model using various components of their domain areas and demonstrate.

Guidelines for same semester Subject Concepts Applied within the Project

- Termwork for each subject will partially reflect how well a student applies subject-specific concepts in their interdisciplinary project.
- The internal assessment panel will collaborate to align project components with subject learning outcomes.

Role of Faculty In-Charges in Nirman Lab Projects

Faculty in-charges play a pivotal role in the success of interdisciplinary, theme-based projects under the Nirman Lab. Their responsibilities extend beyond technical supervision to include academic alignment, innovation facilitation, and active student engagement. Their key roles include:

- Motivating and Inspiring Students
    a. Encourage students to take ownership of their learning and projects.
    b. Cultivate a mindset of curiosity, exploration, and social relevance.
    c. Foster an environment where students feel empowered to take creative risks.

- Conducting Brainstorming and Ideation Sessions
    d. Organize structured brainstorming sessions at the start of the semester to help students define their problem statements and solution pathways.
    e. Promote collaborative thinking, design exploration, and interdisciplinary integration.

- Arranging Guest Lectures and Expert Talks
    f. Identify and invite industry experts, researchers, and innovators for guest lectures aligned with the semester's theme or subject areas.
    g. Facilitate exposure to real-world challenges, current trends, and future opportunities.

- Ensuring Uniqueness and Originality of Projects
    h. Actively review proposed ideas to ensure **no duplication of solutions** across students.
    i. Encourage students to explore novel approaches, technologies, and perspectives.

- Promoting Discussion and Collaborative Learning

- Create platforms for students to present, discuss, and receive peer and mentor feedback.
    j. Facilitate idea refinement through regular discussions and group engagement.

- Monitoring and Documentation
    k. Oversee project logbook maintenance, milestone tracking, and submission of progress reports.
    l. Provide ongoing feedback and ensure project alignment with learning outcomes.

- Coordination with Subject Faculty
    m. Work in collaboration with other subject faculty to help students embed theoretical and practical aspects of their coursework into the project.
    n. Facilitate subject-term mapping and contribute to termwork assessment based on evidence.

Guidelines for Assessment:

Presentation-Based Assessment Structure (Total: 25 Marks)

Assessment Month	**Weightage**	**Marks**
Month 1 (Formative 1)	20%	5 marks
Month 2 (Formative 2)	40%	10 marks
Month 3 (Formative 3)	40%	10 marks

Rubric-Based Evaluation Criteria

Criteria	**Month 1 (5)**	**Month 2 (10)**	**Month 3 (10)**
Problem Understanding	Connects problem to subjects	Defines interdisciplinary scope	Demonstrates deep conceptual grasp
Subject Knowledge Application	Identifies relevant concepts	Applies principles in design	Integrates multiple subject areas correctly
Innovation & Design Thinking	Proposes creative idea	Develops and tests feasible solution	Final solution shows originality and utility
Documentation & Presentation	Logbook initiated, plan presented	Mid-design log and visuals	Final report and demo completed
Progress & Ownership	Meets deadlines, shows planning	Demonstrates self-motivation	Completes project independently with reflection

Total Assessment Structure

Component	Marks	Assessed By
Termwork – Project Execution	25 Marks	Project Guide
Viva Voce (Final Evaluation)	25 Marks	External Examiner

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	02
1111401	Workshop-I Lab	--	--	02*+02	--	--	60	--	60	

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111401	Workshop Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This laboratory course is designed in alignment with the objectives of the **NEP 2020**, emphasizing experiential learning, multidisciplinary education, skill development, and industry-oriented training. The course provides hands-on exposure to computer hardware, software installation, electrical wiring, networking, digital productivity tools, and 3D printing technologies. It helps students develop practical skills in system assembly, troubleshooting, networking, digital collaboration, and rapid prototyping while enhancing problem-solving ability, teamwork, and technical competency. The curriculum also supports industry readiness through exposure to modern engineering tools and technologies used in IT, electronics, and manufacturing sectors. The course aligns with **UNSDG 4, UNSDG 8** and **UNSDG 9** by promoting skill-based technical education, innovation, and employability.

Lab Objectives:

Sr. No.	Lab Objectives
1	To familiarize students with the identification and functions of major internal and external computer hardware components
2	To train students in installation and configuration of essential software utilities and applications
3	To provide knowledge of domestic electrical connections and circuit operation
4	To introduce basic principles of network planning and structured network design.
5	To develop skills in creating technical documents, spreadsheets, forms, and presentations using cloud platforms
6	To develop understanding of major components and working principles of a 3D printer.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Identify and describe the functions of common computer hardware components	L1, L2

2	Install and configure essential software tools and utilities required for system operation	L3
3	Demonstrate understanding of domestic electrical wiring practices and circuit connections	L2, L3
4	Demonstrate understanding of domestic electrical wiring practices and circuit connections	L2, L3
5	Organize and present technical information effectively using digital productivity platforms.	L3
6	Demonstrate the ability to create basic 3D printed models for engineering applications.	L3, L6

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Desktop Computer Systems, Motherboards, Processors, RAM Module, HDD and SSD Drives, SMPS/Power Supply Units, Graphics Cards, Computer Tool Kit, Anti-static Wrist Straps, Bootable USB Drives	OS(Windows and Ubuntu), Packet Tracer, Pen drive Bootable software, 3D Printing software

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
1	**Hardware**	1.1 Identification of internal and external computer components (CPU, RAM, HDD, SSD, GPU, PSU, Motherboard) 1.2 Disassembly and reassembly of a desktop computer/Laptop 1.3 BIOS/UEFI settings — boot sequence, hardware configuration 1.4 Simple hardware diagnostics — RAM testing, HDD health check 1.5 Assembling a working PC from components	15	LO1
2	**Software**	2.1 Partitioning a hard drive (primary, logical, extended) 2.2 Installation of Windows OS — step-by-step 2.3 Installation of Linux (Ubuntu) — dual boot setup 2.4 Basic Linux commands — file system navigation, permissions 2.5 Installation and configuration of essential software tools 2.6 Network configuration — IP settings, connectivity testing	15	LO2
3	**Electrical House Wiring**	3.1 Study of electrical tools — wire strippers, pliers, testers 3.2 Series and parallel wiring of bulbs 3.3 Wiring a plug, switch board, and fluorescent lamp circuit	06	LO3

		3.4 Earthing and safety demonstration		
4	**Computer Networking**	4.1 Networking Devices — Identification, Study & Mini Network Setup 4.2 Cable Crimping, IP Configuration & Connectivity Testing 4.3 Network Design — Lab & Campus Network Planning	06	LO4
5	**Digital Tools & Productivity for Engineers**	5.1 Google Workspace (Docs, Sheets, Forms, Sites), Canva, Figma basics, GitHub — practical toolbox used in every theme project; Google Sheets as lightweight backend, GitHub for team collaboration	06	LO5
6	**3D Printing**	6.1 Introduction to 3D Printing 6.2 Types of 3D Printing Technologies 6.3 3D Printer Components and basics	12	LO6
Total			**60**	

References Books:

1.Fundamentals of Additive Manufacturing and 3D Printing – Md. Enamul Hoque
2.Computer Organization and Architecture – William Stallings

Online References

https://learn.microsoft.com?utm_source=chatgpt.com
https://www.allaboutcircuits.com?utm_source=chatgpt.com
https://support.google.com/a/users?utm_source=chatgpt.com

Assessment:

2. **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list.
3. **Term Work Marks:** 25 Marks (Total) = 20 Marks (Experiments) + 05 Marks (Attendance).
4. **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2991501	Professional Communication Skills for Engineers in the Age of AI	--	-	02*+02	--	–	60		60	2

Course Code	Course Name	Theory					TW	PR/OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2991501	Professional Communication Skills for Engineers in the Age of AI	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

In the present academic and professional world, communication is no longer limited to speaking and writing alone. Engineering students are expected to communicate effectively in classrooms, laboratories, projects, digital platforms, technical environments and AI-enabled workplaces.

The rise of Artificial Intelligence has changed the way students draft, edit, summarize, research, present and document information. Therefore, Computer Engineering students must learn not only the fundamentals of communication but also the responsible and ethical use of AI tools.

This course develops students' ability to understand communication processes, overcome barriers, listen actively, speak confidently, read technical and non-technical texts, write professional and technical documents, and use AI tools ethically. It also prepares students for future academic and professional pathways such as Civil Services, MBA, MS abroad, GRE/TOEFL/IELTS, M.Tech, research and entrepreneurship.

The course follows a practical and student centred approach through classroom tasks, writing practice, listening activities, AI-assisted exercises, self-learning assignments and reflection.

Course Objectives:

Sr. No.	Course Objectives
1	Understand the fundamentals and dynamics of communication in academic, professional, technical and digital contexts. Analyse communication barriers, audience, purpose and context for appropriate communication.
2	Develop effective listening, speaking, reading and writing skills.
3	Comprehend technical and non-technical texts for academic and professional tasks.
4	Prepare professional and technical documents, such as letters and emails, and efficiently organize and create purposeful technical writing for professional transactions.
5	Critically evaluate communication content and improve it using digital tools and AI-based applications while maintaining clarity, accuracy, originality, and ethical standards.
6	Use AI tools responsibly and ethically to communicate effectively with AI agents, obtain accurate information, and maintain originality and authenticity in professional communication.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain and analyze communication processes, barriers, audience needs and workplace communication requirements in academic and professional contexts.	L4
2	Demonstrate effective listening, speaking, interpersonal and presentation skills using appropriate verbal and non-verbal communication strategies.	L3
3	Apply reading, note-making, summarization and information-processing techniques to comprehend technical and non-technical texts.	L3
4	Evaluate and enhance communication outputs using digital technologies and AI-assisted tools while ensuring clarity, accuracy and effectiveness.	L3
5	Prepare professional, business and technical documents using appropriate formats, language conventions and communication principles.	L5
6	Apply ethical principles and AI literacy skills for responsible communication, content creation and professional decision-making in digital environments.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Basic Knowledge of English Language	-	-
1	Module I: Fundamentals of Communication	**Content:** 1.1 Communication: Meaning, and Importance 1.2 Process of Communication 1.3 Types of Communication: Verbal and Non-verbal Communication 1.4 Barriers to Communication 1.5 Communication at Workplace- 1. Communication for self-development and goal setting 2. Dialogs and short group presentations 1.6 Introduction to AI Tools for Communication ● Human verification of AI-generated content ● Ethical use of AI in academic communication (real life examples on ethics) ● Plagiarism, misinformation, bias and fake references ● Data privacy and responsible sharing ● Digital communication etiquette (Mobile,Video conferencing etiquettes) **Self-Learning Topics:** Self-introduction sessions >.< Case studies and role play on fundamentals of communication, >.< Situations to identify barriers to communication and strategies to overcome them, >.< Identify five communication barriers faced by students in classroom or online learning. >.< Case studies on miscommunication in the Workplace >.< Effective team communication Interactive Activities. **Link :** ● (Standard courses URL)OpenLearn – Effective Communication in the Workplace: https://www.open.edu/openlearn/money-business/effective-communication-the-workplace ● MIT Communication Lab: https://mitcommlab.mit.edu	10	**CO 1**

2	**Verbal Aptitude and Language Proficiency**	**Content:** **2.1. Verbal Aptitude and Vocabulary Development** Root Words >.< Prefixes and Suffixes >.< Word forms >.< Synonyms and Antonyms >.< Collocations >.< Idioms and Expressions >.< Vocabulary Building Strategies >.< Academic and Technical Vocabulary >.< Contextual Usage of Words at a similar difficulty level of entrance tests like CAT/GRE/GMAT & proficiency tests like TOEFL/IELTS **2.2. Grammar for Professional Communication** Subject-Verb Agreement >.< Articles and Determiners >.< Prepositions >.< Sentence Structure and Sentence Correction >.< Modifiers >.< Punctuation >.< Common Errors in Professional Writing >.< Editing and Proofreading Skills >.< Redundancies, Idioms, Cliches at a similar difficulty level of entrance tests like CAT/GRE/GMAT & proficiency tests like TOEFL/IELTS **2.3 Interactive & AI-assisted Activities:** AI-assisted Grammar Checking and Editing >.< Comparing Human-written and AI-generated Responses >.< Vocabulary Enhancement using AI Tools >.< Online Aptitude Mock Tests and Quiz Activities >.< Group Competitions on Grammar and Verbal Skills >.< Peer Review and Error Analysis Activities **Self-Learning Topics:** Advanced English Grammar for Competitive Examinations >.< Functional Grammar in Professional Communication >.< Vocabulary Enhancement through Reading and Digital Tools >.< Word Power Techniques and Memory Strategies >.< Common Errors in Workplace Communication >.< AI-based Grammar Checking and Writing Enhancement Tools **Link:** • British Council – English Grammar and Vocabulary • BBC Learning English – Grammar and Vocabulary • Cambridge English – Grammar and Vocabulary Activities • ETS – TOEFL English Learning	10	**CO 5**

		Resources • Vocabulary.com • Grammarly Handbook		
3	**Module 3:** **Listening and Speaking Skills**	**Content:** **3.1 Listening Skills:** Importance of listening in academic and professional contexts >.< Types of listening >.< Listening barriers **3.2 Speaking Skills** Fluency and accuracy >.< Phonetics >.< Voice modulation >.< Pronunciation & articulation (paralanguage) **3.3 Enhancing Listening and Speaking Proficiency through Language Labs, Open-Source Platforms, and AI-Based Learning Tools.** AI based speech recognition tools and listening practice using podcasts, audio books and Youtube. >.< Introduction to AI tools for enhancing peaking skills with the help of chatbot and irtual assistant. Voice modulation and accent raining applications.eg.LingoChatAI: / Microsoft opilot >.<Limitations and Ethical Considerations >.< ccuracy and reliability of AI-generated responses **3.4 Conversational Activities :** Monologues, Dialogues >.< Creating a professional self-introduction >.< Personal branding through verbal communication >.< Elevator pitch fundamentals **Self-Learning Topics:** Listen to a short educational video and prepare five key points. >.< Practise a one-minute explanation of a simple technical concept.>.< Record your own speech and note pronunciation, speed and clarity. >.< Prepare five polite ways of asking doubts in class.>.< Deliver a 60-second professional introduction. **Link :** • British Council – Listening Skills Practice • British Council – Speaking Skills Practice • BBC Learning English • TED Talks Official Website • Coursera – Introduction to Public Speaking	10	**CO 2**

4	**Module 4: Reading Comprehension and Summarization**	**Content:** **4.1. Importance of Reading in Engineering Education-** Reading technical and non-technical texts **4.2. Types of Readings** Intensive Reading >.< Extensive Reading >.< Skimming >.< Scanning >.< SQ5R Method *(Survey, Question, Reading, Recording, Recall, Review and Revise)* **4.3.Reading & Summarisation Skills** Summarising text to Graphic Organisers (GO) and visa-versa. Venn diagrams, Radial Diagrams (*Mindmaps*), Tree Diagrams, Cyclic Diagrams, Flow Charts, Timelines, Matrix (*Tables*), Pyramids >.< Summarising text in point form >.< Summarising text in one-sentence central idea **4.4. AI integration: Reading & Summarisation Skills** AI-assisted summarization and paraphrasing>.< comparison of AI-generated and student-generated summaries >.< verification of AI-generated references and avoidance of false citations >.< AI for summarizing and paraphrasing. **Self-Learning Topics:** Read a technical article and identify its main idea, keywords and conclusion. >.< Prepare notes from a given passage using headings and bullet points. >.< Write a 100-word summary of a non-technical article. >.< Use an AI tool to summarize a passage, then compare it with your own summary and identify errors or missing points **Link :** 1. British Council – Reading Skills Practice 2. BBC Learning English – Reading Skills 3. Coursera – Improve Your English Communication Skills Specialisation 4. Coursera – Learning How to Learn 5. edX – English Grammar and Style	10	**CO 4**
5	**Module 5: Professional Correspondence.**	**Content:** **5.1 Principles of Business Writing: (7C's)** Completeness >.< Conciseness >.< Consideration, Concreteness >.< Clarity >.< Courtesy, Correctness. **5.2 Format & Types of Formal Letters** Parts of a Formal Letter in **Complete Block Style** >.< Request/Permission Letter >.< Claim and Adjustment Letter >.< Sales Letter >.< E-mails >.< Professional email writing	10	**CO 5**

		5.3 AI for drafting, editing and proofreading **5.4 Professional Digital Presence** Introduction to LinkedIn (samples) >.< Components of a professional LinkedIn profile >.< Professional profile headline >.< Digital networking etiquette. **Self-Learning Topics** Draft a professional email to a professor requesting academic guidance. >.< Use AI to edit your email/letters, then manually check tone, accuracy and originality. **Link:** Purdue OWL – Writing the Basic Business Letter: https://owl.purdue.edu/owl/subject_specific_writing/professional_technical_writing/basic_business_letters/index.html Purdue OWL – Email Etiquette for Students: https://owl.purdue.edu/owl/general_writing/academic_writing/email_etiquette_for_students.html		
6	**Technical Writing**	**Content:** **6.1 Basic Technical Writing** 15. What is Technical Writing? 16. Importance and Principles of Technical Writing 17. Framing Definitions **6.2 Writing User Instructions** Writing User Instructions >.< Writing Special Notices (*Note, Caution, Precaution, Warning and Danger*) **6.3 Technical writing and user instruction preparation with the help of AI agents:** Introduction to AI-Assisted Technical Writing >.< Prompt Writing Techniques for Technical Documentation >.< Evaluating the Reliability and Limitations of AI-Generated Technical Content >.< Using Conversational AI for Clarifying Technical Processes >.< Practical Activities Using AI Chatbots for Technical Writing Exercises >.< Case Studies on AI Applications in Documentation and User Support >.< Reviewing and Validating AI-Generated Instructions **Self-Learning Topics:**	10	**CO 5**

		Prepare user instructions for operating a household or laboratory device. >.< Draft safety notices (Note, Caution, Warning, Danger) for a given technical scenario. >.< Use AI chatbots to generate user instructions and compare them with manually written instructions. >.< Practice writing prompts to obtain accurate technical documentation from AI tools. >.< Analyze AI-generated technical content and identify errors, ambiguities, or inaccuracies. **Link:**Technical Writing Resources ● https://lp.hurleywrite.com/?utm_source=chatgpt.comPurdue OWL – Technical Writing ● Technical Communication Center		
			Total Hours	60

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Sanjay Kumar and Pushp Lata, *Communication Skills*, Oxford University Press, 2nd Edition, 2018.
2. Meenakshi Raman and Sangeeta Sharma, *Technical Communication: Principles and Practice*, Oxford University Press, 3rd Edition, 2015.
3. Ashraf Rizvi, *Effective Technical Communication*, McGraw Hill Education, 2nd Edition, 2017.
4. Barun K. Mitra, *Personality Development and Soft Skills*, Oxford University Press, 2nd Edition, 2016.
5. Raymond Murphy, *English Grammar in Use*, Cambridge University Press, 5th Edition, 2019.
6. Norman Lewis, *Word Power Made Easy*, Goyal Publishers, Revised Edition, 2014.
7. Andrea J. Rutherford, *Basic Communication Skills for Technology*, Pearson Education, 2nd Edition, 2007.
8. Krishna Mohan and Meera Banerji, *Developing Communication Skills*, Macmillan Education, 2nd Edition, 2017.

Online References:

Sr. No.	Website Name	URL
1	Communication process and workplace communication	https://www.open.edu/openlearn/money-business/effective-communication-the-workplace
2	Email etiquette and formal digital communication	https://owl.purdue.edu/owl/general_writing/academic_writing/email_etiquette.html
3	Active listening	https://www.open.edu/openlearn/mod/oucontent/view.php?id=87005&section=2
4	Listening improvement	https://www.open.edu/openlearn/mod/oucontent/view.php?id=166863&section=_unit4.5
5	Technical communication guides	https://mitcommlab.mit.edu/eecs/use-the-commkit/
6	Email writing	https://owl.purdue.edu/owl/general_writing/academic_writing/email_etiquette_for_students.html
7	Resume writing	https://mitcommlab.mit.edu/eecs/use-the-commkit/
8	Business and technical writing	https://owl.purdue.edu/owl/subject_specific_writing/professional_technical_writing/basic_business_letters/index.html
9	AI prompting	https://www.skills.google/paths/2337/course_templates/1229
10	Responsible AI	https://mitcommlab.mit.edu/cee/commkit-resources/
11	Career and academic communication	https://owl.purdue.edu/owl/job_search_writing/job_search_letters/cover_letters_1_quick_tips/index.html

Assessment: .

- **Self Learning (SL):** Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.
- **Term Work:** Term Work shall consist of at least 08-10 Activities based on the above syllabus.
- **Term Work Marks:** 25 Marks (Total) = 20 Marks (Activities) + 05 Marks (Attendance)
- **Practical & Oral Exam:** An Oral & Practical Exam will be held based on the above list of Practicals

F.E. Semester – II

Bachelor of Computer Engineering

Syllabus

Semester II

Theme: Smart Vikasit Bharat: A Multidisciplinary Ecosystem

Aligned with UNSDG

- SDG 8: Decent Work and Economic Growth
- SDG 9: Industry, Innovation and Infrastructure
- SDG 11: Sustainable Cities and Communities
- SDG 12: Responsible Consumption and Production
- SDG 17: Partnerships for the Goals

Keywords

Smart Cities, Sustainable Development, Innovation Ecosystem, Digital Transformation, Infrastructure Development, Resource Management, Community Development, Industry 4.0, Citizen-Centric Solutions, Multidisciplinary Engineering, Viksit Bharat 2047

Theme Description

India's vision of **Viksit Bharat 2047** aims to transform the nation into a developed, self-reliant, innovative, and sustainable economy. Achieving this vision requires the integration of technology, infrastructure, governance, environment, education, healthcare, industry, and citizen participation into a cohesive ecosystem.

The theme **"Smart Vikasit Bharat: A Multidisciplinary Ecosystem"** encourages students to understand how multiple disciplines work together to address real-world challenges. It promotes the development of innovative and sustainable solutions that improve quality of life, strengthen communities, optimize resources, and support national development goals.

For rapidly growing regions such as **Navi Mumbai**, multidisciplinary approaches are essential to address urbanization, transportation, sustainability, infrastructure expansion, environmental conservation, digital services, and citizen welfare.

Generalized Problem Statements

1. Sustainable Urban Infrastructure and Community Development

Rapid urban growth creates increasing demands on transportation, housing, utilities, public amenities, and social infrastructure. There is a need to develop integrated approaches that support sustainable and inclusive urban development in Navi Mumbai.

2. Resource Optimization for Smart and Sustainable Communities

Growing populations place pressure on water, energy, land, transportation, and public services. Efficient utilization and management of resources remain critical for long-term sustainability and economic growth.

3. Citizen-Centric Services for a Developed India

Modern communities require accessible, efficient, and responsive public services. Understanding citizen needs and improving service delivery mechanisms is essential for building inclusive and people-focused ecosystems.

4. Innovation and Entrepreneurship for Local Development

Many societal and industrial challenges present opportunities for innovative products, services, and entrepreneurial ventures. Identifying and addressing local needs can contribute to employment generation and economic development.

5. Environmental Sustainability and Climate Resilience

Urban development must be balanced with environmental protection, climate adaptation, and responsible resource utilization. Communities require sustainable approaches that support ecological and economic well-being.

6. Integrated Smart Ecosystems for Future-Ready Cities

Future cities require coordinated planning across transportation, energy, environment, governance, healthcare, education, and infrastructure. Developing interconnected and resilient ecosystems remains a key challenge for achieving the vision of Viksit Bharat.

Mapping to First-Year Semester II Learning

These problem statements are intentionally broad and multidisciplinary so they can be linked to:

- Engineering Mathematics
- Engineering Physics
- Engineering Chemistry
- Basic Mechanical Engineering
- Basic Civil Engineering
- Engineering Graphics
- Programming Fundamentals
- Workshop Practice
- Environmental Studies
- Innovation and Project-Based Learning

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112101	Computational Engineering Mathematics - II	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112101	Computational Engineering Mathematics II	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The course provides fundamental knowledge of engineering mathematics essential for analytical modeling, computational techniques, optimization, and problem-solving in modern engineering applications. It develops mathematical proficiency through multiple integration, vector calculus, numerical methods, linear programming, and nonlinear optimization techniques used in intelligent systems, data analysis, scientific computing, and algorithmic problem-solving. The course further strengthens logical reasoning, quantitative analysis, and computational thinking required in the fields of Artificial Intelligence, Computer Engineering, and Information Technology.

Course Objectives:

Sr. No.	Course Objectives
1	To apply multiple integration techniques for solving engineering and computational problems involving area, volume, and modeling.
2	To analyze vector fields and vector calculus operations used in scientific computing and engineering systems.
3	To implement numerical methods for solving differential equations and numerical integration problems.
4	To formulate and optimize linear programming models for engineering decision-making and resource allocation problems.
5	To evaluate nonlinear optimization problems using constrained optimization techniques.
6	To develop analytical, logical, and computational problem-solving skills required for engineering and intelligent systems applications.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Apply double integration techniques in Cartesian and polar coordinates to solve engineering and computational problems. *(Apply)*	L3
2	Evaluate double and triple integrals for applications involving area, volume, and mathematical modeling. *(Evaluate)*	L5
3	Analyze vector fields using gradient, divergence, curl, Green's theorem, and Stoke's theorem. *(Analyze)*	L4
4	Implement numerical techniques for solving ordinary differential equations and numerical integration problems. *(Implement)*	L3
5	Formulate and solve linear programming problems using Simplex and Dual Simplex methods. *(Apply)*	L3
6	Evaluate nonlinear programming problems using Lagrange multipliers and Kuhn–Tucker conditions. *(Evaluate)*	L5

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Algebraic operations and equations, Differential and integral calculus, Coordinate geometry and vectors, Functions and graphical interpretation, Elementary matrices and determinants, Basic logical and analytical problem-solving techniques.	02	-
1	**Multiple Integration- I**	1.1 Basic of Integration, Double integration - definition, Evaluation of Double Integrals.(Cartesian & Polar). 1.2 Evaluation of double integrals by changing to polar coordinates. **Self- Learning topic-**Tracing of curves, Change the order of Integration. **Link:** https://nptel.ac.in/courses/111104229?utm_source https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/?utm_source	**04**	**CO1**
2	**Multiple Integration II**	2.1 Application of Double integration to Compute Area. 2.2 Triple Integration definition and Evaluation. **Self- Learning topic-**Application of Triple Integration to Compute Volume. **Link :**	**05**	**CO2**

		https://onlinecourses.nptel.ac.in/noc24_ma20/preview?utm_source		
3	**Vector Calculus**	3.1Gradiant, Divergence and Curl of Vector Field , Solenoidal and Irrotational (Conservative vector field), Line Integral (definition and simple problems) 3.2 Green's theorem in a plane(Without Proof), Stoke's theorem (Without proof) **Self- Learning topic-** Work done, Angle between surfaces, Divergence theorem. **Link :** https://nptel.ac.in/courses/111106101?utm_source https://ocw.mit.edu/courses/18-02-multivariable-calculus-fall-2007/?utm_source	**04**	**CO3**
4	**Numerical solution of ordinary differential equations of first order and first degree, and Numerical Integration**	4.1 Numerical solution of ordinary differential equation using (a) Euler's method (b) Modified Euler method, (c) Runge-Kutta fourth order method 4.2 Numerical integration-by (a) Trapezoidal (b) Simpson's 1/3rd (c) Simpson's 3/8th rule (all without proof) **Self- Learning topic-** Adaptive step size method,Application in computer science and engineering. **Link :** https://nptel.ac.in/courses/111105090?utm_source https://www.coursera.org/learn/numerical-methods-engineers?utm_source	**05**	**CO4**
5	**Linear Programming Problems**	5.1 Types of solutions, Standard and Canonical of LPP, Basic and Feasible solutions, slack variables, surplus variables. 5.2 Simplex method. 5.3 Duality, Dual of LPP and Dual Simplex Method. **Self-learning Topics**: Artificial variable method, Big M method, Sensitivity Analysis, Two-Phase Simplex Method,Revised Simplex Method. **Link :** https://nptel.ac.in/courses/110107081?utm_source https://www.coursera.org/learn/operations-research-modeling?utm_source	**05**	**CO5**

6	**Non Linear Programming Problems**	6.1 NLPP with one equality constraint (two or three variables) using the method of Lagrange's multipliers 6.2 NLPP with one inequality constraint (two variables): Kuhn-Tucker conditions **Self-learning Topics:** Problems with two equality constraints, Problems with two inequality constraints, Unconstrained optimization: One, dimensional search method (Golden Search method, Newton's method). Gradient Search Method. **Link :** https://nptel.ac.in/courses/112108298?utm_source https://ocw.mit.edu/courses/15-084j-nonlinear-programming-spring-2004/?utm_source	**05**	**CO6**
			Total Hours	30

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References Books:

1. Higher Engineering Mathematics by Dr. B.S. Grewal.
2. Advanced Engineering Mathematics by R.K. Jain and S.R.K. Iyengar.
3. Fundamentals of Mathematical Statistics by S.C. Gupta and V.K. Kapoor.
4. Numerical Methods for Scientific and Engineering Computation ,M. K. Jain, S. R. K. Iyengar and R. K. Jain , New Age International Publishers.
5. Elementary Linear algebra with application by Howard Anton and Christ Rorres. 6th edition.
6. Bali N.P and Goyal M. "A textbook of Engineering Mathematics" Laxmi Publication, 10th Ed. 2022.
7. Gupta C.B. Singh S.R and Mukesh Kumar : "Engineering Mathematics for semester I and II", Mc- Graw Hill education ,2015Pal S. & Bhunia S.C: "Engineering Mathematics " Oxford University Press, 3rd Ed, 2016.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://nptel.ac.in/
2	MIT OpenCorseWare	https://ocw.mit.edu/courses
3	Wolfram Mathworld	http://mathworld.wolfram.com

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112102	Data Structure for Engineering Science	3	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR/OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112102	Data Structure for Engineering Science	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Data Structures are fundamental for building intelligent and scalable digital systems. The course supports multidisciplinary applications in healthcare, smart cities, transportation, agriculture, cybersecurity, governance, and Industry 4.0 ecosystems. The syllabus aligns with UNSDG goals and NEP 2020 by integrating real-world problem solving, smart system applications, and emerging technologies.

Course Objectives:

Sr. No.	Course Objectives
1	Understand fundamentals of data structures and abstract data types.
2	Apply suitable data structures for efficient problem solving.
3	Develop engineering solutions using linear and non-linear data structures.
4	Apply data structures to multidisciplinary engineering applications.
5	Explore applications of data structures in AI, IoT, and smart systems.
6	Develop computational thinking for sustainable digital ecosystems.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain concepts of linear and non-linear data structures.	L2
2	Implement stack operations and recursion techniques for computational applications.	L3
3	Apply queue operations for scheduling and real-time system applications.	L3
4	Develop linked list-based solutions for dynamic data management.	L3
5	Apply tree structures and traversal techniques for hierarchical data processing.	L3
6	Apply graph traversal and hashing techniques for complex data representation and management.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Basics of programming using C.		-
1	**Introduction to Data Structures**	1.1 Introduction to Data Structures, Types of Data Structures –Primitive and Non-Primitive data structure, Linear and Nonlinear, 1.2 Operations on Data Structures, Concept of ADT, Smart Governance and IoT Systems. **Self-Learning Topics:** Sustainable computing concepts, Aadhaar-scale data systems, Data organization in smart cities. Applications in AI, Healthcare. Online reference: **https://youtu.be/zWg7U0OEAoE?si=Pxg2AEf5tthEA08j**	06	CO1
2	**Stacks**	2.1 ADT of Stack, Stack Operations, Array Implementation of Stack, Recursion, 2.2 Well-formedness of Parenthesis, Recursion, Infix to Postfix Conversion, Postfix Evaluation, Applications in Browser Navigation. **Self-Learning Topics:**Call stack visualization, Stack applications in cloud systems and cybersecurity, AI Chatbots and Smart Systems. Online reference: **https://youtu.be/g1USSZVWDsY?si=xIJS853LrEo586Oi**	07	CO2
3	**Queues**	3.1 ADT of Queue, Queue Operations, Array Implementation of Queue, 3.2 Circular Queue, Priority Queue, Double Ended Queue, Applications of Healthcare Queue Management. **Self-Learning Topics:**Queue scheduling in cloud computing and real-time systems, Railway Reservation Systems, Queue in Smart Traffic Systems. Online reference: https://youtu.be/PGWZUgzDMYI?si=CKfs4rMbcygMnviT	06	CO3
4	**Linked Lists and Dynamic Data Management**	4.1 Introduction to Linked List, Representation of Linked List, Linked List vs Array, 4.2 Singly Linked List, Circular Linked List, Doubly Linked List, 4.3 Implementation of Stack and Queue, Applications in Social Media Feeds. **Self-Learning Topics:** Dynamic memory allocation, Linked lists in blockchain and IoT systems. Online reference: **https://youtu.be/PGWZUgzDMYI?si=CKfs4rMbcygMnviT**	07	CO4

5	**Trees and Hierarchical Intelligent Systems**	5.1 Introduction to Trees, Tree Terminologies, Binary Tree, Representation of Binary Tree, 5.2 Types of Binary Trees, Binary Tree Traversals, Binary Search Tree, 5.3 AVL Tree, B Tree, B+ Tree, Applications – Expression Tree, Huffman Encoding. **Self-learning topics**: Decision trees in machine learning, Hierarchical healthcare databases, AI Decision Systems and Recommendation Systems. Online reference: https://youtu.be/tORLeHHtazM?si=eYJYbhvnw8zMNdMZ https://youtu.be/eWeqqVpgNPg?si=oMB7O6oZ38GJbIC https://youtu.be/mRGQylRWAsI?si=oBumxHftv_tf938j https://youtu.be/TbvhGcf6UJU?si=wmEBh6ylEgFPxAXx	10	CO5
6	**Graphs and Searching Techniques**	6.1 Introduction to Graphs, Graph Terminologies, Graph Representation, 6.2 DFS and BFS Traversals, Linear Search, Binary Search, 6.3 Hashing, Hash Functions, Collision Handling Techniques- Linear Probing, Chaining. Applications in Smart Transportation. **Self-learning topics**: Graph databases, Search optimization in AI systems, Smart network analysis. Cybersecurity Networks, Social Network Analysis and Smart City Systems. Online reference: https://youtu.be/9zpSs845wf8?si=57_LieJkyyVFEkv3 https://youtu.be/CIm6RzdoPCI?si=y19yubJ1mH76VEmA https://youtu.be/r1-p11fSPw?si=BPZBVppkWT1N6pZn https://youtu.be/KW0UvOW0XIo?si=M1UhJhwiGGREuCq_	09	CO6
		Total Hours	45	

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

6. E. Balagurusamy, “Data Structure Using C”, Tata McGraw-Hill Education India.
7. Richard F. Gilberg and Behrouz A. Forouzan, “Data Structures: A Pseudocode Approach with C”, 2ndEdition, CENGAGE Learning.
8. Sahni Horowitz, “Fundamentals of data structures in C”, computer science press, 2008.
9. Jean Paul Tremblay, P. G. Sorenson, “Introduction to Data Structure and Its Applications”, McGraw-Hill Higher Education

10. Narasimha Karumanchi, “Data Structures And Algorithms”, 5th Edition,CareerMonk, 2016.
11. Robert Kruse, C. L. Tondo, Bruce Leung, “Data Structures and Program Design in C”, Pearson Publication. Moseley, Jürgen Garche

Online References:

Sr. No.	Website Name	URL
1	NPTEL Swayam	https://onlinecourses.swayam2.ac.in/e-learning/preview/nou24_cs15
2	NPTEL Swayam	https://nptel.ac.in/courses/106105085?
3	Coursera	https://www.coursera.org/specializations/data-structures-algorithms?

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112103	Engineering Mechanics and Graphics	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112103	Engineering Mechanics and Graphics	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Engineering Mechanics is a branch of science that deals with the behavior of solid bodies subjected to external forces and the effects of these forces on the bodies. In the modern era of Computer Engineering, Information Technology, and Artificial Intelligence & Data Science (AI&DS), the subject has gained increasing relevance through its applications in intelligent systems, automation, simulation, and digital technologies. These engineers are now actively involved in the development of robotics, autonomous systems, smart manufacturing, virtual and augmented reality, gaming physics engines, digital twin technology, IoT-enabled devices, and AI-driven simulation platforms. Engineering Mechanics provides the fundamental principles of force, motion, equilibrium, and structural behavior that support the modelling and analysis of such systems. As the UNSDG Quality Education & Sustainable Cities and Communities goals are mapped.

Course Objectives:

Sr. No.	Course Objectives
1	Analyze coplanar force systems using the concepts of force, moment, resultant, and equilibrium for engineering problem-solving.
2	To acquaint with the basic concept of friction and to understand the parameters required to quantify the Kinetics of rigid body.
3	Understand and apply the basic concepts of robot kinematics for analyzing the motion of robotic mechanisms.

4	To impart and inculcate proper understanding of the theory of projection.
5	To acquaint students with representing internal features of a three-dimensional object by way of section that accurately conveys their internal orientation.
6	Demonstrate proficiency in visualizing and representing three-dimensional objects through standard engineering drawing techniques.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Determine the equivalent force-couple system and illustrate the concept of force, moment and apply the same along with the concept of equilibrium with the help of FBD.	L3
2	Analyze frictional forces of rigid bodies on rough horizontal and inclined planes and how an accelerating body can be analyzed as a static system by introducing an inertial force in the opposite direction of acceleration.	L4
3	Establish the relation between robot joints and parameters.	L3
4	Apply the basic principles of Projection of Lines and Planes.	L3
5	Apply the basic principles of projections in converting pictorial views into orthographic Views.	L3
6	Apply the basic principles of projections in converting orthographic Views into isometric drawing.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
	Prerequisites	1.1 Resolution of a force. Use of trigonometry functions. Parallelogram law of forces. Law of triangle. Polygon law of forces, Lami's theorem. Concepts of Vector Algebra. Uniformly accelerated motion along a straight line, motion under gravity. 1.2 To draw basic geometric shapes like pentagon, hexagon and square (in different orientation). Divide a circle into equal number of parts. (Note: There will be no questions from the prerequisite in the theory examination)	01	–
1	System of Forces and Equilibrium of Force system	Classification of force systems, Principle of transmissibility, composition and resolution of forces. Resultant coplanar force system (Concurrent forces, parallel forces and general system of forces). Moment of force about a point, Couples, Varignon's Theorem. 1.1 Equilibrium: Conditions of equilibrium for concurrent forces, parallel forces and general forces, Couples. Equilibrium of rigid bodies, free body diagrams. 1.2 Equilibrium of Beams: Types of beams, simple and compound beams, type of supports and reaction: Determination of reactions at supports for various types of loads on beams. (Excluding problems on internal hinges)	**07**	CO1

		Self-learning topic: - Analyze force systems in digital kiosks and smart classroom infrastructure using principles of equilibrium. - Explore beam behavior and support reactions in household furniture, balconies, shelves, and simple civil engineering structures. Online Reference: https://youtu.be/6u_rjLjv-MY?si=T_pIJ9FeMJSjkcn2		
2	Kinetics of particles and Friction	**2.1**Force and Acceleration: -Introduction to basic concepts, D'Alembert's Principle, concept of Inertia force, Equations of dynamic equilibrium. **2.2** Laws of friction. Cone of friction. angle of repose, angle of friction, equilibrium of bodies on a horizontal and inclined plane. **Self-learning topic:** - Analyze stability and motion of objects on inclined planes in real applications like ramps, loading platforms, and slopes using angle of friction and angle of repose concepts. - Apply D'Alembert's Principle to Study movement and stability of portable smart learning carts. Online Reference: https://youtu.be/VQRcChR9IkU?si=2_FLPFlrPWQ-TGnT https://youtu.be/X4_K-XLUIB4?si=I90xxgoz_8DfBtUP	**05**	CO2
3	Introduction to Robot Kinematics	Fundamental of Robot Mechanics,Degree of Freedom,D-H Parameters,robot kinematics (Forward), Homogeneous transformation (limited to 2 DOF Serial robot) **Self-learning topic:** - Analyze forward kinematics of simple 2-DOF robotic arms using real-life examples like assistive robotic devices used in education. Online Reference: https://youtu.be/xrwz9IxpMJg?si=PZZIyDmvTGpNbeWj https://youtu.be/sk9A9gntBnk?si=T2-C3l8Z0JhxLFPi	**02**	CO3
4	Projections of Points, Lines and Planes	**4.1 Projections of Points** Projections of points in any quadrants as well as resting on planes. **4.2 Projections of Lines** Projections of lines inclined to both the reference planes (Excluding Traces of lines). Simple application-based problems on projection of lines. **4.3 Projections of Planes** Projections of planes (Triangular, Square, Rectangular, Pentagonal, Hexagonal and Circular) inclined to both the Reference Planes. (Exclude composite planes). **Self-learning topic:**	**05**	CO4

		- Study how projections of planes are used in real engineering drawings of machine parts like plates, brackets, and sheet metal components. - Use CAD software to practice projections of lines inclined to both reference planes and compare with manual drawing methods. Online Reference: https://www.youtube.com/watch?v=sUDQExiEh8U&list=PLp6ek2hDcoNCjoRLQ4rjpCozisCACBxKA&index=15		
5	Orthographic Projections	**5.1 Orthographic Projections** Fundamentals of orthographic projections like concept of quadrants, observer position, horizontal, vertical and profile plane, symbol etc. Different orthographic views, First and Third angle method of projection. **5.2 Sectional Orthographic Projections** Fundamentals of sectional projections like concept of section plane, its representation, section lines and its features, need of sectional views, rib and web in section. Types of section and its representation. Different views of a simple machine part as per the first angle projection. **Self-learning topic:** - Practice converting simple real objects (mobile charger, bottle, tool parts) into orthographic projections using both manual drawing and CAD software. - Learn how ribs, webs, and hidden details are represented in industrial sectional drawings used in manufacturing and design industries. Online Reference: https://youtu.be/OgyP6OgHEIo?si=FlXARLLkka26agCm https://www.youtube.com/watch?v=cpFGfg0f7xM&list=PLLy_2iUCG87Bw9XPfEF3r3EW5UlAOv8iz&index=12	**06**	CO5
6	Isometric Views	Introduction to isometric projection, Isometric lines, Planes, Isometric lines and Planes, isometric scale, Isometric projection and isometric views. Construction of isometric views from given orthographic views. (Excluding sphere) **Self-learning topic:** - Analyze differences between isometric projection and isometric view using real CAD models and understand their practical significance. - Develop spatial visualization skills by mentally converting everyday objects into isometric forms and identifying isometric and non-isometric lines. Online Reference: Lecture 1: Introduction to isometric projection and isometric Scale nptel.ac.in/courses/112105294	**04**	CO6
			Total Hours	30

Note:

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References Books:

1. Engineering Mechanics by R. C. Hibbeler.
2. Engineering Mechanics by F. L. Singer, Harper& Raw Publication
3. Engineering Mechanics by Macklin & Nelson, Tata McGraw Hill
4. Introduction to Industrial Robotics by Ramchandran Nagrajan, Pearson publication
5. Narayana, K.L. & P Kannaiah (2008), Textbook on Engineering Drawing, Scitech Publisher.
6. Prof. Sham Tickoo (Purdue University) & Gaurav Verma, "(CAD Soft Technologies).
7. Auto CAD 2012 (For engineers and Designers)", Dreamtech Press New Delhi.
8. Engineering Mechanics by A K Tayal, Umesh Publication.
9. Engineering Mechanics by Beer &Johnston, Tata McGraw Hill.
10. N.D. Bhatt, "Engineering Drawing (Plane and solid geometry)", Charotar Publishing House Pvt. Ltd.
11. N.D. Bhatt & V.M. Panchal, "Machine Drawing", Charotar Publishing House Pvt. Ltd.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://onlinecourses.nptel.ac.in/noc21_me70/preview
2	NPTEL	https://nptel.ac.in/courses/112103019
3	NPTEL	https://nptel.ac.in/courses/112105236

Assessment:

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End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.

- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112104	Foundation of Network Science	03	--	--	45	--	--	45	90	3

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112104	Foundation of Network Science	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This course introduces the fundamentals of computer networking, communication systems, IP connectivity, wireless technologies, and network security for first-year engineering students of Computer Engineering, AI & DS, and IT disciplines. Designed as per NEP 2020,, the course emphasizes smart and connected digital ecosystems under the theme “Smart Viksit Bharat: A Multidisciplinary Ecosystem.”

The curriculum enables students to understand technologies supporting smart cities, intelligent transportation, telemedicine, digital governance, rural connectivity, and IoT-enabled sustainable systems. It aligns with UNSDGs including SDG 4,9,11,13 & 16 by promoting innovation, sustainability, secure communication, and digital inclusion.

The course develops strong technical foundations, problem-solving skills, cybersecurity awareness, and multidisciplinary thinking required for future-ready communication networks. Overall, the autonomous syllabus transforms the traditional Computer Networks subject into a “Smart Connected Systems Foundation Course.”

Course Objectives:

Sr. No.	Course Objectives

1	To introduce the fundamentals of computer networks, communication models, internet architecture, and smart digital networking infrastructure.
2	To provide knowledge of physical and data link layer technologies, transmission media, error detection, error correction, and wireless communication systems.
3	To develop understanding of network layer services, IPv4/IPv6 addressing, subnetting, routing protocols, and intelligent networking concepts such as SDN
4	To familiarize students with transport and session layer services, reliable communication protocols, congestion control, and secure session management techniques
5	To impart knowledge of presentation and application layer protocols, cloud networking, CDN concepts, multimedia communication, and data security fundamentals
6	To introduce emerging technologies such as IoT, AI-enabled networking, cybersecurity, edge/fog computing, and sustainable smart networking practices.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the fundamentals of computer networks, communication models, smart digital infrastructure, and OSI/TCP-IP architectures used in modern networking systems.	L2
2	Apply physical layer and data link layer concepts, transmission media, error control techniques, and wireless communication principles for reliable data transmission.	L3
3	Analyze IPv4/IPv6 addressing schemes, subnetting, routing mechanisms, and intelligent network layer technologies including SDN for scalable internet communication.	L4
4	Implement transport and session layer services such as TCP/UDP communication, congestion control, socket programming concepts, and secure session management techniques.	L3
5	Evaluate presentation and application layer protocols, cloud networking services, CDN concepts, encryption methods, and multimedia communication systems used in digital applications.	L5
6	Demonstrate the concepts of IoT, AI-enabled networking, cybersecurity mechanisms, sustainable networking, and ethical communication practices in smart digital ecosystems.	L3

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	Basics of computer systems, internet usage, binary concepts, and introduction to communication systems.	-	-
I	**Fundamentals of Smart Networking and OSI Architecture**	1.1 Introduction to computer networks: Types of networks: LAN, MAN, WAN , Network topologies, Internet architecture, OSI and TCP/IP models	6	CO1

		1.2 Data communication basics: Analog and Digital communication, Modes of Communication: simplex,duplex 1.3 Network devices: Hub, Switches, Router, Gateways, Bridges, Repeaters etc. 1.4 Smart Digital Infrastructure in India, BharatNet and Digital India initiatives **Self-learning Topics:** Evolution of Internet in India,5G, overview, Smart campus networking , Real-Time Applications of Computer Networks in Healthcare and Education, etc, Client-server vs peer-to-peer networks, Digital payment systems, Impact on villages and e-governance Online reference: https://onlinecourses.swayam2.ac.in/nou26_cs03/preview https://onlinecourses.nptel.ac.in/noc21_ee12/preview https://onlinecourses.swayam2.ac.in/cec19_cs07/preview https://pmc.ncbi.nlm.nih.gov/articles/PMC3128395/		
II	**Physical and Data Link Layer Technologies**	2.1 Guided and unguided Transmission Media: Twisted pair cables, Coaxial Cables, Optical Fiber, EM Spectrum 2.2 DLL Design issues: Services, MAC Addressing, Switching techniques, Framing, Error Control, Flow control 2.3 Error Detection and Correction: Checksum, CRC, Hamming Code 2.4 Elementary Data Link protocols: Stop and Wait, Sliding Window :Go Back N, Selective Repeat 2.5 Wireless LAN technology,Introduction to green communication systems **Self-learning Topics:** Performance Analysis of Twisted Pair, Coaxial and Fiber Optic Cables , Optical Fiber Communication and Its Industrial Applications , next-Generation Wi-Fi 6 and Wi-Fi 7 Technologies for Smart Campuses , Li-Fi Technology and Optical Wireless Communication Systems Smart transportation communication systems	9	CO2

		Online reference: https://academy.qualcomm.com/course-catalog/Wi-Fi-6-Technical-Training https://nptel.ac.in/courses/115107095 https://archive.nptel.ac.in/content/storage2/courses/105108073/module3/lecture8.pdf https://fr.xhfiber.com/types-and-industrial-applications-of-optical-fiber_n179		
III	**Network Layer and Intelligent Internet Routing**	3.1 Network layer service, Communication primitives : Unicast, Multicast, Broadcast 3.2 Introduction to IPv4 and IPv6. ,IPv4 Addressing:classful /classless, Subnetting and supernetting 3.3 Routing algorithms, Routing Protocols:ARP, RARP, ICMP,IGMP 3.4 Congestion Control Algorithms:Opel Loop and Closed Loop congestion control, QoS, Token and Leaky bucket algorithms 3.5 NAT, Introduction to Software Defined Networking(SDN) **Self-learning Topics:** VLSM (Variable Length Subnet Masking) in Modern Networks,IPv6 adoption in India , SDN applications , AI-assisted routing , Packet Forwarding and Routing Table Management Online reference: https://www.coursera.org/in/articles/ipv6 https://www.coursera.org/learn/sdn https://www.coursera.org/learn/ai-infrastructure-networking-techniques	9	CO3
IV	**Transport and Session Layer Services**	4.1 Transport Layer: Services, Port Numbers and Socket Addressing 4.2 TCP and UDP,Flow and Congestion Control 4.3 Session Establishment and Termination Session Synchronization, Dialog Control, Authentication Concepts 4.4 Smart Banking Communication Sessions **Self-learning Topics:** Video Conferencing Communication Protocols , Real-Time Session Management Systems , Digital Payment Communication Sessions , Client-Server Communication using TCP and UDP Sockets , Secure Communication Sessions in Digital	7	CO4

		Payment Systems , Future Trends in Intelligent and Secure Transport Layer Communication Online reference: https://onlinecourses.nptel.ac.in/noc24_cs19/preview https://onlinecourses.nptel.ac.in/noc24_cs118/preview https://www.coursera.org/learn/paytech		
V	**Presentation and Application Layer Services**	5.1 Presentation Layer Services:Data Representation, Data Encryption Basics, Data Compression Techniques, Multimedia Data Handling Character Encoding Standards 5.2 Application Layer Protocols: DNS, HTTP/HTTPS, FTP,SMTP, POP, IMAP 5.3 Cloud Networking Basics, CDN Concepts **Self-learning Topics:** Symmetric and Asymmetric Encryption Techniques , Lossless and Lossy Compression Algorithms, AI-Based Data Compression Techniques for Smart Communication Systems Content Delivery Networks (CDN) for High-Speed E-Learning and OTT Platforms Online reference: https://www.coursera.org/learn/cloud-networking https://www.coursera.org/projects/googlecloud-cloud-cdn-vovre https://onlinecourses.nptel.ac.in/noc24_cs120/preview	7	CO5
VI	**IoT, AI, Cybersecurity and Sustainable Smart Networking**	6.1 Introduction to IoT, Sensors and Smart Devices,Wireless Sensor Networks,AI-enabled Networking, Edge Computing, Fog Computing 6.2 Basics of Network Security,Firewalls and IDS,VPN Concepts,Cyber Ethics 6.3 Green Computing, Sustainable Networking **Self-learning Topics:** IoT protocols (MQTT, CoAP) , Cybersecurity frameworks, Green data centers, Ethical AI communication systems , Machine Learning Applications in Network Optimization , Intrusion Detection Systems (IDS) and Threat Monitoring Online reference:	7	CO6

		https://onlinecourses.swayam2.ac.in/cec20_cs16/preview https://onlinecourses.swayam2.ac.in/e-learning/preview/nou19_cs08 https://www.greendatacenters.com/ https://www.coursera.org/learn/ai-ethics		
			Total Hours	45

Note:

3. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
4. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References :

1. S.Keshav, "An Engineering Approach To Computer Networking", Pearson
2. Natalia Olifer & Victor Olifer, "Computer Networks: Principles, Technologies & Protocols for Network Design", Wiley India, 2011.
3. Larry L.Peterson, Bruce S.Davie, "Computer Networks: A Systems Approach", Second Edition ,The Morgan Kaufmann Series in Networking
4. A.S. Tanenbaum, "Computer Networks",4th edition Pearson Education
5. B.A. Forouzan, "Data Communications and Networking", 5th edition, TMH
6. James F. Kurose, Keith W. Ross, "Computer .Networking, A Top-Down Approach Featuring the Internet",6th edition, Addison Wesley

Online References:

Sr. No.	Website Name	URL
1	Computer Networks , IIT Madras	https://nptel.ac.in/courses/106106091
2	NOC : Computer Networks and Internet Protocol , IIT Kharagpur	https://nptel.ac.in/courses/106105183
3	Computer Networks	https://onlinecourses.swayam2.ac.in/e-learning/preview/cec19_cs07
4	Network Security	https://www.coursera.org/learn/network-security

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term

work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112105	Web Technology	02	--	--	30	--	--	30	60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112105	Web Technology	20	20	40	60	2	---	---	100

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The proposed Web Development curriculum is closely aligned **(85–90%) with the vision of "Smart Vikasit Bharat: A Multidisciplinary Ecosystem" and the objectives of NEP 2021**. It equips students with practical skills to design, develop, deploy, and maintain modern web applications that address real-world societal, business, and governance challenges. The curriculum strongly supports **UNSDG goals 9, 8 and 4.** Through hands-on learning, students gain expertise in responsive design, web security, cloud deployment, PWAs, and collaborative development using Git and GitHub. The syllabus fosters critical thinking, creativity, teamwork, and problem-solving through project-based learning and real-world case studies. Overall, it prepares learners to become skilled digital innovators capable of contributing to India's growing digital economy and sustainable development goals.

Course Objectives:

Sr. No.	Course Objectives
1	To introduce the fundamentals of Internet technologies and web architecture.
2	To develop skills for creating structured web pages using HTML5.
3	To apply CSS3 concepts for designing responsive and user-friendly web interfaces.
4	To understand JavaScript for creating interactive web applications.
5	To familiarize students with web hosting, security, and deployment concepts.
6	To encourage self-learning of modern web development trends and frameworks.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the fundamentals of Internet technologies.	L2
2	Design and develop structured web pages using HTML5.	L3

3	Apply CSS3 concepts and responsive design techniques to create visually appealing web interfaces.	L3
4	Develop interactive web applications using JavaScript and DOM manipulation techniques.	L4
5	Describe web hosting, deployment methods, cookies, sessions, and web security mechanisms.	L2
6	Explore emerging web technologies, APIs, accessibility standards, and modern web development trends.	L2

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	**Prerequisites**	Basic computer operation and file management, Fundamental understanding of Internet usage ,Basic programming logic and problem-solving skills	-	-
1	**Introduction to Web Technology**	1.1 Internet basics: • WWW, • Website and Web Application, • Web browsers, • Web servers, • URL, • DNS, • Client-Server architecture, • HTTP and HTTPS protocols **Self-Learning Topics:** Evolution of Web Technology, Web 1.0 to Web 3.0 Popular Web Browsers and Developer Tools **Online reference:** **https://www.webfx.com/blog/internet/the-evolution-of-the-web/** https://developer.chrome.com/docs/devtools/	5	CO1
2	**HTML5 Fundamentals**	2.1 Structure of HTML document, 2.2 Formatting tags, 2.3 Lists, 2.4 Tables, 2.5 Hyperlinks, 2.6 Images, 2.7 Forms, 2.8 Semantic tags, 2.9 Audio and Video embedding **Self-Learning Topics:** HTML5 Canvas, SVG Graphics SEO basics using HTML semantic elements **Online reference:** https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API https://developers.google.com/search/docs/fundamentals/seo-starter-guide	5	CO2
3	**CSS3 and Responsive Design**	3.1 CSS syntax, 3.2 Selectors, 3.3 Colors, 3.4 Backgrounds, 3.5 Borders, 3.6 Box model, 3.7 Fonts,	5	CO3

		3.8 Flex box, 3.9 Media Queries, 3.10 Responsive Web Design, 3.11 Introduction to Bootstrap **Self-Learning Topics:** Tailwind CSS CSS Animations Responsive UI design trends **Online reference:** **https://tailwindcss.com/docs/installation/using-vite** **https://developer.mozilla.org/en-US/docs/Web/CSS/Guides/Animations** https://www.smashingmagazine.com/category/responsive-design		
4	**JavaScript Fundamentals**	4.1 Introduction to JavaScript, 4.2 Variables, Data Types, Operators, 4.3 Conditional statements, 4.4 Loops, 4.5 Functions, 4.6 Arrays, 4.7 Events, 4.8 DOM manipulation, 4.9 Form validation. **Self-Learning Topics:** ES6 Features Introduction to TypeScript JavaScript Framework overview **Online reference**: **https://www.javascripttutorial.net/es6/** **https://www.typescriptlang.org/docs/** https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Frameworks_libraries	5	CO4
5	**Web Hosting and Security**	5.1 Website hosting concepts, 5.2 Domain registration, 5.3 FTP basics, 5.4 Deployment process, 5.5 Cookies and Sessions, 5.6 Introduction to Web Security, 5.7 Common Web Vulnerabilities **Self-Learning Topics:** HTTPS Certificates Secure Authentication methods OWASP Top 10 overview **Online reference:** **https://letsencrypt.org/** **https://oauth.net/2/** https://owasp.org/www-project-top-ten/	5	CO5
6	**Modern Web Technologies**	6.1 Introduction to APIs, 6.2 Cloud-based web applications, 6.3 Progressive Web Applications (PWA), 6.4 Web accessibility standards, 6.5 Introduction to Front-End Frameworks **Self-Learning Topics:** AI-powered websites JAMstack Web3 and Blockchain integration Serverless Web Applications **Online reference:**	5	CO6

		https://www.tensorflow.org/js **https://jamstack.org/** **https://web3js.readthedocs.io/en/v1.10.0/** https://aws.amazon.com/serverless/		
			Total Hours	30

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References:

1. Jon Duckett, "HTML and CSS: Design and Build Websites", Wiley Publications.
2. Jon Duckett, "JavaScript and JQuery", Wiley Publications.
3. Deitel & Deitel," Internet and World Wide Web – How to Program", Pearson.

Online References:

Sr. No.	Website Name	URL
1	Web Design Basics	https://web.dev/responsive-web-design-basics/
2	Progressive Web Apps	https://web.dev/progressive-web-apps/
3	NPTEL course	https://onlinecourses.swayam2.ac.in/e-learning/preview/ntr25_ed123
4	IBM Skill build	https://skillsbuild.org/students/course-catalog/web-development

Assessment:

Internal Assessment Test (IAT) for 20 marks each: IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of the syllabus content must be covered in the IAT-I, and the remaining 40% to 50% of the syllabus content must be covered in the IAT-II.

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%**.** This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

End Semester Theory Examination:

Question paper format:

- Question Paper will comprise a **total of six questions** each carrying 15 marks. Q.1 will be compulsory and should cover the maximum contents of the syllabus.
- Remaining questions will be mixed in nature (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)

- A **total of four questions** need to be answered.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112106	Computational Engineering Mathematics II	--	02	--	--	30	--	–	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112106	Computational Engineering Mathematics II	---	---	---	---	---	25	---	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This Mathematics Tutorial course introduces students to advanced calculus, vector calculus, numerical methods, and optimization techniques that are fundamental to Computer Engineering, IT, and AI & Data Science applications. The course develops analytical, computational, and optimization skills required for scientific computing, machine learning, operations research, simulation, and engineering problem-solving.

Tutorial Objectives:

Sr. No.	Tutorial Objectives
1	Apply double and triple integration techniques in Cartesian and polar coordinate systems to evaluate areas, volumes, and engineering quantities.
2	Analyze vector fields using gradient, divergence, curl, and identify solenoidal and conservative fields.
3	Apply Green's and Stoke's theorems to transform and evaluate line and surface integrals.
4	Implement numerical methods for solving ordinary differential equations using Euler, Modified Euler, and Runge-Kutta methods.
5	Apply numerical integration techniques such as Trapezoidal and Simpson's rules for approximating definite integrals.
6	Formulate and solve optimization problems using Simplex Method, Dual Simplex Method, Duality concepts, Lagrange Multipliers, and Kuhn-Tucker conditions.

Tutorial Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Tutorial Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy

1	Evaluate double and triple integrals in Cartesian and polar coordinates and apply them to area and volume calculations.	L3
2	Analyze vector fields using gradient, divergence, curl, and determine conservative and solenoidal properties.	L4
3	Apply Green's and Stoke's theorems to solve engineering problems involving line and surface integrals.	L3
4	Obtain numerical solutions of ordinary differential equations using Euler, Modified Euler, and Runge-Kutta methods.	L3
5	Approximate definite integrals using Trapezoidal and Simpson's numerical integration techniques.	L3
6	Solve constrained and unconstrained optimization problems using Linear Programming, Duality, Lagrange Multipliers, and Kuhn-Tucker conditions.	L4

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Algebraic operations and equations, Differential and integral calculus, Coordinate geometry and vectors, Functions and graphical interpretation, Elementary matrices and determinants, Basic logical and analytical problem-solving techniques.	01	-
1	**Multiple Integration- I**	1.1 Basic of Integration, Double integration - definition, Evaluation of Double Integrals.(Cartesian & Polar). 1.2 Evaluation of double integrals by changing to polar coordinates. **Self- Learning topic-**Tracing of curves, Change the order of Integration. **Link:** https://nptel.ac.in/courses/111104229?utm_source https://ocw.mit.edu/courses/18-02sc-multivariable-calculus-fall-2010/?utm_source	**05**	LO1
2	**Multiple Integration II**	2.1 Application of Double integration to Compute Area. 2.2 Triple Integration definition and Evaluation.	04	LO2

		Self- Learning topic-Application of Triple Integration to Compute Volume. **Link :** https://onlinecourses.nptel.ac.in/noc24_ma20/preview?utm_source		
3	**Vector Calculus**	3.1Gradiant, Divergence and Curl of Vector Field , Solenoidal and Irrotational (Conservative vector field), Line Integral (definition and simple problems) 3.2 Green's theorem in a plane(Without Proof), Stoke's theorem (Without proof) **Self- Learning topic-** Work done, Angle between surfaces, Divergence theorem, **Link :** https://nptel.ac.in/courses/111106101?utm_source https://ocw.mit.edu/courses/18-02-multivariable-calculus-fall-2007/?utm_source	04	LO3
4	**Numerical solution of ordinary differential equations of first order and first degree, and Numerical Integration**	4.1 Numerical solution of ordinary differential equation using (a) Euler's method (b) Modified Euler method, (c) Runge-Kutta fourth order method 4.2 Numerical integration-by (a) Trapezoidal (b) Simpson's 1/3rd (c) Simpson's 3/8th rule (all without proof) **Self- Learning topic-** Adaptive step size method,Application in computer science and engineering. **Link :** https://nptel.ac.in/courses/111105090?utm_source https://www.coursera.org/learn/numerical-methods-engineers?utm_source	04	LO4
5	**Linear Programming Problems**	5.1 Types of solutions, Standard and Canonical of LPP, Basic and Feasible solutions, slack variables, surplus variables. 5.2 Simplex method. 5.3 Duality, Dual of LPP and Dual Simplex Method.	05	LO5

		Self-learning Topics: Artificial variable method, Big M method, Sensitivity Analysis, Two-Phase Simplex Method,Revised Simplex Method. **Link :** https://nptel.ac.in/courses/110107081?utm_source https://www.coursera.org/learn/operations-research-modeling?utm_source		
6	**Non Linear Programming Problems**	6.1 NLPP with one equality constraint (two or three variables) using the method of Lagrange's multipliers 6.2 NLPP with one inequality constraint (two variables): Kuhn-Tucker conditions **Self-learning Topics:** Problems with two equality constraints, Problems with two inequality constraints, Unconstrained optimization: One, dimensional search method (Golden Search method, Newton's method). Gradient Search Method. **Link :** https://nptel.ac.in/courses/112108298?utm_source https://ocw.mit.edu/courses/15-084j-nonlinear-programming-spring-2004/?utm_source	05	LO6
		Total	**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. Dr. B.S. Grewal, "Higher Engineering Mathematics"
2. R.K. Jain and S.R.K. Iyengar" Advanced Engineering Mathematics"
3. S.C. Gupta and V.K. Kapoor," Fundamentals of Mathematical Statistics"
4. M. K. Jain, S. R. K. Iyengar and R. K. Jain , "Numerical Methods for Scientific and Engineering Computation "New Age International Publishers.
5. Howard Anton and Christ Rorres., Elementary Linear algebra with application" 6th edition.

6. Bali N.P and Goyal M. “A textbook of Engineering Mathematics” Laxmi Publication, 10th Ed. 2022.

7. Gupta C.B. Singh S.R and Mukesh Kumar : “Engineering Mathematics for semester I and II”, Mc- Graw Hill education ,2015
8. Pal S. & Bhunia S.C: “Engineering Mathematics “ Oxford University Press, 3rd Ed, 2016.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://nptel.ac.in/
2	MIT OpenCorseWare	https://ocw.mit.edu/courses
3	Wolfram Mathworld	http://mathworld.wolfram.com

List of Tutorials:

Sr No	List of Tutorials	Hrs
01	Problems on Double Integration in Cartesian Coordinates, Polar Co-ordinates.	01
02	Applications of Double and Triple Integration.	02
03	Vector Operations: Gradient, Divergence, and Curl, Solenoidal and Conservative vector fields.	01
04	Problems based on Green’s and Stoke’s Theorems.	02
05	Numerical Solution using Euler and Modified Euler Methods	01
06	Numerical Solution using Runge–Kutta Method	02
07	Numerical Integration using Trapezoidal and Simpson’s Rules	01
08	Linear Programming Problems using Simplex Method	02
09	Dual Simplex Method and Duality Problems	01
10	Problems on Lagrange Multipliers and Kuhn-Tucker Conditions	02
	Total	15

Assessment:

- **Term Work:** Term Work shall consist of at least 10 Tutorials based on the above list.
- **Term Work Marks:** 25 Marks (Total) = 10 Marks (Tutorials) + 10 Marks (Mini Project or Quiz) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112107	Data Structure Lab	--	--	02	--	--	30		30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112107	Data Structure Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The **Data Structures Laboratory** provides hands-on experience in implementing and analyzing fundamental data structures for efficient data organization, storage, and retrieval. Through practical problem-solving activities, students develop computational thinking, programming proficiency, and algorithmic skills essential for software development and emerging digital technologies.

Aligned with **NEP 2020**, the course promotes experiential learning, critical thinking, and application-oriented education. The laboratory contributes primarily to **SDG 9** by enabling learners to design efficient and innovative software solutions. It also supports **SDG 4** through skill-based learning and **SDG 8** by developing industry-relevant programming and problem-solving competencies.

Lab Objectives:

Sr. No.	Lab Objectives
1	Develop programming skills for implementing fundamental data structures using appropriate programming constructs.
2	Design and implement stack data structures for managing sequential data efficiently.
3	Develop and implement queue-based data structures and their variants to manage and process data efficiently in real-world applications.
4	Construct linked-list-based solutions for dynamic memory allocation and data organization.
5	Implement tree and graph structures to represent and process hierarchical and networked data.
6	Analyze and compare the efficiency of various data structures to select suitable solutions for real-world applications.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Implement array and string operations to solve basic computational problems.	L3
2	Develop stack-based solutions for expression evaluation and related applications.	L3
3	Implement queue data structures and their variants for efficient data processing applications.	L3
4	Implement linked list techniques for efficient dynamic data storage and management.	L3
5	Analyze and apply tree structures and traversal techniques for hierarchical data representation and processing.	L4
6	Analyze and apply graph traversal and hashing techniques for efficient data organization and retrieval	L4

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:
• Computer System with Intel i3 or higher processor • Minimum 4 GB RAM and 100 GB Storage • Keyboard, Mouse, and Internet Connectivity	• Windows/Linux Operating System • GCC Compiler / Code::Blocks / VS Code • Web Browser for Online Learning Resources

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Basics of C Programming, Functions, Arrays, Pointers, Structures, and Dynamic Memory Allocation.	02	-
1	**Introduction to Data Structure**	1.1 Insertion and deletion in an array using functions, 1.2 1D arrays, 1.3 Recursion, 1.4 2D Array **Self-learning Topics:** Dynamic Arrays, Applications of Arrays in Databases and Operating Systems Online reference: **https://youtu.be/zWg7U0OEAoE?si=Pxg2AEf5tthEA08j** https://leetcode.doocs.org/en/lc/22/?utm_source=	05	LO1
2	**Stack**	2.1 Stack operations, 2.2 string reversal, 2.3 Infix to postfix conversion, 2.4 Recursive applications 2.5 Postfix Notation 2.6 Expression Evaluation	05	LO_

		Self-learning Topics: Baseball Game, Tower of Hanoi, Stack Memory, Undo/Redo Operations Online reference: **https://youtu.be/g1USSZVWDsY?si=xIJS853LrEo586Oi** https://leetcode.doocs.org/en/lc/682/?utm_source=		
3	**Queue**	3.1 Linear queue operations, 3.2 circular queue operations, 3.3 priority queue applications, 3.4 Double ended queue operations. **Self-learning Topics:** Message Queues in Distributed Systems Online reference: https://youtu.be/PGWZUgzDMYI?si=CKfs4rMbcygMnviT https://leetcode.doocs.org/en/lc/933/?utm_source=	05	LO_
4	**Linked List**	4.1 Singly linked list operations, 4.2 doubly linked list insertion and deletion, 4.3 doubly linked list sorting operations. 4.4 Linked List using Stack 4.5 Linked List using Queue **Self-learning Topics:** Memory Management Techniques garbage collection concepts. Online reference: **https://youtu.be/PGWZUgzDMYI?si=CKfs4rMbcygMnviT** https://leetcode.doocs.org/en/lc/2/?utm_source=	06	LO_
5	**Trees**	5.1 Binary Search Tree creation and searching, 5.2 preorder traversal, inorder traversal and postorder traversal. 5.3 AVL Tree, B Tree, B+ Tree, 5.4 Expression Tree, 5.5 Huffman Encoding. **Self-learning Topics:** Heap Trees, Red black tree Online reference: https://youtu.be/tORLeHHtazM?si=eYJYbhvnw8zMNdMZ https://youtu.be/eWeqqVpgNPg?si=oMB7O6oZ38GJbIC https://youtu.be/mRGQylRWAsI?si=oBumxHftv_tf938j https://youtu.be/TbvhGcf6UJU?si=wmEBh6ylEgFPxAXx https://leetcode.doocs.org/en/lc/98/?utm_source=	05	LO_
6	**Graphs and Hashing**	6.1 Graph representation using adjacency matrix, 6.2 BFS traversal, 6.3 DFS traversal 6.4 Linear Search and Binary Search **Self-learning Topics:** Shortest Path Algorithms, Social Network Analysis, Cryptographic Hash Functions, Hash Maps in Modern Programming Languages Online reference: https://youtu.be/9zpSs845wf8?si=57_LieJkyyVFEkv3 https://youtu.be/CIm6RzdoPCI?si=y19yubJ1mH76VEmA https://youtu.be/r1-p11fSPw?si=BPZBVppkWT1N6pZn https://youtu.be/KW0UvOW0XIo?si=M1UhJhwiGGREuCq_	04	LO_

		https://leetcode.doocs.org/en/lc/752/?utm_source=		
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. Reema Thareja, " Data Structures using C", Oxford Press.
2. E. Balagurusamy, "Data Structure Using C", Tata McGraw-Hill Education India.
3. Richard F. Gilberg and Behrouz A. Forouzan, "Data Structures: A Pseudocode Approach with C", 2ndEdition, CENGAGE Learning.
4. Sahni Horowitz, "Fundamentals of data structures in C", computer science press, 2008.
5. Jean Paul Tremblay, P. G. Sorenson, "Introduction to Data Structure and Its Applications", McGraw-Hill Higher Education
6. Narasimha Karumanchi, "Data Structures And Algorithms", 5th Edition,CareerMonk, 2016.
7. Robert Kruse, C. L. Tondo, Bruce Leung, "Data Structures and Program Design in C", Pearson Publication. Moseley, Jürgen Garche

Online References:

Sr. No.	Website Name	URL
1	NPTEL – Data Structures and Algorithms Course	https://nptel.ac.in/courses/106102064?utm_source=chatgpt.com
2	Coursera – Data Structures (UC San Diego)	https://www.coursera.org/learn/data-structures?utm_source=chatgpt.com
3	LeetCode – Data Structures Problems & Practice	https://leetcode.com/problemset/all/?utm_source=chatgpt.com

List of Experiments.

Sr No	List of Experiments	Hrs
01	Implementation of Insertion and deletion in a specific position in an Array using Function.	02
02	Develop a program to store marks of students in a 1D array and perform operations such as searching, finding highest/lowest marks, and calculating average.	02
03	Implement a program to store daily electricity consumption data in a 1D array and analyze maximum, minimum, and average usage.	02
04	Implement a menu driven program for the following operations on STACK of integers. a. Push an element on to stack. b. Pop an element from the stack. c. Peek operation d. Display the status of the stack.	02
05	Implementation of recursive program (Factorial/ Fibonacci series)	02
06	Write a program to reverse a string using stack operations.	02
07	Develop a program to convert infix expressions into postfix expressions using stack.	02
08	Implement a Linear Queue using arrays for suitable application.	02
09	Implement Priority Queue for handling emergency patients.	02
10	Implement a Circular Queue using arrays to simulate smart traffic signal management.	02
11	Implement insertion and deletion from both ends using Double Ended Queue for Browser Navigation System.	02

12	Implement Singly Linked List using Music Playlist Management System o Add Song o Delete Song o Display Playlist	02
13	Implement a menu driven program for the following operations on Doubly Linked List (DLL) of Employee Data with the fields: Empid., Name, Department, Salary. a. Perform Insertion and Deletion at the End of DLL b. Perform Insertion and Deletion at Front of DLL c. Perform Insertion and Deletion in the middle of the DLL based on position entered by the user. d. Display status of DLL e. Sort DLL	02
14	Students in a class are arranged according to their heights. The teacher wants to quickly find whether a student with a particular height is present in the class. Implement this scenario using a Binary Search Tree (BST) to store and search student heights efficiently.Consider sample data is Student Heights: [150, 145, 160, 142, 148, 155, 165] Implement tree traversal techniques (Preorder, Inorder, and Postorder) using a simple family tree.	02
15	Implement a graph representing connected cities using an adjacency matrix and perform BFS / DFS traversals to find all reachable cities from a given starting city.	02
16	Implement password storage and verification using hashing techniques.	02

Sr No	List of Assignments	Hrs
01	Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses. Example 1: Input: n = 3 Output: ["((()))","(()())","(())()","()(())","()()()"]	02
02	Design a system that counts how many requests were received in the last 3000 milliseconds. Create a function ping(t) where: t represents the current request time in milliseconds. Every new request time will be greater than the previous one. For each ping(t): Add the new request. Return the number of requests received between: t−3000 to t (including both values).	02
03	Differentiate recursion and stack-based execution.	02
04	You are given two non-empty linked lists representing two non-negative integers. The digits are stored in reverse order, and each node contains a single digit. Add the two numbers and return the sum as a linked list. You may assume: The two numbers do not contain leading zeroes except the number 0 itself.	02
05	Problem Statement: Given the root of a binary tree, determine if it is a valid binary search tree (BST). A valid BST is defined as follows: - The left subtree of a node contains only nodes with keys less than the node's key. - The right subtree of a node contains only nodes with keys greater than the node's key. - Both the left and right subtrees must also be binary search trees.	02
06	Analyze the suitability of different data structures for various applications.	02

07	You have a lock represented by 4 wheels. Each wheel has digits from 0 to 9. Return the minimum number of turns required to open the lock using Breadth First Search (BFS).	02

Assessment:

Term Work: Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

Term Work Marks: 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).

Practical & Oral Exam: An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112108	Engineering Mechanics and Graphics Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112108	Engineering Mechanics and Graphics Lab	---	---	---	---	---	25	–	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The practical component of Engineering Graphics and Mechanics develops analytical thinking, problem-solving, visualization, and computer-aided design skills essential for Computer Engineering, Information Technology, and Artificial Intelligence & Data Science students. Through experiments on force systems, equilibrium, friction, and motion, students gain an understanding of the physical principles underlying engineering systems. Drawing and CAD-based practicals enhance spatial visualization, technical communication, and digital drafting skills required for product design, robotics, automation, smart systems, and interdisciplinary engineering applications. The course also strengthens proficiency in modern software tools, supporting future work in modeling, simulation, digital design, and technology-driven engineering solutions.

Lab Objectives:

Sr. No.	Lab Objectives
1	To familiarize with the concepts of force, moment, Resultant and Equilibrium of system of coplanar force.
2	To acquaint with the basic concept of friction and its application in real-life problems.
3	To understand the parameters required to quantify the Kinetics of rigid body.
4	To impart the knowledge to read and interpret a drawing.
5	To acquaint students with representing internal features of a three-dimensional object by way of section that accurately conveys their internal orientation.
6	To enable students to represent three-dimensional objects on a two-dimensional surface in a way that accurately conveys their shape, size, and orientation.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr.No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Determine the equivalent force-couple system for a given system of forces.	L3
2	Illustrate the concept of force, moment and apply the same along with the concept of equilibrium in two- and three-dimensional systems with the help of FBD.	L4
3	Calculate position, velocity and acceleration etc of particle and rigid body using principles of kinematics.	L3
4	Apply the basic principles of projections in converting pictorial views into orthographic Views.	L3
5	Apply the basic principles of projections in converting orthographic views into isometric drawing.	L3
6	Apply basic AutoCAD skills in construction of views and objects.	L4

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:AutoCAD

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Basic knowledge of geometry and engineering drawing instruments. Understanding of force, vector representation, and trigonometric functions. Basic concepts of motion, velocity, and acceleration.	02	-
1	Force Systems and Equilibrium	Polygon law of coplanar forces, principle of moments, resultant and equilibrium of force systems, support reactions in simply supported beams, free body diagrams and applications in engineering systems. Practical List: 1)Verification of Polygon law of coplanar forces. 2)Verification of the Principle of Moments (Bell crank lever). 3)Determination of support reactions of a Simply Supported Beam. **Self-learning Topics:** Study force systems acting on real-life structures such as cranes, bridges, shelves, and signboards. Explore support reactions in household furniture and civil engineering	08	LO1

		structures.		
2	Friction and Dynamics	Coefficient of friction using inclined plane, equilibrium of bodies on rough surfaces, analysis of motion of particles and rigid bodies using kinematics and kinetics principles. Practical List: 1)Determination of coefficient of friction using inclined plane. **Self-learning Topics:** Observe the effect of friction in automobiles, conveyor belts, brakes, and inclined roads. Analyze motion of elevators, escalators, bicycles, and rotating machinery using basic dynamics concepts.	02	LO2
3	Projection of Lines and Planes	Drawing projections of lines and planes in different orientations using engineering drawing principles, interpretation of geometrical views and projection techniques. Practical List:1)Minimum four problems on Projection of Lines to be drawn on a drawing sheet. 2)Minimum four problems on Projection of Planes to be drawn on a drawing sheet. **Self-learning Topics:** Study the applications of projection of lines in engineering structures such as staircases, ladders, roof trusses, crane booms, transmission towers, and pipelines.	04	LO3
4	Orthographic Projection and Sectional Views	Conversion of pictorial views into orthographic projections, sectional orthographic views of engineering objects using drafting software, visualization of internal features of components. Practical List: 1)One problem on Orthographic Projections and one problem on Sectional Orthographic Projection using drawing sheets. **Self-learning Topics:** Prepare orthographic sketches of simple machine components and household objects. Study sectional views of machine parts such as bearings, brackets, couplings, and pipe joints.	04	LO4
5	Isometric Projection and Visualization	Construction of isometric views from orthographic drawings, representation of three-dimensional engineering objects on two-dimensional surfaces using drafting techniques and CAD tools.	04	LO5

		Practical List: 1) Minimum of two problems on Isometric Projections to be drawn using drawing sheets. **Self-learning Topics:** Draw isometric views of simple engineering objects like blocks, brackets, and machine bases. Compare orthographic and isometric representations of engineering components.		
6	Computer Aided Drafting (AutoCAD Applications)	Basic AutoCAD commands, creation of engineering views, drafting of projections and objects using software tools, application of CAD in engineering graphics and visualization. Practical List: 1)One problem on Orthographic Projections and one problem on Sectional Orthographic Projection using drafting software. 2) Minimum of two problems on Isometric Projections to be drawn using drafting software. **Self-learning Topics:** Explore basic AutoCAD tools for engineering drafting and dimensioning. Study CAD applications in mechanical design, civil engineering drawings, and architectural planning.	06	LO6
	Total		**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References Books:

1. Engineering Mechanics by R. C. Hibbeler.
2. Engineering Mechanics by F. L. Singer, Harper& Raw Publication
3. Engineering Mechanics by Macklin & Nelson, Tata McGraw Hill
4. Introduction to Industrial Robotics by Ramchandran Nagrajan, Pearson publication
5. Narayana, K.L. & P Kannaiah (2008), Textbook on Engineering Drawing, Scitech Publisher.
6. Prof. Sham Tickoo (Purdue University) & Gaurav Verma, "(CAD Soft Technologies).
7. Auto CAD 2012 (For engineers and Designers)", Dreamtech Press New Delhi.
8. Engineering Mechanics by A K Tayal, Umesh Publication.
9. Engineering Mechanics by Beer &Johnston, Tata McGraw Hill.
10. N.D. Bhatt, "Engineering Drawing (Plane and solid geometry)", Charotar Publishing House Pvt. Ltd.
11. N.D. Bhatt & V.M. Panchal, "Machine Drawing", Charotar Publishing House Pvt. Ltd.

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://onlinecourses.nptel.ac.in/noc21_me70/preview
2	NPTEL	https://nptel.ac.in/courses/112103019

3	Mechanics and Robot kinematics	https://nptel.ac.in/courses/112105236

List of Experiments.

Sr No	List of Experiments	Hrs
01	Verification of Polygon law of coplanar forces.	**02**
02	Verification of the Principle of Moments (Bell crank lever).	**02**
03	Determination of support reactions of a Simply Supported Beam.	**02**
04	Determination of coefficient of friction using inclined plane.	**02**
05	Minimum four problems on Projection of Lines to be drawn on drawing sheet.	**02**
06	Minimum four problems on Projection of Planes to be drawn on drawing sheet.	**02**
07	One problem on Orthographic Projections and one problem on Sectional Orthographic Projection using drafting software.	**02**
08	Minimum of two problems on Isometric Projections to be drawn using drafting software.	**02**

List of Assignments.

Sr No	List of Assignments / Tutorials	Hrs
01	Resultant of the Coplanar force system.	**02**
02	Equilibrium of the System of Coplanar Forces including support reaction of beams.	**02**
03	Equilibrium of bodies on an inclined plane in Friction.	**02**
04	Homogeneous transformation, Direct Kinematics of robot.	**02**
05	Minimum four problems on Projection of Lines.	**02**
06	Minimum four problems on Projection of Planes.	**02**
07	Two problems on Orthographic Projections and one problem on Sectional Orthographic Projection.	**02**
08	Two problems on Isometric Projections.	**02**

Sr No	List of Assignments	Hrs
01	Q1.Explain the client-server architecture of the web. Draw a diagram showing the request-response cycle. Then, write the complete HTML5 structure of a basic webpage (doctype, html, head, body) and explain the purpose of each major section. Q2. Create an HTML5 document for a "Course Registration Form" that includes: - Text fields (name, email, phone) - Radio buttons (gender) - Checkboxes (subjects: HTML, CSS, JS) - Dropdown (course level: Beginner, Intermediate, Advanced) - Submit and reset buttons - HTML5 form validation (required, email type, pattern for phone) - Semantic elements (<form>, <fieldset>, <legend>, <label>)	**02**

	Q3. Using CSS3, style a product catalog page that includes: - Flexbox for a navigation bar (Home, Products, Contact) - CSS Grid for displaying at least 4 product cards in a responsive layout - Media queries to change from 4 columns (desktop) → 2 columns (tablet) → 1 column (mobile) - CSS animation on hover for each product card (scale or shadow transition) Q4. Explain the key differences between CSS Flexbox and CSS Grid layout systems. Consider: The primary use case for each, One-dimensional vs two-dimensional layout control, How content vs layout-driven approaches differ, A code example showing when to choose Flexbox over Grid, and vice versa Q5. Compare the following using a table: - HTML5 Canvas vs SVG (self-learning) — include use cases, rendering model, and performance differences. - Block-level vs Inline vs Inline-block elements with examples. - LocalStorage vs SessionStorage vs Cookies Q6. Design a responsive personal portfolio webpage that includes: - Semantic HTML5 (<header>, <nav>, <main>, <section>, <article>, <footer>) - Embedded video/audio using HTML5 tags - Hyperlinks with appropriate target attributes - A responsive image with srcset for different screen sizes - SEO basics (self-learning): Add meta tags, alt attributes, and semantic structure for SEO - Submit the HTML/CSS code and explain how it adapts to mobile/tablet/desktop	
02	Q1. Write JavaScript code for the following tasks on a webpage: (a) Declare variables using var, let, and const. Explain their scope differences. (b) Create a function that takes an array of temperatures (in Celsius) and returns a new array converted to Fahrenheit using the map() method. Q2.Write JavaScript code for the following : (a) Add an event listener to a button that, when clicked, changes the background color of the page and displays the current timestamp. (b) Write a form validation script that checks if a password field has at least 8 characters, one uppercase, and one number. Q3.Explain the following web hosting and deployment concepts: (a) Domain registration vs web hosting — how they work together. (b) FTP basics and the typical deployment process for a static website. (c) Cookies and sessions — compare them in terms of storage location, lifespan, security, and use cases. (d) What is an HTTPS certificate and why is it essential? Q4. a) Explain the OWASP Top 10. List the top 3 risks and mitigation strategies. (b) What is SQL Injection? Write a vulnerable code snippet and then fix it using parameterized queries. (c) Explain OAuth 2.0 and JWT — how they are used for secure authentication. (d) Describe Git and GitHub basics (self-learning): clone, commit, push, pull, branch. Q5. What is an API? Write a JavaScript fetch() request to GET data from a public API (e.g., JSONPlaceholder) and display it in an HTML table. Q6. Discuss Performance optimization techniques, list 5 techniques (e.g., lazy loading, minification, CDN) and how they improve Core Web Vitals	**02**

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112109	Foundation of Network Science Lab	--	--	02	--	--	30	--	30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112109	Foundation of Network Science Lab	---	---	---	---	---	25	--	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This course introduces the fundamentals of modern communication networks through experiments on network configuration, connectivity, troubleshooting, and cybersecurity. It supports multidisciplinary applications such as smart cities, IoT, digital governance, and sustainable infrastructure while promoting innovation and digital inclusion.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand the fundamentals of computer networks, networking devices, and communication models..
2	To develop practical knowledge of IP addressing, subnetting, and network configuration techniques.
3	To analyze network connectivity and routing concepts using network simulation tools.
4	To study transport and application layer protocols through packet analysis and socket programming.
5	To explore wireless networking, IoT systems, and smart connected applications.
6	To understand cybersecurity concepts, secure communication techniques, and sustainable networking practices.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy

1	Execute and evaluate network administration commands and demonstrate their use in different network scenarios. .	L2
2	Design & Configure IP addressing, LAN topology, and routing using networking tools	L3
3	Analyze network packets and troubleshoot connectivity issues.	L4
4	Implement client-server communication using socket programming.	L3
5	Demonstrate IoT-based smart networking applications.	L5
6	Apply basic security and green networking practices in modern networks.	L6

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Desktop/Laptop Computers (Core i3/i5, 8 GB RAM minimum)	CPT, Wireshark,NS-2 / NS-3 Network Simulator,Wokwi Simulator,Visual Studio Code,Python 3.x

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	Prerequisite	Study of Networking Devices, Cables, connectors, Crimping of LAN Cable and Testing Connectivity.	02	-
1	Fundamentals of Computer Networks and Networking Devices.	1.1 Basic networking commands in Unix/Linux operating system: ifconfig,ip,traceroute,tracepath,ping, netstat, ss, dig, nslookup, route, host, arp, hostname, curl or wget, mtr, whois, tcpdump **Self learning Topic:** Data Networks - from Circuit Switching Network to Packet Switching Network, Network Protocol Stack and Services at the Different Layers of the Protocol Stack Online reference: https://nptel.ac.in/courses/106105183 https://nptel.ac.in/courses/106105183 https://nptel.ac.in/courses/106105183	04	LO1
2	IP Addressing and Topologies	2.1 Configuration of IPv4 addressing, subnetting, creation of LAN, 2.2 Implementation of Star, Bus, and Ring topologies. NS2/ cisco packet **self learning Topic:** IP Addressing (IPv4) III - Network Address Translation (NAT),IPv6 Addressing. Online reference: https://nptel.ac.in/courses/106105183 https://nptel.ac.in/courses/106105183	02	LO2

3		3.1 Introduction to Network Simulation Tools, Cisco Packet Tracer Environment, 3.2 Analysis of network packets using Wireshark and study of HTTP, DNS, and DHCP protocols. **self learning Topic:** ARP-RAPP-BOOTP-DHCP Online reference: https://nptel.ac.in/courses/106105183 https://nptel.ac.in/courses/106105183	04	LO3
4	Wireless Networking and Routing	4.1 Wireless networking setup, Wi-Fi configuration 4.2 Static routing simulation using Cisco Packet Tracer/NS2. **Self learning Topic:**IP Routing Table,VLAN (Virtual Local Area Network) Online reference: https://nptel.ac.in/courses/106105183 https://www.youtube.com/watch?v=zVdyKTQyvVY	08	LO4
5	Socket Programming and IoT	5.1 Basic TCP/UDP socket programming 5.2 IoT device demonstrations using Arduino/Tinkercad/Wokwi **Self learning Topic:** Design for IOT,Application multiplexing in TCP/IP Online reference: https://www.youtube.com/watch?v=urUBLmXFKl0&list=PLgMDNELGJ1CaBrefq-0eYatfOnoncW0y- https://nptel.ac.in/courses/106105183	08	LO5
6	Security and Green Networking	6.1 Firewall configuration, network security basics 6.2 Green networking and energy-efficient devices. **Self learning Topic:** .Sustainability through Green Manufacturing systems and applied approach,Cloud Security, Security of the Internet of Things (IoT), Hardware Security. **Online reference:** https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_mg85 https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_ee54	04	LO6
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. Behrouz A. Forouzan, “TCP/IP Protocol Suite”,4th Edition, McGraw Hill Education, 2017
2. James F. Kurose, Keith W. Ross, “Computer Networking: A Top-Down Approach”, 8th Edition, Pearson, 2022.
3. Larry L. Peterson, Bruce S. Davie,” Computer Networks: A Systems Approach”, 6th Edition, Morgan Kaufmann, 2021.
4. Alberto Leon-Garcia, Indra Widjaja, “Communication Networks: Fundamental Concepts and Key Architectures”, 2th Edition, McGraw Hill Education, 2017..
5. Stallings William, “Data and Computer Communications”, 10th Edition, Pearson, 2017.

Online References:

Sr. No.	Website Name	URL
1	NPTEL Swayam	https://nptel.ac.in/courses/106106091
2	NPTEL Swayam	https://nptel.ac.in/courses/106105183
3	NPTEL Swayam	https://onlinecourses.swayam2.ac.in/e-learning/preview/cec19_cs07
4	Coursera	https://www.coursera.org/learn/network-security

List of Experiments

Sr No	List of Experiments	Hrs
01	Basic Networking Commands in Windows/Linux (ifconfig ,ip, traceroute, tracepath, ping, netstat, ss, dig, nslookup, route, host, arp, hostname, curl or wget, mtr, whois, tcpdump) .	2
02	Installation and configuration of Network Simulator Software (eg. NS-2 / Cisco Packet Tracer) and To implement number of nodes and physical layer configuration using NS2/CPT. To implement the given network topology and transmit data over the nodes.	2
03	To Simulate and study stop and Wait protocol ,Sliding Window protocol using Network Simulator Software (eg. NS-2 / Cisco Packet Tracer)	2
04	Design and Configure IP Address with Subnetting	2
05	Analyze Network Packets data using Wireshark.	2
06	Design and analyse Star, Bus and Ring Topologies.	2
07	Design Simple LAN using network simulator.	2
08	Design a network server like file application or dhcp/dns.	2
09	Implementation of Socket Programming using TCP/UDP.	2
10	Demonstration & implementation of Wireless Networking and Wi-Fi Configuration.	2
11	Simulation of Static Routing in Small Network.	2
12	Demonstration of Smart IoT Devices using Sensors. (Temperature/Light/Motion).	2

13	Design a Smart Home Automation Simulation.	2
14	Setting of Firewall and Basic Network Security.	2
15	Case Study of Green Networking and Energy Efficient Devices.	2

Assessment:

Sr No	List of Assignments / Tutorials	Hrs
01	Study the networking technologies and digital infrastructure used in BharatNet and Digital India initiatives. Analyze their role in providing internet connectivity, e-governance, and smart digital services across India.	2
02	Explore how Artificial Intelligence is used in modern networking for intelligent routing, traffic management, and automated network monitoring. Study its applications in smart cities, healthcare, and communication systems.	2
03	Analyze common cybersecurity threats such as malware, phishing, and hacking attacks. Study various protection mechanisms including firewalls, encryption, VPNs, and intrusion detection systems.	2
04	Design a smart connected system using IoT and networking technologies for applications such as smart homes, agriculture, healthcare, or transportation. The solution should support sustainable and intelligent digital infrastructure.	2

Term Work: Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.
Term Work Marks: 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).
Practical & Oral Exam: An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112110	Nirman Lab - II	--	--	02	--	--	30	-	30	1

		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)	TW	PR / OR	Total
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112110	Nirman Lab -II	--	--	--	--	--	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional L

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Rationale: Aligned with the National Education Policy (NEP) 2020, the institution emphasizes experiential, interdisciplinary, and project-based learning through the Nirman Lab, a central hub for hands-on innovation.

To strengthen the undergraduate research ecosystem, the institution has adopted a theme-based academic model aligned with UNSGD. Each semester features six curated problem statements based on local need and aligned with core subjects in the same semester, enabling students to apply classroom knowledge to real-world challenges. Every student selects one problem and develops an individual, subject-integrated solution enhancing both academic understanding and research skills.

The Nirman Lab supports this initiative with facilities for design thinking, prototyping, and product development. Students maintain a project logbook throughout the semester to track their progress and reflections

Course Objectives:

Sr. No.	Course Objectives
1	To promote experiential and project-based learning that bridges theoretical knowledge with real-world problem-solving.
2	To encourage interdisciplinary integration by enabling students to apply concepts from multiple subjects within a single cohesive project.
3	To develop innovation and design thinking skills through hands-on activities and iterative solution development.
4	To foster critical thinking and creativity by engaging students in open-ended problems with multiple solution pathways.
5	To enhance communication, collaboration, and documentation skills essential for professional engineering practice.
6	To build an entrepreneurial and research mindset by guiding students to develop scalable socially-relevant, and technically viable prototype.

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Recall and articulate key concepts from core and allied subjects relevant to the assigned project	L1
2	Explain the interdisciplinary nature of the problem and the role of each subject in addressing it	L2
3	Apply appropriate tools, techniques, and theoretical knowledge to develop project components	L3
4	Analyze problem constraints and user requirements to structure a feasible and efficient solution	L4
5	Evaluate multiple design options and justify the chosen solution based on technical and practical considerations	L5
6	Create a functional prototype or solution that demonstrates innovation, utility, and integration of interdisciplinary knowledge	L6

Project Guidelines (Interdisciplinary Project Execution in Nirman Lab)

- Each student works on an individual interdisciplinary project aligned with the semester theme.
- Faculty in-charges mentor both the academic and technical aspects, and track weekly progress.
- Project assessment will be rubric-based, ensuring depth, innovation, documentation, and ownership.
- Students shall convert the best solution into a working model using various components of their domain areas and demonstrate.

Guidelines for same semester Subject Concepts Applied within the Project

- Termwork for each subject will partially reflect how well a student applies subject-specific concepts in their interdisciplinary project.
- The internal assessment panel will collaborate to align project components with subject learning outcomes.

Role of Faculty In-Charges in Nirman Lab Projects

Faculty in-charges play a pivotal role in the success of interdisciplinary, theme-based projects under the Nirman Lab. Their responsibilities extend beyond technical supervision to include academic alignment, innovation facilitation, and active student engagement. Their key roles include:

- Motivating and Inspiring Students
    a. Encourage students to take ownership of their learning and projects.
    b. Cultivate a mindset of curiosity, exploration, and social relevance.
    c. Foster an environment where students feel empowered to take creative risks.
- Conducting Brainstorming and Ideation Sessions
    d. Organize structured brainstorming sessions at the start of the semester to help students define their problem statements and solution pathways.
    e. Promote collaborative thinking, design exploration, and interdisciplinary integration.

- Arranging Guest Lectures and Expert Talks
    f. Identify and invite industry experts, researchers, and innovators for guest lectures aligned with the semester's theme or subject areas.
    g. Facilitate exposure to real-world challenges, current trends, and future opportunities.

- Ensuring Uniqueness and Originality of Projects
    h. Actively review proposed ideas to ensure **no duplication of solutions** across students.
    i. Encourage students to explore novel approaches, technologies, and perspectives.

- Promoting Discussion and Collaborative Learning
  Create platforms for students to present, discuss, and receive peer and mentor feedback.
  Facilitate idea refinement through regular discussions and group engagement.

- Monitoring and Documentation
  Oversee project logbook maintenance, milestone tracking, and submission of progress reports.
  Provide ongoing feedback and ensure project alignment with learning outcomes.

- Coordination with Subject Faculty
  Work in collaboration with other subject faculty to help students embed theoretical and practical aspects of their coursework into the project.
  Facilitate subject-term mapping and contribute to termwork assessment based on evidence.

Guidelines for Assessment:

Presentation-Based Assessment Structure (Total: 25 Marks)

Assessment Month	Weightage	Marks
Month 1 (Formative 1)	20%	5 marks
Month 2 (Formative 2)	40%	10 marks
Month 3 (Formative 3)	40%	10 marks

Rubric-Based Evaluation Criteria

Criteria	Month 1 (5)	Month 2 (10)	Month 3 (10)
Problem Understanding	Connects problem to subjects	Defines interdisciplinary scope	Demonstrates deep conceptual grasp
Subject Knowledge Application	Identifies relevant concepts	Applies principles in design	Integrates multiple subject areas correctly
Innovation & Design Thinking	Proposes creative idea	Develops and tests feasible solution	Final solution shows originality and utility
Documentation & Presentation	Logbook initiated, plan presented	Mid-design log and visuals	Final report and demo completed
Progress & Ownership	Meets deadlines, shows planning	Demonstrates self-motivation	Completes project independently with reflection

Total Assessment Structure

Component	Marks	Assessed By
Termwork – Project Execution	25 Marks	Project Guide
Viva Voce (Final Evaluation)	25 Marks	External Examiner

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
1112111	Web Technology Lab	--	--	02	--	--	30		30	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1112111	Web Technology Lab	---	---	---	---	---	25	25	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This Web Technology syllabus is directly aligned with the "Smart Vikasit Bharat: A Multidisciplinary Ecosystem" theme**,** **NEP** 2021, and UNSDG goals by equipping students to build accessible, data-driven, and secure digital solutions for real-world challenges. Covering HTML5 semantics, CSS3 Flexbox/Grid, JavaScript, responsive design, and deployment on GitHub Pages/Netlify, the curriculum enables the creation of e-governance portals, business landing pages, and complaint systems—**addressing UNSDG 9 (Industry & Innovation), 4 (Quality Education), and 10 (Reduced Inequalities).** NEP 2021 is fulfilled through experiential project-based learning and self-learning topics. **Over 90% of the syllabus directly supports "Smart Vikasit Bharat"** by fostering holistic, innovative, and inclusive web ecosystems for national progress.

Lab Objectives:

Sr. No.	Lab Objectives
1	To understand and apply HTML5 for creating structured web pages.
2	To design responsive and visually appealing web interfaces using CSS3 and frameworks.
3	To develop interactive web applications using JavaScript and DOM manipulation.
4	To explore web hosting, deployment, and basic security concepts.
5	To build a mini project integrating front-end technologies.
6	To encourage self-learning of modern web technologies and trends.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Design and develop static web pages using HTML5 elements.	L6
2	Create responsive layouts using CSS3 Flexbox, Grid, and Media Queries.	L6

3	Build interactive web features using JavaScript events and DOM manipulation.	L3 -> L6
4	Deploy a static website using a cloud hosting platform.	L3
5	Implement form validation and basic client-side logic.	L3-L4
6	Explore and present on emerging web technologies (PWA, Web3, AI, etc.).	L2,L4,L5

Hardware & Software Requirements:

Hardware Requirement:	**Software Requirement:**
Intel i3+ / 8GB RAM / 50GB free space Internet connection	VS Code / Sublime Text / Notepad++ Google Chrome / Firefox with Developer Tools Git, GitHub account, Netlify / GitHub Pages account Node.js (optional for advanced topics) Live Server extension (VS Code)

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	LO Mappings
0	**Prerequisites**	Basic computer operation, file management, internet usage, basic programming logic. Installation and configuration of VS Code, Live Server, and browser developer tools.	02	-
1	**HTML5 Fundamentals**	1.1 Structure of HTML document, 1.2 formatting tags, 1.3 lists, 1.4 tables, 1.5 hyperlinks, 1.6 images, 1.7 forms, 1.8 semantic tags, 1.9 audio/video embedding. 1.10 Explain web fundamentals, client-server **,** Responsive design features. **Self-Learning Topics:** HTML5 canvas and SVG graphics,SEO considerations **Online reference:** - HTML5 Canvas and SVG Graphics - SEO basics using HTML semantic elements - HTML5 Forms Advanced Features	4	LO1
2	**CSS3 and Responsive Design**	2.1 CSS syntax, 2.2 selectors, 2.3 colors, 2.4 backgrounds, 2.5 borders, 2.6 box model, 2.7 fonts, 2.8 animation, 2.9 Flexbox, 2.10 Grid, 2.11 Media Queries, 2.12 responsive design, 2.13 introduction to Bootstrap. **Self-Learning Topics:**	4	LO2

		Tailwind CSS**,** CSS animations,Responsive UI design,CSS grid vs Flexbox **Online reference:** - Tailwind CSS - CSS Animations - Responsive UI design trends - CSS Grid vs Flexbox		
3	**JavaScript Fundamentals**	3.1 JavaScript Basics:Variables, data types, operators, 3.2 conditional statements, 3.3 loops, 3.4 functions, 3.5 arrays, 3.6 events, 3.7 DOM manipulation, 3.8 form validation. **Self-Learning Topics:** TypeScript,ES6 features,JavaScript frameworks,Fetch API and AJAX **online reference:** ES6 Features (Arrow functions, let/const, destructuring, spread operator) - Introduction to TypeScript - JavaScript Framework overview (React, Vue, Angular) - Fetch API and AJAX	5	LO3, LO5
4	**Web Hosting & Deployment**	4.1 Website hosting concepts: domain registration, FTP basics, deployment process, 4.2 cookies, 4.3 sessions, 4.4 introduction to web security. **Self-Learning Topics:** HTTPS Certificates (cLet's Encrypt),Secure authentication methods (OAuth, JWT),OWASP Top 10 overview,Git and GitHub Basics **Online reference:** Getting Started - Let's Encrypt RFC 6749: The OAuth 2.0 Authorization Framework | RFC Editor RFC 7519: JSON Web Token (JWT) | RFC Editor OWASP Top 10:2025 https://www.analyticsvidhya.com/blog/2021/09/git-and-github-tutorial-for-beginners/	4	LO4
5	**Mini Project**	5.1 Integration of HTML5, CSS3, JavaScript, and deployment. **Self-Learning Topics:** Progressive Web Apps (PWA),Web Accessibility (WCAG standards) Performance optimization techniques **Online reference:** **https://developer.mozilla.org/en-US/docs/Web/Progressive_web_apps/Tutorials?utm_source=copilot.com** **https://www.w3.org/TR/WCAG21/?utm_source=copilot.com** **https://developer.mozilla.org/en-US/docs/Web/Performance**	6	LO1, LO2, LO3, LO4, LO6
6	**Modern Web Technologies**	6.1 Introduction to APIs, 6.2 cloud-based web apps, 6.3 PWA, 6.4 web accessibility standards, 6.5 introduction to front-end frameworks. **Self-Learning Topics:** AI-powered websites (TensorFlow.js),JAMstack Architecture Web3 and Blockchain integration,Serverless Web Applications Voice-enabled Web (Web Speech API) **Online reference:**	2	LO6

		Building an AI-Powered Web Application with Next.js and TensorFlow.js - DEV Community https://www.cloudflare.com/learning/performance/what-is-jamstack/ https://executive.mit.edu/course/blockchain-and-crypto-applications--from-decentralized-finance-to-web-3/a054v00000rHFLVAA4.html https://www.coursera.org/learn/create-serverless-applications https://groups.csail.mit.edu/mug/course/6.835_resource.html		
Total			**30**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References:

1. Jennifer Robbins ,"Learning Web Design", 6th Edition, ISBN: 978-1098137687 (July 2025)
2. BPB Publications, "Responsive Web Development", ISBN: 978-9355516749 (June 2024)
3. Jennifer Robbins ,"Web Design in a Nutshell", O'Reilly Media

Online References:

Sr. No.	Website Name	URL
1	NPTEL	https://onlinecourses.swayam2.ac.in/e-learning/preview/nou24_cs09
2	NPTEL	https://onlinecourses.nptel.ac.in/e-learning/preview/noc22_cs125
3	Udemy	https://www.udemy.com/course/the-web-developer-bootcamp/?srsltid=AfmBOorENzL5dxNIqOWqkVCuxPk0x4KfzbnLBK5voYHduwE0eKTcdsVn&couponCode=CP260518SUMMXIN
4	NPTEL	https://viit.digimat.in/nptel/courses/video/106106156/L01.html

List of Experiments

Sr No	List of Experiments	Hrs
01	Study of Internet and Web Architecture	02
02	Analyze existing applications such as: Digital India portals, UMANG, MyGov, Identify: URL structure and HTTP/HTTPS request flow. Responsive design features.	02
03	Design a personal profile webpage using HTML5 including: - Header, navigation, main content, aside, footer (semantic tags) - Personal details in a table - A contact form with text, email, and submit fields - Embedded audio/video	02
04	Design Animated Login Form by applying CSS transition and animation transform keyframes focus effect	02

05	Create a responsive business landing page using CSS3 with: - Navigation bar using Flexbox - Grid layout for product/service cards - Media queries for mobile (max-width: 768px) and tablet (max-width: 1024px) - Flexible Layout - Responsive images - Hover effects and transitions - Alternative version using Bootstrap framework	02
06	Implement CSS Grid for image gallery design with: CSS Grid Grid-template-columns Gap Responsive grids	02
07	Develop a JavaScript-based calculator	02
08	Implement form validation with: - Name, email, phone, password, and confirm password fields - Real-time validation using JavaScript events - Display error messages dynamically using DOM manipulation Practical 8: Create an interactive to-do list application with: - Add, edit, delete tasks - Mark tasks as completed - Store tasks in browser localStorage	02
09	Build a dynamic complaint registration system with show/hide	02
10	Deploy a static website using GitHub Pages or Netlify: - Create a GitHub repository - Upload HTML/CSS/JS files - Enable GitHub Pages from repository settings - (Alternative) Deploy using Netlify drag-and-drop or CLI - Configure a custom domain (optional)	02
11	Design a mini website for a local business, startup, or personal portfolio with: - Minimum 4 interconnected pages (Home, About, Services/Projects, Contact) - Responsive design (mobile, tablet, desktop) - Interactive features (image slider, form validation, dynamic content) - Deployment on GitHub Pages or Netlify - Project documentation and presentation	02
12	Prepare a seminar/presentation on any emerging web technology: - Artificial Intelligence in Web Development - Progressive Web Applications (PWA) - Web Accessibility (WCAG Standards) - Tailwind CSS and Utility-first Design - Serverless Web Applications - Web3 and Blockchain Integration - JAMstack Architecture - Voice-enabled Web Applications - Cloud-based Web Hosting Platforms - Cyber security in Web Applications	02

Assessment:

Term Work: Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

Term Work Marks: 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).

Practical & Oral Exam: An Oral & Practical Exam will be held based on the above list of Practicals.

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	02
1111401	Workshop-II (FAB Lab)	--	--	02*+02	--	--	60	--	60	

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
1111401	Workshop-II (FAB Lab)	---	---	---	---	---	50	–	50

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

This laboratory course is designed in accordance with the vision and recommendations of the **NEP 2020**, emphasizing experiential learning, skill-based education, multidisciplinary exposure, innovation, and industry-oriented technical training. The course provides hands-on experience in electronic components, circuit design, breadboard prototyping, PCB fabrication, soldering, testing, debugging, and basic embedded and sensor-based systems, thereby enabling students to connect theoretical concepts with practical engineering applications. The curriculum promotes critical thinking, creativity, problem-solving ability, teamwork and technical competency.

The course is structured to develop industry-relevant skills by introducing students to modern engineering practices and tools such as EDA software (KiCad/EasyEDA), PCB layout design, electronic testing instruments, and prototyping techniques widely used in electronics, automation, and embedded system industries. It also supports innovation-driven learning and encourages students to explore emerging technologies and real-world engineering solutions.

Further, the course aligns with relevant **United Nations Sustainable Development Goals (UNSDGs),** particularly **SDG 4, SDG 8** and **SDG 9.**

Lab Objectives:

Sr. No.	Lab Objectives
1	Develop basic understanding of electronic components, their identification, and practical usage in circuits.
2	Hands-on skills in assembling and testing electronic circuits using breadboard and basic instruments.
3	Learn schematic design techniques using software tools such as KiCad / EasyEDA
4	Understand PCB fabrication process from design to final working board.
5	Acquire practical skills in soldering, desoldering, and troubleshooting electronic circuits.
6	Understand and implement basic sensor-based circuits for real-world applications and to introduce students to the fundamentals and applications of drone technology.

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Identify and classify basic electronic components including passive, active, and sensor devices	L1, L2
2	Interpret circuit diagrams and implement them on breadboard for functional verification	L2, L3
3	Design and draw electronic schematics using EDA tools such as KiCad / EasyEDA	L3, L6
4	Apply PCB fabrication techniques to convert schematic designs into working hardware	L3
5	Perform soldering, testing, and debugging of electronic circuit	L3, L4
6	Design and analyze basic sensor-based circuits for automation and to assemble and configure a basic drone system using appropriate tools and techniques	L4, L6

Hardware & Software Requirements:

Hardware Requirement:	Software Requirement:
Electronics Workbench with DC Power Supply, Digital Multimeter, Soldering Station / Soldering Iron, Desoldering Pump and Desoldering Wick, Breadboards, Electronic Component Kits (Resistors, Capacitors, Inductors, Diodes, LEDs, Transistors), Sensor Kit (LDR, IR Sensor, Temperature Sensor, etc.), Connecting Wires and Jumper Wires, PCB Fabrication Kit (Copper Clad Boards, Etching Tray, Ferric Chloride Solution, Marker Pens), PCB Drilling Machine / Mini Drill, Wire Stripper and Cutting Pliers, Magnifying Lamp / PCB Inspection Tools, Regulated DC Power Supply Module, Continuity Tester	KiCad /EasyEDA

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
1	**Discrete Components and Testing**	1.1 Introduction to electronic components used in basic circuits 1.2 Identification and classification of components into passive (R, L, C) and active (Diode, Transistor) 1.3 Study of physical appearance, symbols, and standard marking of components 1.4Introduction to basic sensors (LDR, temperature sensor, IR sensor) and their applications.	12	LO1

		1.5 Datasheet Reading and Interpretation ● Importance of datasheets in component selection ● Understanding key parameters: ○ Maximum ratings (voltage, current, power) ○ Electrical characteristics ○ Pin configuration and package types 1.6 How to identify component specifications from datasheets.		
2	**Circuit Making and Testing on Breadboard**	2.1 Introduction to breadboard and its internal connections 2.2 Understanding of circuit schematic diagrams and their conversion into practical wiring on breadboard. 2.3 Hands-on practice of assembling simple electronic circuits such as series and parallel resistor networks, LED blinking circuits, diode rectifiers, and basic transistor switching circuits. Emphasis on circuit testing, troubleshooting, and measurement using multimeter and power supply.	12	LO2
3	**Schematic Software-KiCad /EasyEDA**	3.1 Introduction to electronic schematic design using open-source software tools such as KiCad / EasyEDA 3.2 Understanding of schematic symbols, component libraries, and circuit representation in digital format.	06	LO3
4	**PCB Fabrication**	4.1 Introduction to Printed Circuit Board (PCB) and its importance in electronics 4.2 Types of PCBs: single-layer, double-layer 4.3 Basics of PCB layout design and component placement 4.4 Conversion of schematic diagram into PCB layout 4.5 Methods of PCB design transfer 4.6 Solder mask concept and basic finishing techniques 4.7 Component mounting and soldering on PCB	12	LO4
5	**Soldering Testing and Debugging**	5.1 Types of soldering tools: soldering iron, solder wire, flux, desoldering pump 5.2 Proper soldering techniques for through-hole components 5.3 Desoldering techniques and component removal from PCB, Introduction to PCB inspection and visual fault detection, Use	06	LO5

		of multimeter for continuity, voltage, and resistance testing.		
6	**Arduino/ Raspberry Pi (Demo) and Drones**	6.1 Introduction to Arduino and Raspberry Pi platforms and their applications in engineering. 6.2 Study of basic components: microcontroller/microprocessor, GPIO pins, sensors, and actuators. 6.3 Demonstration of simple Arduino programming environment (IDE) and basic coding structure. 6.4 Hands-on demo of LED blinking, switch input, and basic sensor interfacing (e.g., LDR or temperature sensor). 6.5 Introduction to Raspberry Pi setup and basic Linux-based operation (GUI/terminal overview). 6.6 Demonstration of simple automation or control task using Arduino/Raspberry Pi (e.g., LED control, fan control, or sensor-based alert system).	12	LO6
Total			**60**	

References : 1.Microelectronic Circuits – Sedra & Smith
2.Basic Electronics Engineering – N. N. Bhargava
3.Embedded Systems: Introduction to ARM Cortex-M Microcontrollers – Jonathan Valvano

Online References

https://www.datasheetarchive.com?utm_source=chatgpt.com
https://docs.arduino.cc?utm_source=chatgpt.com
https://droneblocks.io/resources?utm_source=chatgpt.com
https://nptel.ac.in?utm_source=chatgpt.com

Assessment:

- **Term Work:** Term Work shall consist of at least 10-12 Practicals based on the above list.
- **Term Work Marks:** 50 Marks (Total) = 45 Marks (Experiments) + 05 Marks (Attendance)

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	2
2992601	NSS & Civil Defense	--	--	02*+02	--	--	60	--	60	

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2992601	NSS & Civil Defense	---	---	---	---	---	25	---	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

The NSS & Civil Services course complements technical education by developing social responsibility, leadership, discipline, teamwork, and communication skills among engineering students. Through community service, environmental conservation, disaster management, and national integration activities, students gain practical exposure to societal challenges and civic responsibilities. The course enhances problem-solving abilities, ethical values, volunteerism, and emergency preparedness, helping students become responsible citizens, competent professionals, and lifelong learners capable of contributing effectively to society and nation-building

Lab Objectives:

Sr. No.	Lab Objectives
1	Understand the philosophy, objectives, structure, and role of NSS/NCC in community and nation building
2	Apply leadership, teamwork, discipline, and communication skills during social service and awareness activities.
3	Analyze social, environmental, and disaster-related problems and identify suitable community-based solutions.
4	Demonstrate participation in area-based projects, national integration activities, and civic responsibilities
5	Develop awareness regarding disaster management, civil defense, fire safety, and first aid practices.
6	Create social responsibility and ethical values through voluntary service, community engagement, and self-learning activities

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the history, objectives, structure, and importance of NSS and NCC in nation building.	L2
2	Participate effectively in awareness programs, rallies, workshops, and community service activities.	L3
3	Analyze social and environmental issues through surveys, village visits, and community projects.	L4
4	Demonstrate leadership qualities, discipline, teamwork, and civic responsibility during NSS/NCC activities.	L3
5	Apply basic concepts of disaster management, civil defense, fire safety, and first aid in emergency situations.	L3
6	Develop ethical values, social commitment, and lifelong learning through self-learning and volunteer activities.	L6

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	Prerequisite	Scout/Guide & Social studies	--	--
1	NSS Orientation and Community Service	1.1 Introduction to NSS 1.2.Code of conduct & discipline 1.3.Importance to community service and National services. 1.4. History, objectives, organizational structure of NSS 1.5. Community identification; Importance of volunteerism and contribution to Nation building 1.6. Social service activities. **Self-learning-Topics**: Digital volunteering, Role of youth in Sustainable Development Goals (SDGs), Case studies of successful NSS projects, Swami Vivekananda's views on youth and service	10	LO1
2	Activity & Awareness based Programs	Activities can be conducted throughout the academic Semester, like 2.1.Swatch Bharat mission..2.2. Digital literacy drive 2.3.Community visit etc. Evaluation will be based on activity record and involvement of student. 2.4.Awareness Programs such as Rally, Seminar, Workshops, Poster making, Theme based Rangoli, Essay writing, Slogan writing, Quizzes 2.5.Celebration of National and International days, (plan various social activities as per the relevance for learners and make a detailed report) **Self-learning Topics**: Social media awareness campaigns, Cyber safety awareness, Mental health awareness, Plastic-free campus	10	LO2

		initiatives, Event management techniques		
3	Area Based Projects	Visit to following areas to promote social awareness: 3.1. Swachhata Programme (Swachh Bharat Mission) 3.2.Socio Economic Survey of village, Nature Camp, Environmental Education 3.3.Women Empowerment Programme 3.4. Health and hygiene programs and Blood donation 3.5. Legal awareness programme 3.6 Literacy Programme 3.7. Cyber theft awareness, 3.8.Water Conservation Programme and RWH. 3.8.One Day Special Camp in a village (preferably in adopted village/Slums/MR Schools etc.). **Self-learning-Topics**: Smart village concepts, Rainwater harvesting, Waste management practices, Organic farming awareness, Rural entrepreneurship and self-help groups	10	LO3
4	NCC Orientation and National Integration	4.1. Aims, Objectives and Organization of NCC 4.2. National integration, Discipline, Patriotism 4.3.Drill basics, Civil defense awareness.4.4.Importance of National Cadet Corps and contribution to Nation building. **Self-learning-Topics**: Armed forces contribution to nation building, Border security awareness, Leadership qualities in defense services, National symbols and constitutional values	10	LO4
5	Introduction to Civil Defense and Disaster Management	5.1. Definition and Objectives of civil defense and Disaster Management 5.2. Structure and functions of the Civil Defense Organization 5.3.Types of disasters: Natural and Man-made 5.4.Phases of disaster management: Prevention, Preparedness, Response, Recovery, NDMA (National Disaster Management Authority) and its role, Role of youth in disaster management **Self-learning-Topics**: Earthquake preparedness, Flood management techniques, Disaster communication systems, Role of NGOs in disaster management, Community resilience planning	10	LO5
6	Basics of Fire Safety and First Aid	6.1. Introduction to basics of Fire Safety and First Aid 6.2.Types of Fire extinguishers, Basic fire-fighting techniques. 6.3. First aid for common injuries:	10	LO6

		Bleeding, Fractures, Burns, CPR, Emergency numbers and response protocol. 6.4. Evacuation techniques. **Self-learning-Topics**: CPR awareness videos, Home electrical safety, Emergency evacuation planning, Industrial safety practices, Psychological first aid and stress management		
Total			**60**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

References Books:

1. Youth Development through NSS – IGNOU Publications.

2. Disaster Management Handbook – NDMA.

3. Community Development and Social Service – Academic Publication

4.National Service Scheme Manual – Government of India.

5. NCC Handbook – Ministry of Defense.

6. Environmental Studies – Erach Bharucha.

Online References:

Sr. No.	Website Name	URL
1	https://nss.gov.in	https://nss.gov.in
2	https://indiancc.nic.in	https://indiancc.nic.in
3	https://www.ndma.gov.in	https://www.ndma.gov.in

List of Experiments.

Sr No	Activity / Task	Hrs
01	NSS & NCC Orientation, Oath Taking and Organizational Structure Study	02
02	Cleanliness Drive and Swachh Bharat Campaign	02
03	Tree Plantation and Environmental Conservation Activity	02
04	Traffic Awareness Rally and Road Safety Campaign	02
05	Community/Village Survey on Social or Environmental Issues	02
06	Water and Energy Conservation Audit of Campus/Community	02
07	Leadership and Team-Building Exercise (Group Task/Outdoor Activity)	02

08	NCC Drill Practice and Discipline Training	02
09	First Aid Demonstration (Bandaging, CPR Awareness, Fracture Handling)	02
10	Disaster Management and Fire Safety Mock Drill	02
11	Visit to Old Age Home/Orphanage/NGO and Community Interaction	02
12	Basics of Fire Safety & First Aid	02
13	Preparation of Activity Report, Reflection Diary and Presentation on NSS/NCC Activities	02

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Sr No	List of Assignments / Tutorials	Hrs
01	Prepare a report highlighting the history, objectives, organizational structure, motto, symbols, and contribution of NSS and NCC in nation building.	02
02	Prepare a report on a cleanliness, traffic safety, health, or environmental awareness campaign conducted in the institute/community, including observations and outcomes.	02
03	Conduct a survey on a social/environmental issue (plastic use, water conservation, waste management, literacy, etc.) and analyze the collected data with suitable conclusions.	02
04	Write a reflective report on leadership roles, teamwork experiences, discipline, communication skills, and challenges faced during NSS/NCC activities.	02
05	Study a natural disaster, accident, or emergency situation and propose appropriate first aid, rescue, evacuation, and disaster management measures.	02
06	Design a small community service project (e.g., plastic-free campus, digital literacy drive, energy conservation campaign) and prepare a reflection on its social impact and future implementation.	02

Assessment:

- **Term Work:** Term Work shall consist of at least 8-10 Activity/Task based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

- **Term Work Marks:** 25 Marks (Total) = 15 Marks (Activity/Task) + 05 Marks (Assignments) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)					Total Credits (C) (Notional Learning Hours/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2992601	Yoga for Adolescents	--	--	02*+2	--	--	60	--	60	1

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2992601	Yoga for Adolescents	---	---	---	---	---	25	---	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale:

Yoga for Adolescents addresses one of the most critical yet underserved dimensions of student development. Adolescence is a phase of immense physical change, psychological turbulence and academic pressure. Students from arts, commerce and science backgrounds are equally susceptible to fear of failure, peer pressure, parental expectations, low self-esteem, poor concentration and psychosomatic disturbances. This course introduces the true essence of Yoga — not merely as physical exercise but as an integrated science of self-transformation drawn from Patañjali Yoga Sūtra, Haṭha Yoga Pradīpikā, Bhagavad Gītā and Yoga Vāsiṣṭha. Through understanding of different paths of Yoga, the nine obstacles (Antarāyāḥ), five Kleśas, Vṛttis, Saṁskāras and the Pañca Kośa framework, students gain both theoretical insight and practical tools. Āsana, Prāṇāyāma, Trataka and Yoga Nidrā practices are designed specifically for adolescents to promote physical health, mental peace and academic concentration. The course aligns with NEP 2020 objectives by fostering experiential learning, ethical values, holistic development and the scientific study of India's classical knowledge systems.

Lab Objectives:

Sr. No.	Lab Objectives
1	Understand the philosophy, foundational texts, definitions, and different paths of Yoga in the context of adolescent development.
2	Apply principles of Karma Yoga, Bhakti Yoga, Jñāna Yoga, and Haṭha Yoga to develop emotional stability, academic focus, and self-awareness.
3	Analyse the nine obstacles, five Vṛttis, five Kleśas, six psychological propensities, and Saṁskāras as causes of adolescent academic and behavioural challenges.
4	Demonstrate correct practice of foundational Āsanas, Sūrya Namaskāra, Prāṇāyāma and Trataka with awareness of breath and body.

5	Develop awareness of the Pañca Kośa framework and its application in designing an integrated yogic lifestyle for concentration and wellbeing.
6	Create a personal disciplined yogic routine integrating Āsana, Prāṇāyāma, meditation concepts and reflective self-study based on classical Sūtras

Lab Outcomes: Upon completion of the course, learners will be able to:

Sr. No.	Lab Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the philosophy of Yoga, its classical definitions, paths and relationship with Indian philosophy and adolescent wellbeing.	L2
2	Apply principles of different paths of Yoga to reflect on personal emotions, academic pressures, and develop equanimity.	L3
3	Analyse the nine obstacles, Vṛttis, Kleśas and Saṁskāras as root causes of distraction, and describe yogic remedies.	L4
4	Demonstrate correct technique and breathing awareness in Āsanas, Sūrya Namaskāra, Prāṇāyāma and Trataka.	L3
5	Apply the Pañca Kośa framework to understand psychosomatic health and design an integrated daily yoga practice.	L3
6	Develop a disciplined personal yoga practice schedule and document its effects on concentration, mood and academic readiness.	L6

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content and Experiment List	Hours	LO Mappings
0	**Prerequisite**	Scout/Guide & Social studies	--	--
1	**Introduction to Yoga and Its Philosophy**	1.1 Why Yoga for Adolescents is a distinct topic: physical changes, psychological effects, ego, peer pressure and academic impact 1.2 What is Yoga? — Definitions from Patañjali (Yogaścittavṛttinirodhaḥ), Bhagavad Gītā, B.K.S. Iyengar, Swami Satyananda Saraswati and Swami Niranjanananda 1.3 How Yoga helps: physical harmony, mental balance, emotional stability and spiritual awareness 1.4 Introduction to Indian Philosophy: The Vedas, Upaniṣads, Mahābhārata, Rāmāyaṇa, Bhagavad Gītā, Yoga Vāśiṣṭha 1.5 Six systems of Indian Philosophy (Shad-Darshana): Nyāya, Vaiśeṣika, Sāṁkhya, Yoga, Mīmāṁsā, Vedānta 1.6 Academic performance in education: skills, Bhāvanās and development of Citta (mind, ego, intellect) **Self-learning Topics:**	10	LO1

		Compare Yoga with modern psychology for adolescent wellness; Read selected verses from Bhagavad Gītā Chapter 2 (Sthitaprajña) and reflect on relevance to student life		
2	**Different Paths of Yoga**	2.1 Karma Yoga (Action): definition, aim — purifying the mind, working without expectation, BG 2-47; relevance for adolescents 2.2 Bhakti Yoga (Devotion): definition, aim — channelizing emotions, developing concentration (Dhāraṇā), emotional culture and stability 2.3 Jñāna Yoga (Knowledge): definition, aim — path of enquiry, clearing dogmas, developing discriminative intelligence, overcoming fear of failure 2.4 Haṭha Yoga (Willpower): definition, aim — control of Prāṇa, purification, biochemical and psycho-physiological science; Yamas, Niyamas, Āsanas, Prāṇāyāma overview (HYP reference) 2.5 Patañjali / Rāja Yoga (Mind): four Pādas — Samādhi, Sādhana, Vibhūti, Kaivalya; Puruṣārthas (Dharma, Artha, Kāma, Mokṣa) in the context of adolescents 2.6 Integration of different paths: Integral Yoga (Swami Sivananda); how paths complement each other for holistic development **Self-learning Topics:** Study the concept of Abhyāsa and Vairāgya (PYS 1-12 to 1-14); write a reflective note on which path of yoga suits your personality and why	10	LO2
3	**Yoga Vāsiṣṭha and Mind: Adolescent Perspective**	3.1 Introduction to Yoga Vāsiṣṭha: ten chapters, parables, stories and their approach to adolescent queries 3.2 Dispassion, qualification and behaviour of seeker; spirit of inquiry and self-control 3.3 Aim of Yoga for Adolescents: removing fixed habits, adaptability, integration at physical-emotional-mental-spiritual levels 3.4 The nine obstacles (Antarāyāḥ) from Patañjali Yoga Sūtra 1-30: Vyādhi, Styāna, Saṁśaya, Pramāda, Ālasya, Avirat, Bhrānti-darśana, Alabdha-bhūmikatva, Anavasthitatvāni — related to adolescent academic behaviour	10	LO3

		3.5 State of mind (Cittabhūmis): Kṣipta, Mūḍha, Vikṣipta, Ekāgra, Nirodha — importance for adolescents comparing themselves with peers 3.6 How mind cognizes external objects: nineteen mouths (Māṇḍūkya Upaniṣad), five Jñānendriyas, five Karmendriyas, five Prāṇas, Manas, Buddhi, Ahaṁkāra, Citta **Self-learning Topics:** Reflect on Yoga Vāsiṣṭha's story of Sukra; study Table of Nine Obstacles and map each to a personal academic challenge		
4	**Vṛttis, Kleśas, Saṁskāras and Academic Performance**	4.1 Yoga: Cessation of Movement in Consciousness (PYS 1-2): Citta, Vṛtti, Nirodha explained for adolescents 4.2 Five Vṛttis (fluctuations of consciousness): Pramāṇa, Viparyaya, Vikalpa, Nidrā, Smṛti — Kliṣṭā and Akliṣṭā 4.3 Five Kleśas (afflictions): Avidyā, Asmitā, Rāga, Dveṣa, Abhiniveśa — three levels: intellectual, emotional, instinctive 4.4 Six psychological propensities (Ṣaḍ-ripu): Kāma, Krodha, Lobha, Moha, Mada, Mātsarya and their manifestation in adolescent behaviour 4.5 Saṁskāras and Vāsanās: cognitive, affective and conative imprinting in Citta; cycle of Vṛtti–Saṁskāra 4.6 Rāga and Dveṣa as root cause of nine obstacles; four additional distractions (PYS 1-31): Duḥkha, Daurmanasya, Aṅgamejayatva, Śvāsa-Praśvāsa **Self-learning Topics:** Maintain a personal 'mind diary' for one week — record a moment of Rāga and Dveṣa each day and observe their effect on concentration	10	LO4
5	**Haṭha Yoga Practices for Adolescents**	5.1 Āsanas for adolescents: Tāḍāsana, Vṛkṣāsana, Trikoṇāsana, Bhujaṅgāsana, Śalabhāsana, Paścimottānāsana, Sarvāṅgāsana, Śavāsana — correct method, benefits and contraindications 5.2 Sūrya Namaskāra: twelve steps, synchronisation with breath, physical and psycho-physiological benefits 5.3 Prāṇāyāma for adolescents: Sūryabhedana (Prāṇa energising), Sītkārī (cooling), Śītalī (calming), Bhāstrikā (purifying) — technique and benefits from Haṭha Yoga Pradīpikā	10	LO5

		5.4 Trataka (Ṣaṭkarma): focused gazing practice for improving concentration and reducing mental fatigue (HYP 2-31, 2-32) 5.5 Mudrā and Bandha overview: Jñāna Mudrā, Chin Mudrā, Mūlabandha — introductory practice for adolescents 5.6 Nāḍī Śodhana (Alternate Nostril Breathing): technique, effect on nervous system, role in calming emotional turbulence **Self-learning Topics:** Practice Sūrya Namaskāra daily for 21 days and maintain an observation journal; watch SVYASA / AYUSH demonstrations of Prāṇāyāma techniques		
6	**Pañca Kośa Framework and Integrated Yoga Practice**	6.1 Pañca Kośa from Taittirīya Upaniṣad: Annamaya, Prāṇamaya, Manomaya, Vijñānamaya, Ānandamaya — framework for integrated yogic practice 6.2 Psychosomatic diseases (Adhija Vyādhi) and adolescents: how agitated mind disturbs all five Kośas; Yoga Vāsiṣṭha's classification 6.3 Aṣṭāṅga Yoga (PYS 2-29): Yama, Niyama, Āsana, Prāṇāyāma, Pratyāhāra, Dhāraṇā, Dhyāna, Samādhi — practical application for adolescents 6.4 Cittaprasādana (calming the mind): Maitrī, Karuṇā, Mudita, Upekṣā (PYS 1-33); Prāṇa regulation (PYS 1-34); Jyotiṣmatī (PYS 1-36) 6.5 Concentration (Dhāraṇā) and academic performance: link between relaxed mind and ability to study; role of yoga in achieving Ekāgratā 6.6 Designing integrated yoga practice for adolescents: Āsana (Annamaya), Prāṇāyāma (Prāṇamaya), Mantra-stories-games (Manomaya), discrimination (Vijñānamaya), relaxation / Yoga Nidrā (Ānandamaya) **Self-learning Topics**: Experience a 20-minute Yoga Nidrā session (NSYSU audio); write a short report on how each of the five Kośas was affected; explore AYUSH Yoga protocol for students	10	LO6
	Total		**60**	
Note: Students are encouraged to explore these topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.				

Reference Books:

1. Iyengar, B. K. S. (2002). Light on Yoga. HarperCollins India.
2. Swami Satyananda Saraswati (2013). Asana Pranayama Mudra Bandha (4th ed.). Bihar School of Yoga, Munger.
3. Swami Niranjanananda Saraswati (2009). Prana and Pranayama. Bihar School of Yoga.
4. Vyas Dev Ji (2019). Hatha Yoga Pradipika (with commentary). Kaivalyadhama, Lonavla.
5. Youth Development through NSS – IGNOU Publications.
6. Disaster Management Handbook – NDMA.
7. Community Development and Social Service – Academic Publication

Online References:

Sr. No.	Website Name	URL
1	Ministry of AYUSH — Yoga	https://yoga.ayush.gov.in
2	Bihar School of Yoga	https://www.biharyoga.net
3	SVYASA Yoga University	https://svyasa.edu.in
4	Kaivalyadhama Yoga Institute	https://kdham.com
5	Patanjali Yogpeeth	https://www.divyayoga.com

Sr No	List of Assignments / Tutorials	Hrs
01	Write a comparative report on the five paths of Yoga (Karma, Bhakti, Jñāna, Haṭha, Rāja) — definition, aim, suitable personality type, and one classical Sūtra for each. Conclude with which path you identify with and why.	02
02	Map the nine obstacles (Antarāyāḥ) from Patañjali Yoga Sūtra 1-30 to real-life challenges faced by adolescents during studies. For each obstacle, suggest one practical yogic remedy (Āsana, Prāṇāyāma, Bhāvana, or Sūtra-based practice).	02
03	Maintain a 21-day Sūrya Namaskāra journal: record date, number of rounds, breath observation, physical sensations, and mental state. Write a summary report on changes observed in energy, focus and mood over the three weeks.	02
04	Study the Pañca Kośa framework (Taittirīya Upaniṣad) and design a one-week integrated yoga schedule addressing all five Kośas. Justify each practice with reference to a classical text (PYS, HYP, BG or Yoga Vāsiṣṭha).	02
05	Research one adolescent psychosomatic challenge (e.g., exam anxiety, sleep disorder, anger, low self-esteem) and prepare a report explaining its Yogic root cause (Rāga/Dveṣa, Kleśa, obstacle) and a recommended integrated Yoga intervention.	02
06	Write a personal reflection essay (500 words) titled 'My Yoga Journey this Semester': describe what changed in your understanding of Yoga, your practice experience, one Sūtra that guided you, and how you plan to continue Yoga in daily life.	02

List of Experiments.

Sr No	Activity / Task
01	Yoga orientation session: discussion on 'What is Yoga?' using classical definitions (Patañjali, Bhagavad Gītā, Iyengar). Students share their expectations and misconceptions about yoga. Introduction to the course structure and personal yoga journal.
02	Sūrya Namaskāra practice: learn all twelve steps with correct breath synchronisation. Practice 3 slow rounds with awareness. Discuss benefits — physical warming, pranic activation and mental alertness. Students document observations in journal.
03	Standing and balancing Āsanas: Tāḍāsana, Vṛkṣāsana, Trikoṇāsana — correct posture, breath, dṛṣṭi (gaze) and duration. Discuss how balancing postures develop focus, patience and self-control relevant to academic discipline.
04	Forward bends and back bends: Paścimottānāsana, Bhujaṅgāsana, Śalabhāsana — technique, breath and modifications for beginners. Discussion on how these postures influence the nervous and endocrine system (pineal and pituitary glands — Yoga Vāsiṣṭha connection).
05	Inversions and relaxation: Sarvāṅgāsana (with wall support for beginners) and Śavāsana — correct method, duration, and contraindications. Practice guided Śavāsana (10 min). Students reflect on difference in mental state before and after.
06	Prāṇāyāma session 1: Nāḍī Śodhana (Alternate Nostril Breathing) — technique, ratio, and rounds. Observe calming effect on the nervous system. Relate to Haṭha Yoga Pradīpikā 2-10 and its role in adolescent emotional regulation.
07	Prāṇāyāma session 2: Sūryabhedana, Sītkārī, Śītalī — technique and therapeutic benefits from HYP. Discuss appropriate contexts: Sūryabhedana for energy, Śītalī/Sītkārī for calming anger and anxiety in adolescents.
08	Trataka practice: steady gazing at a candle flame or black dot for 3–5 minutes. Observe Aśru (tears) onset. Discuss HYP 2-31 and 2-32 — benefit for eyes, concentration and mental withdrawal (Pratyāhāra). Students journal their experience.
09	Mind mapping workshop — Nine Obstacles: students individually map the nine Antarāyāḥ (PYS 1-30) to personal challenges in academic life. Group discussion on how Yoga addresses each obstacle. Prepare a visual mind-map poster.
10	Karma Yoga activity: students perform a selfless service task in the campus (classroom cleaning, garden care, helping peers) with awareness and without expectation. Post-activity reflection connecting the experience to BG 2-47 and purification of mind.
11	Yoga Nidrā (guided relaxation): 30-minute guided Yoga Nidrā session (Ānandamaya Kośa). Students experience deep relaxation and visualisation. Post-session journal: describe the state of mind, body sensations and comparison with normal rest.
12	Integrated practice session and personal yoga schedule design: students design a 30-minute personalised daily yoga routine covering all five Kośas — Āsana, Prāṇāyāma, mantra/affirmation, reflective reading and relaxation. Present and discuss routines in pairs.
13	Sutras reflection and closing session: students select one Sūtra (from PYS, HYP or BG) that resonated with them during the course. Prepare a 3-minute spoken reflection. Compile personal yoga journal (observations, sketches, reflection entries) for submission.

Assessment:

1. **Term Work:** Term Work shall consist of at least 8-10 Activity/Task based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

2. **Term Work Marks:** 25 Marks (Total) = 15 Marks (Activity/Task) + 05 Marks (Assignments) + 05 Marks (Attendance).

Course Code	Course Name	Teaching Scheme (Contact Hours Per week)			Teaching Scheme (Contact Hours Per semester)				Total Credits (C) (Notional Learning Hour/30)
		L	T	P	L	T	P	SL	Notional Learning Hour	
2992501	Indian Wisdom for Modern Challenges	--	--	2*+2	30	--	60		60	2

Course Code	Course Name	Theory					TW	PR / OR	Total
		Internal Assessment			End Semester Exam	Exam Duration (in Hrs)			
		IAT-I	IAT-II	(IAT-I + IAT-II) =TOTAL					
2992501	Indian Wisdom for Modern Challenges	--	--	--	--	--	25	---	25

L: Lecture, P: Practical, OR: Oral, T: Tutorial, IAT: Internal Assessment Test, NL: Notional Learning

Rationale: The course **"Indian Wisdom for Modern Challenges"** has been designed to bridge the gap between ancient Indian knowledge and present-day global issues. In today's rapidly changing world, society is facing challenges related to stress, environmental degradation, ethical decline, unhealthy lifestyles, loss of cultural identity, and unsustainable consumption patterns. This course introduces students to the timeless wisdom embedded in Indian traditions, sciences, philosophies, and sustainable practices, and demonstrates their relevance in solving modern challenges.

The syllabus aligns with the vision of the **National Education Policy (NEP) 2020**, which emphasizes holistic, multidisciplinary, experiential, and value-based education. It also supports the **United Nations Sustainable Development Goals (UNSDGs)** by promoting sustainable living, ethical leadership, wellness, innovation, and social responsibility.

The course has been structured into six interdisciplinary modules that integrate Indian culture, science, mathematics, architecture, leadership, healthcare, and sustainability with practical applications and modern problem-solving approaches.

Course Objectives:

Sr. No.	Course Objectives
1	To help students understand the roots of Indian Knowledge Systems and the importance of holistic and value-based education in today's world.
2	To create awareness about Indian arts, literature, traditions, festivals, architecture, and cultural practices as pathways to sustainable and socially responsible living.
3	To explore the scientific and mathematical contributions of ancient India, including Vedic Mathematics and astronomy, to develop logical thinking, creativity, and innovation.

4	To introduce students to ancient Indian science, Ayurveda, Charaka, Sushruta, and wellness practices for achieving healthy and balanced lifestyles.
5	To inspire ethical leadership, emotional intelligence, strategic thinking, and responsible decision-making through lessons from Chanakya, Ramayana, and Mahabharata.
6	To encourage students to apply indigenous knowledge, sustainable practices, and eco-friendly innovations to solve modern environmental and societal challenges

Course Outcomes: On successful completion of the course, the learner will be able to:

Sr. No.	Course Outcomes	Cognitive Levels of Attainment as per Bloom's Taxonomy
1	Explain the foundations, evolution, and relevance of Indian Knowledge Systems and holistic education practices in contemporary society.	L2
2	Analyze the role of Indian arts, literature, traditions, festivals, architecture, and cultural values in promoting sustainable and socially responsible living.	L4
3	Apply concepts of Vedic Mathematics, astronomy, and ancient Indian scientific heritage to enhance logical thinking, analytical ability, and current problem-solving skills.	L3
4	Demonstrate understanding of Ayurveda, ancient Indian science, and wellness systems for promoting preventive healthcare and balanced lifestyles for today.	L2
5	Evaluate leadership, ethics, emotional intelligence, and governance principles through teachings from Chanakya, Ramayana, and Mahabharata for responsible decision-making.	L5
6	Develop sustainable and innovative solutions using indigenous technologies, eco-friendly practices, sustainable materials, and traditional Indian environmental knowledge systems.	L6

Detailed Syllabus:

Sr. No.	Name of Module	Detailed Content	Hours	CO Mapping
0	Prerequisites	Basic understanding of society, environment, ethics, and contemporary lifestyle challenges with an interest in exploring Indian knowledge, traditions, and sustainable practices	10	-
1	**Foundations of Indian Knowledge Systems (Holistic Education & Value Systems)**	1.1 Meaning, scope, and significance of IKS Holistic learning and NEP 2020 1.2 Evolution of Indian education system ((Nalanda and Takshila University) 1.3 Importance of values and value education -- Gurukul system Vs modern education, Guru-Shishya tradition and ethics(Discipline, empathy, gratitude, and self-learning)	10	CO1

		1.4 Role of Indian traditions and family systems in personality development **Self-Learning Topics: :** <Macaulay's Education Policy and Max Mullar Ideology in destroying Indian Tradition>, <Relevance of Gurukul Education in today's modern world>. Online reference: https://vajiramandravi.com/current-affairs/lord-macaulay-minute/?.com https://www.britannica.com/biography/Max-Muller https://www.ijirmf.com/wp-content/uploads/IJIRMF202503020-min.pdf		
2	**Indian Arts, Culture , Architecture and Scientific Heritage**	2.1. Introduction to Indian classical arts and literature (Sangeet, Natyashastra and Indian folk origin and form) and their relevance in today's world. 2.2. Importance of Indian traditions, culture, and festivals (Scientific significance of Indian festivals) 2.3. Scientific heritage, Temple architecture: geometry, acoustics, ventilation, sustainability 2.4. Traditional eco-friendly construction materials: Mud, and cow dung, Bamboo and wood, Lime plaster, Terracotta etc, Metallurgy,Sustainable architectural practices in ancient India, used to build sustainable homes today **Self-Learning Topics: :** <Origin and Evolution of Indian Classical Music and Dance>, <Natyashastra by Bharata Muni>, <Ancient Heritage sites>, <Indian Handicrafts and Decorative Arts> Online reference: https://www.culturalindia.net/indian-music/?.com https://www.exoticindiaart.com/article/bharat-munis-natyashastra/ https://www.britannica.com/art/Sanskrit-literature https://whc.unesco.org/en/list/	10	CO2
3	**Vedic Mathematics & Astronomy**	3.1. Introduction to Vedic Mathematics: <Meaning and origin of Vedic Mathematics>, <Fast calculation techniques and logical thinking>,< Importance of mental calculation techniques>,	10	CO3

		<Applications of Vedic Mathematics in modern learning> 3.2. Ancient Indian contributions to mathematics and astronomy: < Development of Zero and Decimal System>, <Contributions of Aryabhatta>, <Brahmagupta, and Bhaskaracharya in Algebra, geometry, and trigonometry in ancient India>, <Applications of Vedic Mathematics in competitive exams and computing> 3.3. Ancient Indian Astronomy and Planetary Calculations: -<Study of planetary motion, eclipses, and calendars>,< Indian calendar system and time calculation>,< Importance of astronomy in navigation and calendars> 3.4. Mathematical Applications in Daily Life and Architecture, Geometry in temple architecture, Measurement systems and time calculation methods **Self-Learning Topics:** Work by Indian scientists and Nobel Laureates like Baudhayana, Aryabhatta, Brahmgupta, Bhaskaracharya, Amartya Sen, C.V. Raman etc. Online reference: https://mathshistory.st-andrews.ac.uk/Biographies/Baudhayana/?.com https://www.britannica.com/biography/Aryabhata-I?.com https://www.nobelprize.org/prizes/physics/1930/raman/biographical/?.com		
4	**Health, Ayurveda & Holistic Health**	4.1. Principles of Yoga and Ayurveda- Tridosha theory in Ayurveda , Holistic health and wellness in modern life. 4.2. Diet, sleep, and lifestyle management 4.3. Mental health and stress management 4.4. Traditional Indian food systems and immunity-building practices **Self-Learning Topics: <**Contribution by Charak and Shushruta>, <Development in medical and Life Science, Physiology>, <Ayurveda significance even in current times> (Ancient Vs Modern science) etc. Online reference:	10	CO4

		https://ayush.gov.in/about-the-systems/ayurveda/introduction?.com https://www.britannica.com/biography/Sushruta?.com https://www.britannica.com/biography/Charaka?.com		
5	**Leadership, Ethics & Good Governance**	5.1. Chanakya's Arthashastra and governance- Economic planning and Financial management, Good leadership values and Diplomacy and Foreign Relations required in Modern Relevance. 5.2. Leadership lessons from Ramayana and Mahabharata -Ethics, integrity, and accountability, Conflict management and strategic thinking. 5.3. Emotional intelligence and value-based leadership 5.4. Corporate ethics and responsible decision-making **Self learning Topics: <**Relevance of Arthashastra in modern corporate management>, ,<Chanakya's principles of strategic planning and diplomacy>, <Leadership qualities of Lord Rama>, <Strategic leadership of Lord Krishna>, <Teamwork and loyalty in Ramayana>, <Comparison of leadership styles: Rama vs Ravana and Yudhishthira vs Duryodhana>. Online reference: https://www.scribd.com/document/87505621/Chanakya-Management?.com https://www.scribd.com/document/642719459/leadership https://ebooks.inflibnet.ac.in/hrmp01/chapter/246/ https://www.researchgate.net/publication/395518936_Principles_of_Leadership_from_the_Indian_Epics https://www.icprindia.com/reports/lord-krishnas-strategic-wisdom-leadership-diplomacy-and-statecraft/	10	CO5

6	**Sustainable Practices, Indigenous Technologies for solving current problems** **(Mini Project)**	6.1. Sustainable Indian traditions and lifestyles-Indigenous technologies and eco-friendly innovations 6.2. Water conservation systems in ancient India 6.3. Organic farming and natural resource management 6.4. Circular economy and minimalism in Indian culture 6.5. Sustainable materials and reusable traditional products **Self- learning Topics**: <Sustainable living practices followed in traditional Indian households>,< Concept of minimalism in Indian culture and it's relevance today>, <Importance of seasonal food and local resources>,< Community-based living and resource sharing in villages>, <Traditional Indian practices promoting environmental balance> Online reference: https://sadbhavnadham.org/sustainable-living-practices-from-indian-villages/ https://ijpsl.in/wp-content/uploads/2021/05/Decluttering-the-Future-An-Analysis-of-Indian-Minimalism_Aastha-Rathour-Gunita-Mankame.pdf https://premiummedicalcircle.com/en/artikel/seasonal-nutrition-the-natural-way-to-better-health https://www.smsfoundation.org/community-led-village-development-in-india/ https://earth.org/ancient-indian-wisdom-timeless-lessons-for-modern-sustainability/	10	CO6
		Total	60	

Note:

1. Students are encouraged to explore self-learning topics to gain a deeper understanding of the course. No questions related to self-study topics will be included in the end-semester examination.
2. **Course Engagement (ARCS-based):** The engagement component aims to motivate active student participation in classroom learning. At the end of selected lectures, students will provide short reflective inputs—such as concepts they feel confident about or aspects that brought clarity or satisfaction.

References Books:

1. B. Mahadevan, Vinayak Rajat Bhat and Nagendra Pavan R. N., *Introduction to Indian Knowledge System*, PHI Learning Pvt. Ltd., 2022.
2. A. K. Bag, *History of Technology in India* (Set of 3 Volumes), Indian National Science Academy, 1997.
3. B. Mahadevan, V. R. Bhat and Nagendra Pavana R. N., *An Introduction to Indian Knowledge Systems: Concepts and Applications*, Prentice Hall of India, 2022.
4. Dr. A. R. Saravanakumar, *Indian Knowledge Systems and NEP 2020*, 2022.
5. Vinayak Rajat Bhat and Tejusvi Shukla, *Kautilya's Arthashastra: Timeless Strategies for Modern Governance*, 2023.
6. Amit Jha, *Traditional Knowledge System in India*, 2021.
7. B. N. Goswamy (Ed.), *Indian Art: Forms, Concerns, and Development in Historical Perspective (History of Science, Philosophy and Culture in Indian Civilization)*, Munshiram Manoharlal Publishers Pvt. Ltd., 1999.
8. Bhag Chand Chauhan, *Textbook on The Knowledge System of Bharata*, Garuda Prakashan, 2023.
9. Pradeep Kohle et al. (Ed.), *Pride of India: A Glimpse of India's Scientific Heritage*, Samskrit Bharati, 2006.
10. Maharshi Charaka, *Charaka Samhita*, Chaukhamba Orientalia, Reprint Edition, 2014.
11. Maharshi Sushruta, *Sushruta Samhita*, Chaukhamba Sanskrit Sansthan, Reprint Edition, 2015.
12. Jagadguru Swami Bharati Krishna Tirthaji, *Vedic Mathematics*, Motilal Banarsidass Publishers, Revised Edition, 2016.
13. Narayan Dongre, *Ancient Indian Science and Technology*, 2019.
14. Percy Brown, *Indian Architecture (Buddhist and Hindu Periods)*, D.B. Taraporevala Sons & Co. Pvt. Ltd., Reprint Edition, 2013.

Online References:

Sr. No.	Website Name	URL
1	https://iks.aicte-india.org/index.php	https://iks.aicte-india.org/ebook.php
2	Swayam Chttps://swayam.gov.in/	Preview: Indian Knowledge System(IKS): Concepts and Applications in Engineering | SWAYAM
3	Swayam Chttps://swayam.gov.in/	https://www.mygov.in/campaigns/indian-knowledge-systems
4	https://siddhantaknowledge.org/	https://siddhantaknowledge.org/2026/04/13/indian-knowledge-system-in-universities/
5	https://www.mygov.in//	https://www.mygov.in/campaigns/indian-knowledge-systems

List of suggested module wise activities to be included in practicals:

Module no:	Suggested list of Activities

1	**A.“Then vs Now” Education Debate. (**Students compare Gurukul and modern education systems through a structured debate.) **B.Value Tree Activity (**Students create a “Tree of Values” showing how Indian values strengthen society and professional life.) **C.Role Reversal Classroom (**Students conduct one mini session in Guru-Shishya style learning.)
2	**A.Eco-Friendly Festival Redesign by making poster** (Students redesign festivals like Diwali/Holi/Ganpati using sustainable practices.) **B.Temple Architecture Challenge** (Teams build miniature temple structures using sustainable materials.) **C.Creating artwork,** like decorative materials, cutlery, etc from Traditional materials.
3	**A.Vedic Maths Speed Battle:** Fun game creation using Vedic calculation techniques. **B. Build Your Own Sundial:** Students create a working sundial model.
4	**A. Ayurvedic Lifestyle Planner** Students design healthy daily routines based on Dinacharya principles. **B. Healthy Traditional Food Day** Students showcase traditional immunity-boosting foods. **C. Home Remedy Innovation Activity** Create awareness posters/videos on traditional remedies.
5	**A.Corporate Mahabharata Role-Play** Convert epic conflicts into modern corporate scenarios. **B. Chanakya Strategy Game** Teams solve business/governance problems using Arthashastra principles. **C. Leadership Style Analysis** Compare leadership qualities of Rama, Krishna, Chanakya, and modern CEOs.
6	**A.Waste to Wealth Challenge** Students create useful products from reusable/traditional materials. **B. Smart Village Design Activity** Teams design eco-friendly self-sustainable villages. **C. Green Festival Innovation Challenge** Students create eco-friendly celebration models/products. **D. Sustainable Product Innovation Fair** Prototype exhibition inspired by ancient Indian sustainable practices.

List of suggested class based written assignments :(Any 4 topics can be chosen)

1. **Science Behind Traditions (**Students explain scientific reasons behind Indian traditions and rituals.)
2. **Cultural Storytelling Circle (**Narrate life management lessons from Ramayana, Mahabharata, Panchatantra.)
3. **Ancient Scientist Case study:** Students present as Aryabhata, Sushruta, Bhaskaracharya, etc.
4. **Night Sky Observation Activity:** Observation and documentation of moon phases/constellations.
5. **Herbal Identification Activity:** Identify medicinal plants used in Indian households.
6. **Ethical Dilemma Simulation:** Students solve real-world ethical workplace situations.
7. **Mock Boardroom Discussion:** Students debate profit vs ethics in business decisions.

Assessment:

Self Learning (SL): Self-learning is integrated into both theory and practical subjects, particularly where practical sessions have only one contact hour per week. It emphasizes student-driven efforts in preparing for assignments, practicals, and examinations. Additionally, self-learning includes topics that go beyond the prescribed syllabus to align with current industry trends and requirements. Assessment of self-learning will follow a formative approach. For theory subjects that include term work, 30% of the assignment questions and student presentations/discussions should be based on self-learning topics. In

subjects without term work, the corresponding laboratory sessions will incorporate self-learning, carrying a weightage of 30%. This may involve lab work using the latest tools or preparing technical write-ups/research papers on advanced topics.

Term Work: Term Work shall consist of at least 10-12 Practicals based on the above list. In addition, Term Work Journal must include at least 02 Assignments.

Term Work Marks: 25 Marks (Total) = 15 Marks (Experiments) + 05 Marks (Assignments) + 05 Marks (Attendance).

Justification for B.E. Computer Engineering

.	Necessity for starting the course:	The rapid growth of digital technologies, artificial intelligence, cloud computing, data science, networking, cybersecurity, web technologies, and software development has created an increasing demand for skilled computer science professionals. The proposed curriculum is designed to equip students with strong foundations in mathematics, programming, computing, engineering sciences, communication skills, ethics, and innovation to meet the requirements of modern industries and society. The programme integrates theoretical knowledge, laboratory practice, The curriculum is designed to provide a balanced combination of scientific foundations, computing skills, engineering knowledge, ethical values, practical competence, and innovation capabilities, preparing students for emerging technologies and future professional challenges.
2.	Opportunities of Employability / Employment available after undertaking these courses:	The programme develops skills in programming, software development, data management, networking, web technologies, problem-solving, and innovation, preparing students for careers in:Software and Application Development, Data Analytics and Data Science, Network and System Administration, Cybersecurity.Cloud Computing and Artificial Intelligence, Internet of Things (IoT) and Smart Technologies, Telecommunications and Digital Infrastructure, Research and Development, Technology Consulting and Product Development, Government, Public Sector, and Start-up Ventures, The programme also provides a strong foundation for higher education, research, professional certifications, and emerging technologies such as AI, Machine Learning, Data Science, Cybersecurity, Cloud Computing, and Industry 4.0 The programme also provides a strong foundation for higher studies, professional certifications, research, innovation, and emerging fields such as Artificial Intelligence, Data Science, Cybersecurity, Cloud Computing, Blockchain Technology, Edge Computing, Smart Systems, Industry 4.0, and Human-Centered Computing.

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BoS-Chairman
Science and Humanities

Dr. Vaishali D. Khairnar
Director IQAC
Member Secretary-
AcademicCouncil

Dr. Deven Shah
Principal
Chairman-Academic Council

Letter Grades and Grade Points:

Semester GPA/ Programme CGPA Semester/ Programme	% of Marks	Alpha-Sign/ Letter Grade Result	Grading Point
9.00 - 10.00	90.0 – 100	O (Outstanding)	10
8.00 - < 9.00	80.0 - < 90.0	A+ (Excellent)	9
7.00 - < 8.00	70.0 - < 80.0	A (Very Good)	8
6.00 - < 7.00	60.0 - < 70.0	B+ (Good)	7
5.50 - < 6.00	55.0 - < 60.0	B (Above Average)	6
5.00 - < 5.50	50.0 - < 55.0	C (Average)	5
4.00 - < 5.00	40.0 - < 50.0	P (Pass)	4
Below 4.00	Below 40.0	F (Fail)	0
Ab (Absent)	-	Ab (Absent)	0